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A survey of intrusion detection datasets for communication networks

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Abstract

This paper presents a comprehensive survey of datasets used for intrusion detection in communication networks, examining 124 datasets and characterizing each across 20 key dimensions, such as attack diversity, temporal coverage, data formats, and availability. We systematically analyze how these datasets support research on intrusion detection systems (IDS) in networking environments by reviewing detection approaches, machine and deep learning models, feature selection techniques, traffic analysis tools, and performance metrics. Beyond descriptive comparison, the survey analyzes datasets according to their application domains, readiness for modern AI-driven IDS architectures, realism, quality and benchmarking risks, and sustainability and reproducibility. Our in-depth comparative analysis reveals critical gaps—such as limited real-world traffic representation, inconsistent documentation, dataset bias, evaluation risks, and underrepresentation of emerging attack vectors—and provides actionable recommendations for dataset standardization, high-fidelity data collection, improved labeling, quantitative dataset assessment, and privacy-preserving sharing. Furthermore, we introduce SHIELD, a continuously evolving online repository that supports centralized dataset discovery, comparison, and selection. This survey aims to guide researchers in selecting appropriate datasets for evaluating IDS in communication networks and to inform future efforts toward more realistic, scalable, reliable, and reproducible intrusion detection research.

Keywords Intrusion detection systems (IDS), Communication networks, Network security, Datasets, Machine learning, Internet of Vehicles, Internet of Things, Cybersecurity, Data evaluation

1 Introduction

Intrusion detection systems (IDS) are essential components of modern cybersecurity frameworks, designed to monitor network traffic and identify potential security threats [1]. By analyzing patterns of normal and malicious behavior, IDSs act as a second line of defense, complementing other mechanisms such as encryption, access control, and authentication [2]. With the increasing sophistication of cyberattacks, IDS technologies must continuously evolve to enhance detection accuracy and adaptability.

In the context of communication networks, the role of IDSs becomes even more critical as these systems must analyze high-throughput, heterogeneous, and often real-time

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data streams. A core element in IDS research and development is the use of high-quality datasets that represent both benign and attack traffic under realistic networking conditions. Despite their critical role, identifying suitable datasets remains a challenge due to variations in attack coverage, accessibility, data formats, and alignment with specific research goals.

To address this gap, this study presents the most comprehensive survey to date, reviewing 124 intrusion detection datasets-the largest number analyzed in any such study so far. In contrast to previous surveys, we provide a detailed comparative analysis of how these datasets have been applied in IDS research, including detection approaches, models used, feature selection techniques, performance metrics, and the tools employed during dataset development and evaluation. These tools span stages like data collection, feature processing, storage, inspection, and labeling. Nearly 20 dataset properties have been considered in the comparative study presented in tabular format, providing researchers with a structured reference for evaluating and selecting appropriate datasets.

To streamline dataset discovery and reuse, we introduce Systematic Hub for Intrusion Event and Log Datasets (SHIELD), an interactive online repository (https://shield-datasets.in/) that allows users to search, filter, and retrieve datasets by arbitrary criteria and is continuously updated. This unified resource enhances the reproducibility and reach of IDS evaluation by offering curated metadata and comparative insights for communication network scenarios.

Paper contributions: This paper delivers three primary contributions to the field of intrusion detection research:

- A systematic survey of 124 intrusion detection datasets—one of the largest collections analyzed to date—characterized across 20 key properties, including attack diversity, data format, collection environment, accessibility, labeling, and class balance.
- A comprehensive multidimensional analysis of IDS datasets that extends beyond descriptive comparison by examining their application-domain suitability, readiness for modern AI-driven IDS architectures, realism, quality, bias and benchmarking risks, and sustainability and reproducibility. The survey further examines how these datasets are used in practice, including detection approaches, machine learning and deep-learning models, feature-selection techniques, traffic-analysis tools, and performance metrics.
- SHIELD: the Systematic Hub for Intrusion Event and Log Datasets (https://shield-datasets.in/), an interactive and continuously evolving online repository designed to support centralized dataset discovery, search, filtering, comparison, and selection based on dataset characteristics.*Organization:* This paper is organized as follows: Sect. 2 provides an overview of Intrusion Detection Systems (IDS), including their fundamental concepts, technologies, detection methodologies, and major approaches. Section 3 reviews related work on intrusion detection datasets and identifies the limitations of existing surveys. Section 4 presents the survey methodology, including the search strategy, dataset eligibility and exclusion criteria, and the screening and verification process. Section 5 provides a comprehensive analysis of the collected datasets, including their characteristics, application-domain taxonomy, readiness for modern AI-driven IDS, realism, quality and benchmarking risks, and sustainability and reproducibility. The subsequent sections examine

the practical use of IDS datasets, including preprocessing and feature-selection techniques, machine learning and deep learning models, performance metrics, and network traffic analysis tools. The paper then presents the SHIELD repository developed to support dataset discovery, comparison, and selection. Finally, the challenges and future directions for IDS dataset development and research are discussed, followed by the conclusion, which summarizes the major findings and contributions of the survey.

2 Overview of IDS

Intrusion Detection Systems (IDS) are vital tools for maintaining the security and resilience of communication networks. They monitor traffic across network infrastructures or host systems to identify malicious activities that could compromise the confidentiality, integrity, or availability of data [3]. IDS are capable of detecting a wide range of threats, including malware infections, unauthorized access, denial-of-service (DoS) and distributed denial-of-service (DDoS) attacks, privilege escalation, and protocol abuses, while differentiating between genuine attacks and benign anomalies such as user or system errors [4, 5].

Functioning as an intelligent layer beyond traditional perimeter defenses like firewalls, IDS are typically composed of multiple components-sensors for data collection, analyzers or servers for traffic inspection, and user interfaces for reporting and response. These systems must offer high detection accuracy, low false-positive rates, and real-time responsiveness to be effective in fast-evolving communication environments.

Core IDS functionalities include threat detection, alert generation, event logging, and incident reporting. Depending on their deployment strategy and analytical approach, IDS can be broadly categorized into signature-based and anomaly-based systems, as well as into host-based and network-based architectures [2]. Emerging IDS techniques also leverage hybrid detection models and machine learning-based methods to enhance adaptability and performance in diverse network scenarios [1].

1. Intrusion detection technologies: According to Ozkan-Okay et al. [3] Intrusion Detection Systems (IDS) are categorized into four types based on the detection technology:

   (a) Network-based IDS (NIDS): NIDS monitors network traffic to detect malicious activities by analyzing protocols at various layers (application, transport, and network) [6]. It uses sensors to monitor multiple systems across the network. While NIDS offers isolation and is effective against external attacks, it struggles with encrypted traffic and can miss internal attacks. It may generate both false positives and false negatives, especially during high traffic loads or encrypted sessions.
   (b) Host-based IDS (HIDS): HIDS focuses on monitoring individual hosts by analyzing logs, file modifications, and system activities [7]. Installed directly on the host, it provides deeper insights into internal processes but is vulnerable if attackers gain access to the host. Though effective in detecting insider threats, HIDS can be resource-intensive and prone to false alarms, particularly during routine system changes like reboots or software updates.

(c) Wireless intrusion detection system (WIDS): WIDS monitors wireless network traffic to detect anomalies at the protocol level. It cannot analyze higher-layer protocols like TCP/UDP and is vulnerable to physical attacks and denial-of-service attacks. WIDS typically combines multiple detection techniques (e.g., signature-based and anomaly-based detection) but may miss passive monitoring or offline attacks.

(d) Network behavioral analysis (NBA): NBA systems detect abnormal traffic patterns by analyzing network statistics. They are effective in identifying large-scale attacks like DDoS but struggle with stealthy, slow-developing attacks. NBA relies on analyzing deviations from normal behavior and may experience delays in detecting fast attacks due to data batching. While sensitive adjustments improve detection speed, they increase the likelihood of false positives.

2. Intrusion detection methodologies: Ozkan-Okay et al. [3] classified Intrusion detection methodologies into three main types:

(a) Signature-based method: Signature-based intrusion detection systems (SIDS) are also known as knowledge-based or misuse detection systems [2]. They operate by comparing incoming traffic against a database of known attack signatures, triggering alerts when a match is found. Examples of signatures include specific usernames associated with attacks (e.g., "root"), suspicious email subject lines, or abnormal system logs. While SIDS effectively detect known threats and typically have low false positive rates, they are computationally intensive and require frequent updates to maintain an up-to-date signature database. They struggle to identify zero-day attacks and sophisticated threats that utilize evasion techniques, such as renaming malicious files.

(b) Anomaly-based method: Anomaly-based intrusion detection systems (AIDS), also called heuristic/behaviour-based detection, addresses the limitations of signature-based systems by focusing on deviations from established behavioural norms. These systems analyze traffic patterns to identify anomalies indicative of potential intrusions, including zero-day attacks. By modeling normal behavior and flagging significant deviations, AIDS can detect previously unknown threats. However, they are prone to high false positive rates due to the variability in benign activities. Creating effective anomaly detection systems requires careful rule formulation, and attackers may attempt to manipulate the system by slowly ramping up malicious activity to blend in with normal behavior. Despite these challenges, AIDS remain valuable for detecting unusual user actions and can be categorized into various training methods, including machine learning and statistical techniques.

(c) Stateful protocol analysis: Stateful protocol analysis, also called specification based detection systems, differentiates itself by understanding and monitoring the state of communication protocols. It compares current protocol activities with established models of normal protocol usage. For example, in an FTP session, the system tracks whether a user is authenticated and restricts actions accordingly. While stateful analysis can identify protocol misuse, it may struggle with certain attacks, such as denial-of-service attacks, which execute valid commands at a high rate. This methodology requires significant resources to

manage multiple sessions simultaneously and may vary in effectiveness based on protocol implementations across different operating systems. A lesser-known detection method called as reputation-based detection, blocks traffic from IP addresses or domains associated with malicious or suspicious activities based on their historical behavior. This approach relies on databases that track and score the reputations of IP addresses, URLs, and domains, flagging those involved in spamming, phishing, or distributing malware. In practice, a hybrid approach that combines these methodologies can enhance detection capabilities and provide more robust security against a wider range of cyber threats.

3. Intrusion detection approaches: Ozkan-Okay et al. [3] classified Intrusion Detection Systems (IDS) into several approaches based on their underlying methodologies:

   (a) Statistics-based IDS: This approach establishes a baseline profile of normal network behavior by monitoring transactions. It assigns scores to transactions based on deviations from this profile, triggering alerts when scores exceed a predefined threshold. Statistics-based IDS are classified as: (i) Univariate Models analyze a single metric to detect anomalies. (ii) Multivariate Models assess the relationships among multiple metrics, which enhances detection accuracy but complicates the modeling process. (iii) Time Series Models evaluate data over time, identifying unusual occurrences based on their historical likelihood.
   (b) Rule-based IDS: These systems rely on predefined rules or patterns to identify potential intrusions. Rules can be generated using artificial intelligence techniques that analyze attack patterns. This approach is efficient as it can detect multiple attack types with fewer rules compared to signature-based systems, which require extensive signature databases.
   (c) Heuristic-based IDS: Heuristic IDS focuses on identifying malicious behaviors by establishing a model of acceptable actions. It relies on expert knowledge to distinguish between normal and suspicious activities. Execution traces are analyzed for unusual patterns, enabling the detection of both known and previously unknown (zero-day) attacks.
   (d) Pattern-based IDS: This method detects intrusions by identifying specific patterns or signatures associated with known attacks. The system extracts strings and character sequences from collected data to recognize these patterns. While effective for known attacks, it is less efficient at detecting zero-day attacks due to the absence of predefined signatures.
   (e) Cloud-based IDS: Leveraging cloud computing, this approach facilitates scalable intrusion detection services. It comprises a data collector that gathers network packets, a cloud service for data processing, and a detection engine that analyzes the standardized data. This model enhances the detection capabilities by analyzing network activities from multiple perspectives and allows for concurrent detection of various attack types.
   (f) Machine learning-based IDS: Machine learning techniques automate the data analysis process, employing various algorithms to identify patterns in network traffic. (i) Supervised Learning uses labeled datasets to train the model, allowing it to classify new data as normal or intrusive based on learned patterns. (ii) Unsupervised Learning analyzes unlabeled data, clustering records to identify

anomalies where smaller clusters may indicate intrusions. (iii) Semi-Supervised Learning combines labeled and unlabeled data, improving detection performance while reducing labeling costs.

(g) Deep learning-based IDS: A specialized area within machine learning, deep learning employs multiple layers of processing to automatically extract features from data without manual engineering. This approach is particularly effective in recognizing complex patterns in large datasets, allowing for the classification of new traffic and identification of various attack types.

3 Related work

Several studies have examined datasets used in Intrusion Detection Systems (IDSs). Table 1 presents a summary of these studies, highlighting key aspects such as the number of datasets reviewed, their focus, detection methods, feature selection, models used, performance metrics, dataset Characteristics, and traffic analysis tools. In 2019, Ring et al. [8] surveyed 35 network-based IDS datasets, categorizing them by traffic formats and attack scenarios while identifying 15 key dataset characteristics. However, their study did not include a detailed evaluation of feature selection, machine learning models, or performance metrics, limiting its applicability for IDS benchmarking. In 2020, Ferrag et al. [9] focused on deep learning-based intrusion detection systems, reviewing 35 cybersecurity datasets categorized into seven types. Their study compared deep learning methods, such as recurrent neural networks and convolutional neural networks, with traditional machine learning models like Naïve Bayes and Support Vector Machines but lacked a structured dataset classification across different IDS methodologies. In 2022, Yang et al. [10] reviewed 119 highly cited studies, emphasizing preprocessing techniques, intrusion detection methods, and performance evaluation strategies. Their study aimed to standardize IDS performance assessment but did not offer a structured classification of datasets based on attributes like attack types or dataset usability. In 2023,

Table 1 Related work

S. no.	Study	Year	Total datasets	Scope	Intrusion detection method			Dataset characteristics	Traffic analysis tools
					Feature selection	Model	Performance metrics		
1	Ring et al. [8]	2019	35	Network traffic	✗	✗	✗	✓	✓
2	Ferrag et al. [9]	2020	35	All traffic	✗	✓	✓	✗	✓
3	Yang et al. [10]	2022	52	All traffic	✓	✓	✓	✗	✗
4	Alex et al. [11]	2023	44	IoT	✗	✓	✗	✓	✗
5	De Keersmaeker et al. [12]	2023	74	IoT	✗	✗	✗	✓	✓
6	Daniela Pinto et al. [13]	2025	41	All traffic	✗	✗	✗	✗	✓
7	Our study	2026	124	All traffic	✓	✓	✓	✓	✓

Alexa et al. [11] reviewed 44 IoT-related IDS datasets, categorizing them by taxonomy, dataset format, feature availability, and attack types. While their study highlighted essential dataset attributes such as normal-to-attack traffic ratio and device heterogeneity, it was limited to IoT datasets and did not address those relevant to other IDS domains. Also in 2023, De Keersmaeker et al. [12] analyzed 74 datasets suitable for IoT network security research, classifying them based on network protocols, data collection methodology, and record detail levels. Despite discussing tools for generating IoT network traffic, the study did not explore feature selection, model evaluation, or dataset applicability in IDS performance assessment. In a more recent study published in 2025, Daniela Pinto et al. [13] reviewed 41 datasets and focused on the role of network traffic analysis tools in the development of IDS datasets. The paper provides a detailed review of these tools, outlining their key strengths and limitations, and introduces a categorization framework for understanding their use. It also explains how different tools are combined in the dataset creation process and presents an updated analysis of existing datasets, highlighting the evolution in dataset production. Additionally, the study discusses the impact of dataset features on the performance of intrusion detection systems, emphasizing the importance of high-quality, well-structured data in improving the effectiveness of machine learning-based IDS models. To the best of our knowledge, our study presents the most comprehensive review to date, examining 124 intrusion detection datasets-the largest number covered in any survey thus far. In contrast to previous studies, we offer an in-depth comparative analysis by assessing how these datasets are utilized in IDS research. This includes evaluating the number of datasets reviewed, their scope and focus areas, detection methods applied, feature selection techniques, models employed, performance metrics assessed, dataset characteristics, and the traffic analysis tools used in dataset development and evaluation.

4 Methodology

This study analyzes publicly available intrusion detection datasets, as well as datasets available upon request via a form or email. It aims to provide a structured and in-depth review of their characteristics, applications, and relevance in intrusion detection systems (IDS) research. The objectives of the study include examining widely used datasets, understanding variations in traffic types and attack diversity, exploring preprocessing techniques, identifying feature selection methods, evaluating machine learning and deep learning models applied to the datasets, highlighting dataset limitations, and recognizing emerging trends in IDS dataset development. Additionally, the study develops a comprehensive repository that consolidates IDS datasets and their characteristics, offering researchers a centralized resource for dataset discovery, selection, comparison, and benchmarking.

To gather relevant literature, a well-defined search strategy was constructed around three core concepts: "intrusion detection," "datasets," and "machine learning/deep learning models." The methodology followed several key steps:

1. Identifying primary terms aligned with the research objectives;
2. Expanding these terms by incorporating synonyms, alternate spellings, and related keywords such as "network intrusion detection," "cybersecurity datasets," "intrusion detection benchmark," "IoT intrusion detection dataset," and "supervised learning;"

3. Refining the search terms iteratively by reviewing retrieved studies, dataset papers, survey articles, and dataset repositories and incorporating additional relevant terminology;
4. Structuring the search queries using Boolean operators such as AND and OR.

Example search queries include: "Intrusion Detection System" AND "dataset," "Cybersecurity" AND "dataset" AND "intrusion detection," and "Deep learning" AND "intrusion detection" AND "dataset." To ensure comprehensive dataset coverage, the study draws from five major digital libraries renowned for indexing publications in cybersecurity and computer science. The selected libraries are:

- IEEE Xplore Digital Library (https://ieeexplore.ieee.org)
- ACM Digital Library (https://dl.acm.org)
- SpringerLink (https://link.springer.com)
- Scopus (https://www.scopus.com)
- ScienceDirect (https://www.sciencedirect.com)

These repositories provide access to journal articles, conference papers, book chapters, and dataset publications relevant to IDS research. Additionally, arXiv (https://arxiv.org/) was used to identify preprint publications describing potential datasets that may not yet appear in peer-reviewed sources. Google Scholar (https://scholar.google.com/) was uti lized as a supplementary discovery and citation-tracing tool to identify dataset papers, technical reports, institutional dataset pages, and studies that may not be indexed in the primary databases.

Backward and forward reference tracing was also performed where appropriate. References of dataset papers and previous IDS dataset surveys were examined to identify additional candidate datasets, while subsequent studies citing or applying particular datasets were used to verify dataset usage, accessibility, characteristics, and alternative authoritative sources. Official institutional repositories, research-group websites, and established public dataset repositories were consulted when the original dataset publication did not provide a persistent download location.

This multi-source search strategy was intended to maximize dataset coverage while allowing candidate datasets to be verified against explicit eligibility and exclusion criteria.

4.1 Dataset selection criteria

To improve the transparency and reproducibility of the dataset-selection process, explicit eligibility criteria, exclusion rules, and screening steps were applied. The unit of inclusion was a distinct dataset developed, released, or made available for intrusion detection research. Publications proposing only IDS algorithms, models, or frameworks were not treated as datasets unless they introduced a distinct data resource.

4.1.1 Eligibility criteria

A candidate dataset was included when it satisfied the following criteria:

1. *IDS relevance:* The dataset was developed, released, or demonstrably used for intrusion detection, anomaly detection, malicious traffic detection, or closely related cybersecurity monitoring tasks.

2. *Distinct dataset identity:* The candidate represented an identifiable dataset, benchmark, or traffic collection rather than only an IDS algorithm, framework, or experimental methodology.
3. *Sufficient documentation:* Adequate information was available from an original publication, official repository, institutional webpage, technical report, or scholarly literature to establish the dataset's provenance and essential characteristics.
4. *Accessibility:* The dataset was publicly accessible, available through registration, an access form, or upon request, or had an authoritative source clearly documenting its access conditions.
5. *Availability of dataset characteristics:* Sufficient information was available to extract and verify relevant attributes considered in the survey, such as application domain, traffic type, collection environment, labeling, attacks, features, and availability.
6. *Independent dataset contribution:* Updated versions or variants were considered separately only when they introduced substantial changes in traffic, attack scenarios, collection environments, features, scale, or intended application.

4.1.2 Exclusion criteria

Candidate datasets were excluded when one or more of the following conditions applied:

1. Duplicate records, mirrors, or multiple publications referring to the same underlying dataset without a substantial extension;
2. Data resources unrelated to intrusion detection or closely related cybersecurity detection tasks;
3. Studies proposing IDS algorithms, frameworks, or models without introducing an independently identifiable dataset;
4. Dataset names or resources whose provenance, authoritative source, or existence as a distinct dataset could not be reliably verified;
5. Datasets with insufficient documentation to establish their basic characteristics and relevance;
6. Inaccessible datasets for which no authoritative source provided sufficient information for reliable characterization;
7. Derived, merged, resampled, or transformed collections that were not released and documented as distinct reusable IDS datasets.

4.1.3 Screening and verification process

The dataset screening and verification process consisted of the following stages:

1. *Identification:* Candidate datasets were identified through the selected digital libraries, Google Scholar, arXiv, previous IDS dataset surveys, citation tracing, institutional repositories, research-group websites, and public dataset portals.
2. *Initial screening:* Retrieved records were examined for relevance to IDS dataset research, and clearly irrelevant studies or resources not identifying a dataset were removed.
3. *Source verification:* Dataset identity and provenance were verified using original publications, official repositories, or institutional sources. When an original dataset publication could not be identified, authoritative dataset sources and subsequent scholarly studies were used for verification.

4. *Duplication and eligibility assessment:* Duplicate records were consolidated, dataset versions and extensions were examined for independent contributions, and the remaining candidates were assessed against the predefined eligibility and exclusion criteria.
5. *Attribute extraction and cross-verification:* Dataset characteristics were extracted primarily from original publications and official sources. Secondary studies and survey papers were used, where necessary, to supplement missing information or cross-check dataset characteristics.
6. *Final inclusion:* Datasets satisfying the eligibility criteria after screening, verification, deduplication, and attribute assessment were retained for the comparative analysis and SHIELD repository.

Identification of a dataset through a survey paper or subsequent application study did not automatically result in inclusion. Each candidate was independently assessed to verify its provenance, IDS relevance, distinct dataset identity, and availability of sufficient supporting information. This procedure reduced the risk of including duplicate datasets, IDS algorithms incorrectly presented as datasets, and unsupported dataset references.

The selected datasets were subsequently analyzed across the characteristics reported in the master comparison table, including application domain, data format, traffic representation, collection environment, network size, metadata availability, accessibility, traffic composition, anonymization, dataset size and duration, features, labeling, predefined splits, attack categories, citation counts, and class balance. Additional analyses examine dataset applications, preprocessing and feature-selection methods, ML/DL approaches, realism, quality and benchmarking risks, modern AI-readiness, and sustainability and reproducibility.

Following the screening and verification process, 124 intrusion detection datasets were retained for the final survey. The collection spans nearly three decades of IDS dataset development, beginning with the DARPA 1998 benchmark [14] and extending to recent datasets published in 2026, including CubeSat Cybersecurity, 5GCID, and BigFlow-NIDS. The final collection covers general networks, IoT, IIoT, vehicular, host-based, wireless, mobile, satellite, 5G, and other specialized cybersecurity environments and forms the basis of the comparative analyses and the SHIELD repository.

5 Datasets and their characteristics

Evaluation datasets are fundamental to the development and validation of intrusion detection systems (IDS), as they provide the traffic, system events, and attack scenarios required to train, test, and compare detection approaches. However, the characteristics of IDS datasets vary considerably in terms of application domain, traffic representation, collection environment, attack coverage, labeling, accessibility, realism, and suitability for modern machine learning and deep learning methods. Furthermore, high-fidelity operational traffic is often difficult to release publicly because of privacy, confidentiality, and security concerns. Consequently, researchers frequently rely on publicly available benchmark datasets collected from simulated, emulated, synthetic, or controlled environments.

This section presents a structured and multidimensional analysis of the 124 intrusion detection datasets retained through the selection and screening process described in Sect. 4. Figure 1 provides an overview of these datasets, illustrating their distribution

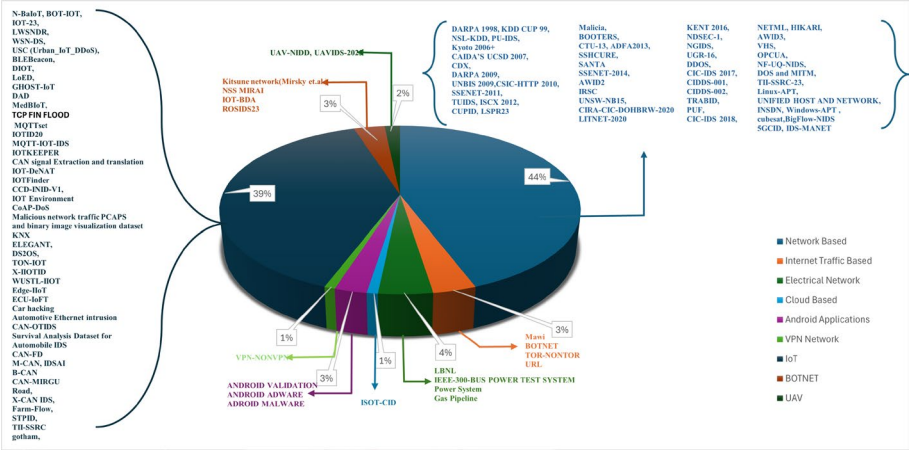


Fig. 1 Dataset categories

across the major application domains considered in this review. Section 5.1 defines the characteristics used to systematically compare the datasets and presents the comprehensive master table. Section 5.2 classifies the datasets according to their application domains and provides detailed descriptions of the individual datasets. The subsequent analyses move beyond descriptive comparison by evaluating representative benchmark datasets from several complementary perspectives. Section 5.3 examines their readiness for modern AI-driven IDS architectures, Section. 5.4 assesses different dimensions of dataset realism, and Section. 5.5 analyzes dataset quality, bias, leakage risks, and benchmarking pitfalls. Finally, Section. 5.6 evaluates dataset sustainability and reproducibility in terms of accessibility, documentation, labeling transparency, feature-generation information, and long-term availability.

Together, these analyses provide a structured progression from the fundamental characteristics and application domains of IDS datasets to their practical suitability, realism, reliability, benchmarking risks, and long-term research usability. This multidimensional assessment is intended to help researchers understand not only what datasets are available, but also their strengths, limitations, and suitability for different intrusion detection research requirements (.

5.1 Dataset characteristics

To make it easier for researchers to evaluate different intrusion detection datasets and select the most appropriate one for their specific needs, it is necessary to define a set of common attributes that can be used as a basis for evaluation. Therefore, we examine the typical properties of the data sets that have been used in previous studies to evaluate the data sets for intrusion detection. To support this analysis, we have created a comprehensive Table 2, which examines approximately 124 datasets against these identified properties, providing a clear and structured comparison for researchers [15].

1. *Year of creation:* The year of creation of a dataset indicates the time when the data was collected or generated for the first time. This information can provide important context and help researchers understand the temporal aspects of the dataset. Additionally, knowing the year of creation can help determine the relevance and usefulness of the dataset for certain applications. It covers the datasets from 1998 to 2026.

Table 2 Dataset properties

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Metadata	Compnet	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
1	1998	DARPA98 [14]	TCP-dump	Network	Packet	Simulated	Small	Yes	Yes	Yes	Yes	Yes	No	4GB	5/7 weeks	43	Yes	Yes	Probe, DoS, R2L, U2R	1573	No
2	1998	KDD-CUP [16]	TCP-dump	Network	Flow	Simulated	Small	No	Yes	Yes	Yes	Yes	No	5M points	N/A	43	Yes	Yes	Probe, DoS, R2L, U2R	948	No
3	1998	PU-IDS [17]	.csv	Network	Flow	Synthetic	Small	No	Yes	Yes	Yes	Yes	No	200K points	N/A	42	Yes	No	Probe, DOS, R2L, U2R	31	No
4	2001	MAWI [18]	TCP-dump	Internet	Packet	Real	Large	Yes	Yes	Yes	Yes	Yes	Yes	30M flows	15 min/day	41	Yes	N/A	DoS, DDoS, Scan,etc	7	No
5	2004	LBNL [19]	.csv	Electrical network	Packet	Real	Enterprise	No	Yes	Yes	Yes	Yes	Yes	160M packets	3 months	12	No	No	Storm Botnet, etc	87	No
6	2006	Kyoto[20]	Tcp-dump	Network	Other	Real	Honeypot	No	No	Yes	Yes	Yes	Yes	93M Points	3 years	24	Yes	No	N/A	115	No
7	2007	CAIDA-UCSD [21]	.pcap	Network	Packet	Real	Large	Yes	No	Yes	No	Yes	Yes	5.3GB	1 h	43	Yes	N/A	DoS	78	No
8	2008	Twente [22]	NetFlow	Network	Flow	Real	Honeypot	Yes	No	Yes	No	Yes	Yes	14.2M flows	6 days	10	Yes	No	Brute force, scan, etc	242	No
9	2009	NSL-KDD [23]	.ARFF, .csv	Network	Flow	Simulated	Small	No	Yes	Yes	Yes	Yes	No	150K points	N/A	43	Yes	Yes	Probe, DOS, R2L, U2R	7677	No
10	2009	CDX [24]	.csv	Network	Packet	Real	Small	Yes	Yes	Yes	Yes	Yes	No	14GB pakets	4 days	N/A	No	No	Scan, DoS, phishing, etc	200	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Metadata	Compnet	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
11	2009	DARPA 2009 [25]	.pcap	Network	Packet	Simulated	Medium	Yes	Yes	Yes	Yes	Yes	Yes	6.5TB	10 days	N/A	Yes	N/A	DoS, malware, scans, etc	34	No
12	2009	UNIBS [26]	.pcap	Network	Flow	Real	University	No	Yes	Yes	Yes	No	Yes	27GB	3 days	N/A	No	No	N/A	270	No
13	2010	ISOT-CID [27]	.pcap	Cloud	Packet	Emulated	Small	Yes	Yes	Yes	Yes	Yes	N/A	55.2GB	1-2 h/day	164	Yes	No	N/A	37	No
14	2010	IEEE-300-BUS POWER TEST SYSTEM [28]	Raw data files	Electrical network	Graph data	Simulated	Large	Yes	Yes	Yes	Yes	N/A	N/A	300 nodes 409 links	N/A	N/A	N/A	No	N/A	312	N/A
15	2010	BOTNET [29]	.pcap	Internet	Packet	Simulated	Large	Yes	Yes	Yes	Yes	Yes	No	14GB	5 h/day	86	Yes	Yes	Spam, DDOS, etc	130	No
16	2010	CSIC-HTTP [30]	.csv	Network	Flow	Simulated	Medium	Yes	No	Yes	Yes	Yes	N/A	223,585 count	N/A	30	Yes	Yes	SQL injection, buffer overflow, etc	105	N/A
17	2010	LWSNDR [31]	.csv, .json	IOT	Sensor data	Real	Small	Yes	Yes	Yes	Yes	Yes	No	N/A	6 h	N/A	Yes	N/A	N/A	208	No
18	2011	SSENET-2011 [32]	N/A	Network	Other	Simulated	Small	No	Yes	Yes	Yes	Yes	No	N/A	4 h	23	Yes	No	Probing, flooding, etc	89	No
19	2011	TUIDS [33]	N/A	Network	Packet, flow	Simulated	Medium	No	Yes	Yes	Yes	Yes	No	250k flows	21 days	N/A	Yes	Yes	Botnet, DDOS, scan, brute force	206	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Meta data	Comp net	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
20	2012	ISCXIDS 2012 [34]	.pcap	Network	Packet, flow	Simulated	Small	Yes	Yes	Yes	Yes	Yes	No	2M	7 days	19	Yes	No	DoS, DDoS, brute force, infiltration	1753	No
21	2012	Malicia [35]	.csv	Network	Flow	Real	Large	Yes	No	Yes	No	Yes	Yes	11,363 samples	1 year	7	Yes	N/A	N/A	142	No
22	2013	Booters [36]	N/A	Network	Packet	Real	Small	No	No	Yes	No	Yes	Yes	250GB	2 days	N/A	No	No	DDOS	280	No
23	2013	CTU-13 [37]	.pcap, Biargus	Network	Packet, flow	Real	University	Yes	Yes	Yes	Yes	Yes	Yes	81M flows	125 h	15	Yes	No	CTU-Malware-Capture-Botnet	1250	No
24	2013	ADFA [38]	.csv	Host	Other	Simulated	Small	Yes	N/A	Yes	Yes	Yes	No	5952 instances	N/A	N/A	Yes	N/A	DoS, DDoS, Brute-force, and Infiltration	541	No
25	2013	SSHCURE [39]	N/A	Network	Flow	Real	University	No	Yes	Yes	Yes	Yes	Yes	2.4GB flows	2 months	N/A	Yes	No	SSH attacks	98	No
26	2013	Gas Pipeline [40]	.csv, .ARFF	Signal	RAW data	Real	Small	Yes	No	Yes	Yes	Yes	N/A	N/A	N/A	6 and 20	Yes	Yes	N/A	262	No
27	2014	SANTA [41]	N/A	Network	Other	Real	ISP	No	Yes	Yes	Yes	Yes	Yes	N/A	N/A	28	Yes	N/A	DDoS, scans, flooding, etc	43	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Meta data	Comp net	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
28	2014	SSENET-2014 [42]	N/A	Network	Other	Simulated	Small	No	Yes	Yes	Yes	Yes	No	200K points	4 h	28	Yes	Yes	Flooding, probe, Bot, Dos, etc.	24	Yes
29	2014	Android Validation [43]	Apk files	Android application	Other	Simulated	N/A	Yes	No	Yes	Yes	N/A	N/A	792 samples	N/A	N/A	Yes	N/A	N/A	97	No
30	2014	Power System [44]	.csv, .ARFF	Signal	RAW data	Real	Large	Yes	Yes	Yes	Yes	Yes	N/A	2M tuples	N/A	29	Yes	Yes	Disabling relay function, SLG fault replay, etc	578	No
31	2014	Gas Pipeline and Water storage tank [45]	.ARFF	Signal	Other	Real	Small	Yes	No	Yes	Yes	Yes	No	214580,60048 pkts	N/A	N/A	Yes	Yes	Injection, DOS etc	578	No
32	2015	AWID2 [46]	.csv	Network	Other	Simulated	Small	Yes	Yes	Yes	Yes	Yes	No	37M packets	1 h	156	Yes	Yes	Availability, MITM, etc.	729	No
33	2015	IRSC [47]	.pcap	Network	Packet, flow	Real	Production N/W	No	Yes	No	Yes	Yes	N/A	N/A	N/A	N/A	Yes	N/A	Controlled and uncontrolled attacks	63	N/A

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Metadata	Compnet	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
34	2015	UNSW-NB15 [48]	.pcap, .Bro, .csv	Network	Packet	Simulated	Small	Yes	Yes	Yes	Yes	Yes	No	100GB/2M points	31 h	49	Yes	Yes	Fuzzers, analysis, backdoors, DoS, etc.	6050	No
35	2016	KENT [49]	N/A	Network	Flow, logs	Real	Enterprise	No	Yes	Yes	Yes	N/A	Yes	130M flows	58 days	N/A	No	No	N/A	55	No
36	2016	NDSEC-1 [50]	.pcap, .csv	Network	Packet, logs	Simulated	Small	No	Yes	Yes	No	Yes	No	3.5M	N/A	N/A	Yes	No	Botnet, brute force, flooding, etc.	42	No
37	2016	NGIDS [51]	.pcap	Network	Packet, logs	Simulated	Small	No	Yes	Yes	Yes	Yes	No	1M	5 days	N/A	Yes	No	DOS, DDOS	277	No
38	2016	UGR-16 [52]	.csv	Network	Flow	Real	ISP	Yes	Yes	Yes	Yes	Yes	Yes	16900M	4 months	N/A	Yes	Yes	Botnet, DoS, scans, brute force, etc	339	No
39	2016	TOR-NONTOR [53]	.pcap, .csv	Internet	Time based	Real	Large	Yes	No	Yes	Yes	Yes	Yes	N/A	N/A	23	Yes	Yes	N/A	1319	No
40	2016	ISCX-URL-2016 [54]	.csv	Internet	N/A	Real	Large	Yes	No	Yes	Yes	Yes	N/A	114400 urls	N/A	79	Yes	Yes	URL attacks	320	No
41	2016	VPN-NONVPN [55]	.pcap	VPN network	Flow	Real	Medium	Yes	No	Yes	Yes	N/A	Yes	28GB	N/A	6 groups	Yes	N/A	N/A	385	Yes
42	2016	DDOS [56]	.csv, .pcap	Network	Packet	Synthetic	N/A	No	N/A	Yes	Yes	Yes	Yes	2.1M	N/A	27	Yes	No	DDOS	1686	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Meta data	Comp net	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
43	2016	WSN-DS [57]	.csv	Network	Flow	Simulated	Small	Yes	No	Yes	Yes	Yes	N/A	374,661 records	1 h	23	Yes	Yes	Flooding etc.	540	No
44	2016	USC (Urban-IoT-DDoS) [58]	.csv, .txt, .avi	IOT	Sensor data	Real	Large	Yes	No	Yes	Yes	Yes	Yes	4060 nodes	1 month	3	Yes	Yes	DDOS	22	No
45	2017	CIC-IDS 2017 [59]	.csv	Network	Packet, flow	Simulated	Small	Yes	Yes	Yes	Yes	Yes	No	3.1M flows	5 days	80	Yes	No	Brute force, portscan, botnet, Dos, DDoS, etc	7071	No
46	2017	CIDDS-001 [60]	.csv	Network	Flow	Simulated + Real	Small	Yes	Yes	Yes	Yes	Yes	Yes	32M Flows	28 days	14	Yes	No	Scan, DoS, brute force	18	No
47	2017	CIDDS-002 [61]	.csv	Network	Flow	Simulated	Small	Yes	Yes	Yes	Yes	Yes	Yes	15M flows	14 days	14	Yes	No	Scan, DoS, brute force	133	No
48	2017	TRABID [62]	.pcap, .ARFF	Network	Packet	Simulated	Small	No	Yes	Yes	Yes	Yes	Yes	460M packets	8 h	50	Yes	Yes	DoS, probing	212	No
49	2017	Unified Host and Network [63]	.csv	Network, host	Logs, flow	Real	Enterprise	No	Yes	Yes	Yes	N/A	Yes	150GB flows	90 days	21	No	No	N/A	214	No
50	2017	Android Adware [64]	.pcap, .csv	Android application	Packet	Real	Small	Yes	Yes	Yes	Yes	Yes	Yes	395828 instances	N/A	9	Yes	Yes	Shaunet, Mobidash etc.	191	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Meta data	Comp net	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
51	2017	Android Malware [65]	.pcap	Android application	Packet	Real	Small to medium	Yes	No	Yes	Yes	Yes	Yes	N/A	7 days	N/A	Yes	N/A	SMS malware, scareware, etc	396	No
52	2017	CAN-OTIDS [66]	.txt	IOV	Packet	Real	Small	Yes	No	Yes	Yes	Yes	N/A	34.8MB	N/A	12	Yes	N/A	DoS, fuzzing, etc.	583	No
53	2018	PUF [67]	N/A	Network	Flow	Real	University	No	No	No	Yes	Yes	Yes	300K flows	3 days	16	Yes	No	DNS attack	35	No
54	2018	CIC-IDS-2018 [68]	.csv	Network	Flow	Real	Large	Yes	Yes	Yes	Yes	Yes	Yes	N/A	N/A	80	Yes	Yes	Brute force, botnet, Dos, etc.	7095	No
55	2018	DS2OS [69]	.csv	IIOT	Packet	Real	Large	Yes	Yes	Yes	Yes	Yes	Yes	357952 samples	24 h	13	Yes	N/A	DoS, malicious control, scan, spying, etc	222	Yes
56	2018	N-BAIOT [70]	.csv	IOT	Packet	Real	Medium	Yes	No	Yes	Yes	Yes	N/A	N/A	N/A	23	Yes	Yes	DoS, DDOS, etc	2134	No
57	2018	Survival Analysis [71]	.txt	IOV	Packet	Real	Small	Yes	No	Yes	Yes	Yes	N/A	5.4MB	20 s	12	Yes	Yes	Flooding, fuzzing, etc	262	No
58	2018	BLEBeacon [72]	.csv	IOT	Signal data	Real	Medium	Yes	No	N/A	Yes	No	N/A	N/A	33 days	5	No	No	N/A	27	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Meta data	Comp net	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
59	2018	Kitsune Network [73]	.csv, .pcap	Botnet	Packet	Real	Medium	Yes	No	Yes	Yes	Yes	Yes	N/A	N/A	23	Yes	Yes	Recon, man in the middle, DoS and botnet malware	2093	No
60	2018	OPCUA [74]	.csv	Network	Flow	Real	Medium	Yes	No	N/A	Yes	Yes	Yes	24.2MB	N/A	32	Yes	Yes	DoS, MITM, etc	11	No
61	2019	TCP FIN FLOOD [75]	.csv, .pcap	IOT	Packet	Real	Small	Yes	No	Yes	Yes	Yes	Yes	823MB	5 min	14	Yes	Yes	TCP FIN flood attack pattern	11	No
62	2019	DIOT [76]	.csv, .pcap	IOT	Packet	Real	Small	Yes	No	No	Yes	Yes	Yes	8.7GB	1 weeks	7	Yes	Yes	Infection, DoS etc.	1032	No
63	2019	BOT-IOT [77]	.csv	IOT	Flow	Real	Large	Yes	Yes	Yes	Yes	Yes	Yes	16.7GB	N/A	32	Yes	Yes	DDoS, DoS, Scan, etc.	2500	No
64	2019	CUPID [78]	.pcap	Network	Packet	Real	Enterprise	Yes	No	Yes	Yes	Yes	Yes	50GB	24 h	84	Yes	Yes	DoS, R2L, U2R, probe	32	No
65	2020	Car Hacking [79]	.csv	IOV	Packet	Real	Small	Yes	No	Yes	Yes	Yes	N/A	Yes	2.3 h	6	Yes	Yes	DoS, fuzzy, etc.	812	No
66	2020	CIRA-CIC-DOH-BRW [80]	.pcap	Network	Packet	Simulated	Medium	Yes	Yes	Yes	Yes	Yes	N/A	N/A	N/A	28	Yes	Yes	MITM, DNS, etc.	337	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Meta data	Comp net	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
67	2020	INSDN [81]	.pcap, .csv	Network	Time based	Simulated	Small	Yes	Yes	Yes	Yes	Yes	N/A	343939 instances	N/A	83	Yes	N/A	DOS, brute force	537	No
68	2020	IOT-23 [82]	.pcap	IOT	Packet	Real	Small	Yes	Yes	Yes	Yes	Yes	N/A	20GB	N/A	21	Yes	Yes	C&C, DDoS, scan, C&C, Okiru, etc.	7	No
69	2020	LITNET-2020 [83]	.csv	Network	Flow	Real	Large	Yes	Yes	Yes	Yes	Yes	Yes	1.3GB	10 months	85	Yes	N/A	Botnet, brute-force, DoS, etc.	168	No
70	2020	NETML [84]	.pcap	Network	Flow	Real	Large	Yes	Yes	Yes	Yes	Yes	Yes	484056 flows	N/A	48	Yes	Yes	N/A	50	No
71	2020	LOED [85]	.csv	IOT	Packet, signal	Real	Large	Yes	Yes	Yes	Yes	No	N/A	11,263,001 packets	4 months	13+	Yes	N/A	N/A	15	No
72	2020	GHOST-IOT [86]	.csv	IOT	Flow, packet	Real	Small	Yes	Yes	Yes	Yes	Yes	Yes	3.8M	10 days	112	Yes	Yes	IP based attacks	45	N/A
73	2020	DAD [87]	.csv	IOT	Flow	Real	Small	Yes	Yes	Yes	Yes	Yes	No	10k: pkts, 67k: flows	7 days	N/A	Yes	Yes	Duplication, interception, etc.	29	N/A
74	2020	MEDBIOT [88]	.pcap	IOT	Flow	Real	Medium	Yes	Yes	Yes	Yes	Yes	Yes	17,845,567 packets	6 days	100+	Yes	N/A	Mirai, Bashlite, etc.	219	N/A
75	2020	MQTTset [89]	.csv	IOT	Flow	Real	Small	Yes	Yes	Yes	Yes	Yes	Yes	11,915,716 packets	1 week	34	Yes	Yes	Brute force, DOS, etc.	387	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Metadata	Compnet	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
76	2020	IOTID20 [90]	.pcap, .csv	IOT	Flow	Real	Small	Yes	Yes	Yes	Yes	Yes	Yes	625783 instances	N/A	83	Yes	Yes	DOS, Mirai, MITM, Scan	87	No
77	2020	MQTT-IOT-IDS2020 [91]	.pcap	IOT	Packet	Simulated	Small	Yes	Yes	Yes	Yes	Yes	Yes	1.66GB	N/A	N/A	Yes	N/A	scan, brute-force, MQTT brute-force, etc.	3	No
78	2020	NF-UQ-NIDS [92]	.csv	Network	Flow	Real	Large	Yes	N/A	Yes	Yes	Yes	Yes	1GB	N/A	12	Yes	N/A	Backdoor, benign, bot, brute force, DoS, etc.	653	No
79	2020	IOT-KEEPER [93]	.pcap	IOT	Packet	Real	Large	Yes	Yes	Yes	Yes	Yes	N/A	15GB	72 h	38	No	N/A	Scanning, MITM, malware, DOS, etc.	248	No
80	2020	IOT-DeNAT [94]	.csv	IOT	Flow	Real	Medium	Yes	Yes	Yes	Yes	N/A	Yes	304MB	37 days	22	Yes	N/A	N/A	107	No
81	2020	IOT-Finder[95]	.pcap	IOT	Flow	Real	Large	Yes	N/A	No	Yes	N/A	Yes	367MB	1.5 months	N/A	Yes	N/A	N/A	193	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Metadata	Compnet	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
82	2020	ROAD [96]	.txt	IOV	RAW CAN	Real	Small	Yes	Yes	Yes	Yes	Yes	Yes	55.67MB	3.5 h	3	Yes	N/A	Fuzzing, spoofing, masquerading	124	No
83	2021	X-IIOTID [97]	.pcap, .csv	IIOT	Flow	Real	Medium	Yes	Yes	Yes	Yes	Yes	Yes	106.71MB	4 months	67	Yes	Yes	Reconnaissance, exploitation, RDoS, etc.	354	N/A
84	2021	CAN signal [98]	.txt	IOV	Packet	Real	Small	Yes	No	Yes	Yes	N/A	No	40 dump files	N/A	4	N/A	N/A	N/A	38	N/A
85	2021	CAN-FD [99]	.csv	IOV	Packet	Real	Small	Yes	Yes	Yes	Yes	Yes	No	96.8MB	1 h	12	Yes	N/A	Flooding, fuzzing, malfunctioning	7	No
86	2021	M-CAN [100]	.csv	IOV	Packet	Real	Small	N/A	N/A	Yes	Yes	Yes	N/A	16.08MB	36 min	12	Yes	N/A	DOS, fuzzing	10	No
87	2021	B-CAN [100]	.csv	IOV	Packet	Real	Small	N/A	N/A	Yes	Yes	Yes	N/A	66.86MB	24 min	12	Yes	N/A	DOS, fuzzing	10	No
88	2021	HIKARI [101]	.csv, .pcap	Network	Packet, flow	Real	Medium	Yes	Yes	Yes	Yes	Yes	Yes	N/A	39 h	86	Yes	N/A	Brute force and probing, SQL injection, Dos	139	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Meta data	Comp net	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
89	2021	TON-IOT [102]	.csv, .pcap	IOT/IIOT	Packet, flow	Real	Large	Yes	Yes	Yes	Yes	Yes	Yes	23 files	N/A	44	Yes	Yes	Scanning attack, DoS, DDoS, MITM, etc.	852	No
90	2021	USB-IDS-1 [103]	.csv	Network and apps	Flow	Real	Medium	Yes	No	Yes	Yes	Yes	N/A	16 csv filse	10min	84	Yes	N/A	DoS	32	No
91	2021	AWID3 [104]	.csv, .pcap	Network	Packet	Real	Enterprise	Yes	Yes	Yes	Yes	Yes	No	2.5M frames	10 min	256	Yes	N/A	Deauth, evil twin, ARP	168	No
92	2021	Automotive Ethernet Intrusion [105]	.csv, .pcap, logs	IOV	Packet	Simulated + Real	Small	Yes	Yes	Yes	Yes	Yes	N/A	N/A	70 min	20+	Yes	Yes	Replay attack	94	No
93	2021	WUSTL-IIOT [106]	.csv, .pcap	IIOT	Packet, flow	Real	Small	Yes	No	Yes	Yes	Yes	N/A	2.7GB	53 h	23	Yes	Yes	Injection, reconnaissance, DoS	646	No
94	2021	CCDINID-V1 [107]	.pcap, .csv	IOT	Flow	Simulated	Small	Yes	No	No	Yes	Yes	N/A	50GB	N/A	83	Yes	Yes	MITM, DoS, DDoS, etc.	73	No
95	2021	IOTEnvironment [108]	.pcap	IOT	Packet	Real	Small	Yes	No	Yes	Yes	Yes	N/A	287MB	34 min	N/A	Yes	Yes	Port scan, flooding etc.	N/A	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Metadata	Compnet	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
96	2021	ELEGANT (DOS AND MITM) [109]	.pcap, .csv	IoT	Packet, flow	Simulated	Small	Yes	No	Yes	Yes	Yes	N/A	289GB	N/A	N/A	Yes	Yes	DOS, MITM	6	No
97	2021	NSS MIRAI [110]	.csv, .pcap	Botnet	Other	Simulated + Real	Large	Yes	No	Yes	Yes	Yes	N/A	181MB	N/A	13	Yes	Yes	Port scan, malware loader, DDos, noise	80	No
98	2021	IOT-BDA [111]	.csv, .json, .pcap	Botnet	Packet	Real	Large	Yes	No	Yes	N/A	Yes	N/A	25.35GB	7 months	N/A	Yes	N/A	Port scanning, DDoS traffic, etc.	74	No
99	2021	IoT-CSIDDS [112]	.csv	IOT	Packet	Real	Small	Yes	Yes	Yes	Yes	Yes	N/A	15.2MB	N/A	20	Yes	N/A	False data injection attack (FDIA) and flooding attack	11	No
100	2021	Malicious Network traffic PCAPS and BIV [113]	.pcap, .csv, .jpg, .zip	IOT	Packet, Flow	Simulated + Real	Small	Yes	No	No	Yes	Yes	N/A	125MB	N/A	N/A	Yes	Yes	DDoS, backdoor, brute-force, etc	72	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Meta data	Comp net	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
101	2022	VHS-22 [114]	.csv, .pcap	Network	Flow	Simulated + Real	Large	Yes	No	Yes	Yes	Yes	No	27.7M flows	8 days	45	Yes	Yes	DoS, botnet, etc.	18	No
102	2022	ECU-IOFT [115]	.pcap, .csv	IOFT	Packet, flow	Real	Small	Yes	No	Yes	Yes	Yes	N/A	54,492 instances	3 min/ session	10	Yes	Yes	Wifi de-authentication, etc.	41	No
103	2022	COAP-DOS [116]	.csv, .pcap	IOT	Packet, flow	Real	Small	Yes	No	Yes	Yes	Yes	N/A	661,304 packets	16 h	17	Yes	Yes	DDOS	16	No
104	2022	Edge-IIOT [117]	.pcap, .csv, .txt, .zip	IIOT	Packet, flow	Simulated + Real	Large	Yes	No	Yes	Yes	Yes	N/A	11 million packets,1GB	50 days	61	Yes	Yes	DoS/ DDoS, MITM, etc.	1343	No
105	2022	KNX [118]	.pcap, .csv	IOT	Packet, flow	Real	Small	Yes	No	Yes	Yes	Yes	N/A	289MB	2 months	21	N/A	N/A	Scan, injection, etc.	28	No
106	2022	IDS-MANET [119]	.pcap	MANET network	Packet	Simulated	Small	Yes	Yes	Yes	Yes	Yes	Yes	9.7MB	N/A	21	Yes	N/A	Black-hole, worm-hole	28	No
107	2023	IDSAI [120]	.pcap, .csv	IOT	Flow	Simulated + Real	Small to medium	Yes	No	Yes	Yes	Yes	N/A	1000,000 samples	N/A	21	Yes	Yes	DDoS, brute force, etc.	64	Yes
108	2023	STPID [121]	Other	IOT	Image frames	Simulated + Real	Small	Yes	No	Yes	Yes	Yes	N/A	17000 frames	15 days	N/A	Yes	Yes	Climbing, tapping, etc.	26	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Metadata	Compnet	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
109	2023	TII-SSRC [122]	.pcap, .csv	Network	Packet	Simulated	Small	Yes	No	Yes	Yes	Yes	N/A	27.5GB	N/A	75	Yes	Yes	DoS, Mirai botnet, etc.	56	No
110	2023	ROSIDS23 [123]	.pcap	Botnet	Packet, flow	Real	Small to medium	Yes	No	Yes	Yes	Yes	N/A	4.1GB	4 h	83	Yes	Yes	DoS, flood, etc.	18	No
111	2024	CICIOV 2024 [124]	.csv	IOV	Packet	Real	Small	Yes	Yes	Yes	Yes	Yes	N/A	62MB	N/A	12	Yes	No	DoS, steering wheel spoofing, RPM spoofing, gas spoofing, speed spoofing	188	No
112	2024	X-CANIDS [125]	.parquet	IOV	Signals	Real	Small to medium	Yes	No	Yes	Yes	Yes	N/A	4.44GB and 1.11GB	N/A	N/A	Yes	N/A	Fuzzing, replay, masquerade, etc.	64	No
113	2024	LINUX-APT [126]	.csv, xml, xlsx	Network	Other	Simulated	Small to medium	Yes	No	Yes	Yes	Yes	N/A	199MB	3 months	17	Yes	N/A	Apache Struts, keyloggers, etc.	35	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Meta data	Comp net	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
114	2024	CAN-MIRGU [127]	.csv	IOV	RAW CAN data	Real	Medium	Yes	No	Yes	Yes	Yes	N/A	N/A	17+ h	5	Yes	N/A	Dos, fuzzing, spoofing, replay	47	No
115	2024	LSPR23 [128]	.csv	Network	Flow	Real	Large	Yes	Yes	Yes	Yes	Yes	N/A	704GB	2 days	94	Yes	Yes	Privilege escalation, system compromise, data theft, website defacement, APT-like attacks, etc	6	No
116	2024	CICIoMT 2024 [129]	.csv	IoMT	Packet	Real + Synthetic	Medium	Yes	Yes	Yes	Yes	Yes	N/A	176MB	N/A	45	Yes	Yes	DDoS, DoS, reconnaissance, MQTT, spoofing	200	No
117	2025	Windows-APT 2025 [130]	.csv	Network	Host and network logs	Real	Small	Yes	Yes	Yes	Yes	Yes	N/A	52.4MB	2 months	N/A	Yes	N/A	MITRE attcks	2	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Meta data	Comp net	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
118	2025	Farm-Flow [131]	.csv	AG-IOT	Flow	Simulated	Small	Yes	N/A	Yes	Yes	Yes	Yes	532MB	5 s intervals	101	Yes	Yes	Spoofing, botnet DDoS, flood, scanning	25	No
119	2025	Gotham [132]	.pcap, .csv	IOT	Packet, feature	Realistic, simulated	large	Yes	Yes	Yes	Yes	Yes	N/A	3,134,004 packets	20 min to 2.5 h	23	Yes	Yes	DoS, brute force, CoAP amplification, Mirai infection, etc.	17	No
120	2025	UAV-NIDD [133]	.pcap	UAV network	Packet, feature	Realistic	Small	Yes	Yes	Yes	Yes	Yes	N/A	1.7GB	5 days	45 (UAV), 51 (AP), 85 (GCS)	Yes	N/A	Scanning, reconnaissance, DoS, DDoS, GPS jamming, etc.	41	No

Table 2 (continued)

S.no.	Year	Dataset	Type	Category	Format	Traffic	Network size	Metadata	Compnet	Availability	Normal	Attack	Anonymity	Count	Duration	Feature	Labeled	Pred. splits	Attacks type	Citations	Balanced
121	2025	UA-VIDS-2025[134]	.csv	UAV network	Flow	Simulated	Small	Yes	Yes	Yes	Yes	Yes	N/A	20.1MB	600 s per simulation	22	Yes	N/A	Black hole, flooding, Sybil, wormhole	19	No
123	2026	5GCID [135]	.csv	Network	N/A	N/A	Large	N/A	Yes	Yes	Yes	Yes	N/A	N/A	N/A	28	N/A	N/A	AMF DoS, UPF DoS, and SMF DoS cases	0	No
124	2026	cubesat cybersecurity [136]	.csv	Satellites	Packet	Simulated	Small	Yes	Yes	Yes	Yes	Yes	N/A	4.23 MB	N/A	31	Yes	N/A	Command flooding, data injection, etc.	1	No

2. *Dataset:* It refers to the official or commonly recognized title of a collection of data. It identifies the specific dataset being used, analyzed, or referenced, helping users distinguish among multiple datasets and track the source or version of the data. We have included active links to the associated research papers for reference, ensuring proper attribution and easy access to the original sources.
3. *Type of dataset:* It refers to the format or structure in which the data is captured, stored, or represented. Common types include PCAP (packet capture files used for network traffic analysis), TCPDump (text-based output of captured packets), and CSV (Comma-Separated Values, a tabular format used for structured data). This column helps identify how the data is formatted for processing, analysis, or conversion.
4. *Network category:* It refers to the domain or environment in which a data instance, event, or attack occurs. It classifies the context into areas such as Network (targeting network infrastructure), Electrical (related to power systems), IoV (Internet of Vehicles, involving connected vehicles), IoT (Internet of Things, covering smart devices), and Host (focusing on endpoints like servers or computers). This categorization helps in understanding the operational scope and impact area of the data or activity.
5. *Format:* This feature describes the traffic format, which is captured either in flow-based or packet-based format. Packet-based traffic includes payload information (pcap), while flow-based traffic only has meta-information about network connections (Netflow). Another type of traffic includes flow-based traces supplemented by additional attributes from packet-based data or host-based log files. The DARPA 1998 dataset is packet-based, UNIBS2009 is flow-based, and NSL-KDD is of another type.
6. *Traffic type:* This characteristic describes the type of environment in which the data was obtained: real, simulated, emulated, or synthetic. Real network traffic refers to data that was captured in an actual operational network environment. Simulated traffic refers to data generated within a controlled experimental environment where network topology, user activities, and attack scenarios are intentionally designed and executed according to predefined configurations. Emulated traffic refers to data captured in an environment that reproduces real network behaviour using virtual machines, containers, cyber ranges, or testbeds while maintaining experimental control. Synthetic traffic refers to traffic that was artificially generated through the use of traffic generators or algorithmic data generation techniques. The DARPA dataset includes simulated/emulated traffic, the LBNL dataset contains real traffic, and the PU-IDS dataset contains synthetic traffic.
7. *Network size:* Various security systems and assessments are required for different types of environments. This characteristic indicates the size of the network in which the dataset was created, whether it is a small network, a medium-sized network, or a large network (such as an Internet Service Provider). For consistency, small networks generally refer to environments containing fewer than 100 devices or nodes, medium-sized networks contain approximately 100–1000 devices or nodes, while large networks contain more than 1000 devices or represent large-scale operational deployments. Enterprise, University, and ISP categories are retained where the original dataset documentation explicitly identifies the collection environment. The

DARPA1998 and PU-IDS datasets were created in small networks, while the MAWI dataset was generated in a large network.

8. *Metadata:* The dataset should include supplementary details such as network topology, attack scenarios, IP addresses, etc., to help third parties interpret the packet and flow-based network. The presence or absence of this additional metadata determines the dataset's comprehensibility.
9. *Complete network:* This characteristic indicates whether the dataset includes comprehensive network traffic from multiple hosts and routers. If the dataset only contains traffic from a single host (honeypot) or a limited number of protocols from the network (such as SSH traffic), then the network is not considered complete. The DARPA 1998 dataset contains a complete network, while the Kyoto dataset does not include a complete network.
10. *Availability:* Datasets should be made available to the public so that researchers can compare different intrusion detection methods. Furthermore, if datasets are available to the public, third parties can verify their quality. Some datasets, such as DARPA, KDDCUP, and NSL-KDD, have been widely used due to their public availability.
11. *Normal user behaviour:* The statement suggests that the dataset contains regular user activity, but there is no evidence of any attacks. The effectiveness of an Intrusion Detection System (IDS) is determined by its ability to detect attacks accurately without generating false alarms. To assess an IDS, it is necessary to have a dataset that includes normal user behaviour. Without such a dataset, the evaluation may not be reliable. However, if the dataset lacks normal user behaviour, it can still be useful, but needs to be combined with other datasets or real-world network traffic. Some datasets, such as DARPA, NSL-KDD, and VHS contain regular traffic, while others, like Twente, do not.
12. *Attack traffic:* The IDS dataset should include attack scenarios, which means that the dataset contains malicious network traffic. This property is indicated by a value of "yes" if the dataset contains at least one attack. Some datasets, such as DARPA, NSL-KDD, VHS, and Twente, contain attack traffic, whereas KENT does not include any attack traffic.
13. *Anonymity:* For privacy reasons, IDS datasets may not be publicly available or may be accessible in an anonymized format. This feature indicates whether the data is anonymized or not.
14. *Count:* This feature indicates the size of the dataset, measured in terms of the number of packets/points/flows or the overall size in Gigabytes (GB). Dataset sizes can range from GBs to TBs, such as the DARPA-1998 dataset with a size of 4 GB or the DARPA-2009 dataset with a size of 6.5 TB.
15. *Duration:* This attribute indicates the duration of the traffic that has been logged or obtained. The information is gathered either for an hour or for longer periods of time, such as days or weeks. The DARPA1998 dataset was compiled over a period of 5 to 7 weeks, while the Mawi dataset was collected for 15 min each day.
16. *Features:* It refer to the individual measurable properties, attributes, or variables used to describe each data instance in a dataset. These are the inputs used for analysis, modeling, or classification tasks and can include characteristics such as packet size, protocol type, duration, source/destination IPs, or behavior patterns. In this column,

we list the key features extracted or used from each dataset to provide insight into what parameters are considered for detecting or analyzing network activities.

17. *Labelled:* In supervised Intrusion Detection Systems, having labelled datasets is crucial for both training and evaluation. However, for unsupervised methods, labelled datasets are only necessary for evaluation purposes. This characteristic specifies whether the dataset is labelled or not. The DARPA 1998 dataset contains labels, while the LBNL dataset does not include labelled data.
18. *Predefined splits:* This attribute specifies whether the dataset contains predefined subsets for training and evaluation purposes. It is important to ensure that the same subsets are used for both training and evaluation, in order to compare the quality of different Intrusion Detection Systems (IDS). The DARPA 1998 dataset includes pre-established splits for this purpose, while the NSL-KDD dataset does not have predefined splits.
19. *Attacks type:* Attack type refers to the classification of a cybersecurity threat or intrusion based on its behavior, intent, and method of execution. It categorizes attacks into general groups, such as Denial of Service (DoS), Probe, or Remote to Local (R2L), to simplify analysis, detection, and response. Each type represents a distinct strategy used by attackers to compromise system integrity, availability, or confidentiality.
20. *Number of citations:* It refers to the total count of times a particular work, publication, dataset, or entry has been referenced or cited by other sources. In the dataset table, the number of citations column typically contains an integer value indicating the influence, popularity, or academic impact of the dataset. Citation counts reported in this survey were obtained from Google Scholar and collected during June 2026. Since citation counts continuously change over time, the reported values should be interpreted as a snapshot corresponding to the data collection period of this study.
21. *Balanced:* In the training phase of an anomaly-based Intrusion Detection System, it is important for the dataset to be balanced in terms of class labels. However, real network traffic is typically unbalanced and contains a higher proportion of normal user behaviour than attack traffic. To address this imbalance, pre-processing techniques are often applied before utilizing data mining or machine learning algorithms. In this survey, balance refers primarily to the distribution between benign (normal) and malicious (attack) traffic classes rather than the distribution among individual attack categories. Datasets with relatively comparable proportions of normal and attack traffic are marked as balanced, whereas datasets exhibiting substantial class imbalance are marked as unbalanced.

To ensure consistency in interpreting Table 2, the values are defined as follows: "Yes" indicates that the corresponding property is available or explicitly supported by the dataset. "No" indicates that the property is absent. "N/A" (Not Applicable/Not Available) indicates that the property is either not applicable to the dataset or could not be determined from the available documentation.

5.2 Application-domain taxonomy of datasets

IDS datasets can be classified along multiple independent dimensions, including application domain, traffic representation, traffic generation environment, and attack type. These dimensions describe different dataset characteristics and should be considered

separately rather than combined within a single classification structure. The present subsection focuses specifically on the application-domain dimension. Intrusion Detection Systems are deployed across a wide range of computing environments, each exhibiting distinct communication protocols, attack surfaces, operational constraints, and security requirements. Consequently, no single benchmark dataset is suitable for evaluating IDS solutions across all application domains. Instead, dataset selection should be driven by the operational requirements of the target IDS deployment, ensuring that the communication characteristics, feature representation, and attack scenarios of the benchmark dataset closely match those of the intended application environment. To facilitate dataset selection, the surveyed datasets are categorized according to their target application environment, as summarized in Table 3. This application-oriented taxonomy enables researchers to identify datasets that best satisfy the requirements of different IDS deployment scenarios rather than selecting datasets solely because of their popularity or widespread adoption.

The taxonomy presented in Table 3 highlights the progressive diversification of IDS benchmark datasets beyond traditional enterprise networks. While early benchmark

Table 3 Application-domain taxonomy of the surveyed IDS datasets

Application domain	Representative datasets	Typical IDS requirements and key dataset characteristics
Traditional networks	DARPA98, KDD-CUP99, NSL-KDD, Kyoto, MAWI, CTU-13, UNSW-NB15, CICIDS2017, CICIDS2018, UGR16, LITNET-2020, NF-UQ-NIDS, BigFlow-NIDS, 5GCID, Windows-APT2025 and related enterprise datasets	High throughput, protocol diversity (e.g., TCP and QUIC), encrypted communications, enterprise traffic, modern attack coverage, and flow-/packet-level feature analysis
IoT	LWSNDR, USC Urban-IoT-DDoS, N-BaIoT, Bot-IoT, IoT-23, MQTTset, MQTT-IoT-IDS2020, MedBIoT, IoT-Keeper, IoT-DeNAT, IoT Environment, Edge-IoT, KNX, Gotham, Farm-Flow, IoT-CSIDDS and related datasets	Resource-constrained devices, heterogeneous communication protocols (e.g., MQTT and CoAP), device heterogeneity, botnet detection, lightweight feature extraction, and real-time inference
Industrial IoT (IIoT)	DS2OS, WUSTL-IIoT, X-IIoTID, Edge-IIoTset, ToN-IoT	Industrial protocols (e.g., Modbus and OPC-UA), cyber-physical processes, SCADA environments, operational technology attacks, and low-latency detection
Vehicular networks (IoV/CAN)	CAN Signal, CAN-OTIDS, Survival Analysis CAN, Car-Hacking, ROAD, CAN-FD, M-CAN, B-CAN, Automotive Ethernet Intrusion, X-CANIDS, CAN-MIRGU, CICIoV2024	CAN message timing, ECU behaviour, spoofing detection, low-latency communication, in-vehicle security, and V2X environments
UAV networks	UAV-NIDD, UAVIDS-2025	Cross-layer UAV communications, MAVLink traffic, wireless telemetry, GPS spoofing/jamming, dynamic topology, UAV-specific routing, and low-latency intrusion detection
Cloud and virtualized systems	ISOT-CID	Cloud workloads, virtual machines, malware and botnet detection
Software defined networks	InSDN	Flow-based traffic monitoring, centralized controller protection, DDoS detection
Android/mobile applications	Android Validation, Android Adware, Android Malware	Application behaviour analysis, malware detection, mobile security
Cyber-physical systems and critical infrastructure	IEEE 300-Bus, Gas Pipeline, Power System, Gas Pipeline and Water Storage Tank	Industrial control systems, sensor integrity, operational technology security, critical infrastructure resilience
Specialized domains	CubeSat Cybersecurity, VPN-NonVPN, Tor-NonTor, USB-IDS, OPCUA and other domain-specific datasets	Application-specific traffic characteristics, emerging communication technologies and specialized threat models

datasets primarily focused on conventional network intrusion detection, modern IDS applications now encompass heterogeneous environments such as IoT, Industrial IoT, vehicular networks, UAV networks, cloud computing, software-defined networking, cyber-physical systems, and mobile platforms. Each application domain exhibits distinct communication protocols, traffic characteristics, computational constraints, attack surfaces, and feature requirements. Consequently, datasets developed for one application domain cannot be assumed to accurately represent the operational characteristics required by another.

From an IDS development perspective, the application domain represents one of the primary factors influencing benchmark dataset selection. Different deployment environments exhibit distinct communication protocols, network architectures, operational constraints, traffic characteristics, attack surfaces, and feature requirements. Consequently, benchmark datasets differ in their suitability for developing, evaluating, and comparing IDS solutions, depending on how closely they reflect the characteristics of the target application domain.

For example, enterprise and campus networks increasingly employ both traditional TCP/TLS and emerging QUIC/UDP communications, together with encrypted traffic, diverse application-layer protocols, and heterogeneous cyberattacks. Recent comparative studies have shown that modern transport protocols exhibit different security and traffic characteristics under attacks, highlighting the importance of benchmark datasets that capture protocol diversity and realistic encrypted network traffic [137]. Representative benchmark datasets for this domain include DARPA98, UNSW-NB15, CICIDS2017, CICIDS2018, LITNET-2020, NF-UQ-NIDS, BigFlow-NIDS, and 5GCID.

In contrast, IoT environments are characterized by resource-constrained devices, heterogeneous sensors, lightweight communication protocols such as MQTT and CoAP, and device-specific traffic patterns. Recent studies on IDSs for IoT-based smart buildings further highlight the importance of lightweight flow-based features, heterogeneous device behaviour, and scalable traffic representations for accurate and efficient intrusion detection in large-scale IoT deployments [138]. Consequently, IoT benchmark datasets place greater emphasis on protocol-specific traffic, device diversity, and botnet-oriented attacks than conventional enterprise datasets. Representative datasets include N-BaIoT, Bot-IoT, IoT-23, MQTTset, MQTT-IoT-IDS2020, IoT-CSIDDS, Gotham, and Edge-IoT.

Industrial IoT environments introduce additional requirements, including industrial control protocols, cyber-physical process behaviour, SCADA communications, and operational technology attacks, making datasets such as ToN-IoT, Edge-IIoTset, WUSTL-IIoT, and X-IIoTID more representative of these deployment scenarios.

Similarly, vehicular IDS research requires datasets containing Controller Area Network (CAN) traffic, in-vehicle communication, message timing characteristics, electronic control unit (ECU) behaviour, spoofing attacks, and increasingly Vehicle-to-Everything (V2X) communications. Representative datasets include Car-Hacking, CAN-OTIDS, CAN Signal, CAN-FD, ROAD, X-CANIDS, CAN-MIRGU, and CICIoV2024. Likewise, UAV intrusion detection introduces additional challenges arising from highly dynamic wireless communication, cross-layer interactions, GPS spoofing and jamming attacks, flight-control behaviour, and rapidly changing network topology. Recently introduced datasets such as UAV-NIDD and UAVIDS-2025 extend benchmark coverage for

securing autonomous aerial platforms and illustrate the continued diversification of IDS datasets into emerging application domains.

Therefore, the proposed application-domain taxonomy extends beyond a simple categorization of benchmark datasets by relating them to the operational characteristics of the environments they represent. It provides a structured framework for understanding the suitability of different datasets for enterprise networks, IoT, IIoT, vehicular systems, UAVs, and other specialized application domains. This application-oriented perspective complements the detailed dataset descriptions presented in the following subsection and the subsequent analyses of traffic representation, attack diversity, AI readiness, realism, quality, and reproducibility, thereby providing a comprehensive framework for selecting appropriate benchmark datasets for modern intrusion detection research.

5.2.1 Detailed dataset description

1. *DARPA98:* The DARPA98 [14] created by MIT Lincoln Laboratory for the U.S. Defense Advanced Research Projects Agency, is one of the earliest and most well-known intrusion detection datasets. It comprises 4GB of packet-level traffic data with payloads for TCP, UDP, and ICMP, including 7 weeks of labeled training data (7 million records) and 2 weeks of unlabeled test data (2 million records). It contains 57 attacks categorized as DoS, Probe, U2R, and R2L, with 41 features plus a class label. Despite its historical importance, DARPA98 is criticized for redundancy and lack of modern attack patterns. Table 4 summarizes representative studies that have utilized the DARPA98 dataset. It was succeeded by KDDCUP99, which grouped features into content, intrinsic, and traffic-based sets, and included 22 attack types in training and 15 in testing. The NSL-KDD dataset further refined KDDCUP99 by removing duplicates and addressing data imbalance to reduce false alarm rates (Table 4).
2. *KDD CUP 99*: The KDDCUP99 dataset [16], created in 1998, is a benchmark dataset for evaluating intrusion detection systems. It was collected over nine weeks using TCP traffic on a simulated LAN environment. The training data consists of approximately 5 million connection records over seven weeks, while the testing data comprises around 2 million connections collected over two weeks. The dataset includes 22 known attack types in training and 39 attack types in testing, with attack categories including: (1) Probe (e.g., port scanning), (2) DoS (e.g., SYN flooding), (3) R2L (e.g., password guessing), and (4) U2R (e.g., buffer overflow). Each record is labeled as normal or attack, with 41 associated features. The training set contains 494,021 records: 391,458 (79.24%) DoS, 97,277 (19.69%) normal, 4,107 (0.83%) Probe, 52 (0.01%) U2R, and 1126 (0.23%) R2L.

Table 4 DARPA98

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[139]	2025	ADAPTIVE-DECAYRANK	N/A	N/A	64.55	N/A	N/A
[140]	2022	DKNN	CRDO	97.22	95.99	95.32	95.66
[141]	2021	CNN+RNN	HAST-NAD	98.68	N/A	97.8	N/A
[142]	2019	Random forest	Hybrid FSM	99.93	99.98	99.51	98.9
[143]	2019	ImCNN	ImGWO	97.92	N/A	98.62	98.92

(a) *Full KDD:* This version contains 4,898,430 instances in a 750 MB file. It is heavily imbalanced, with Neptune (1,072,017 records) and Smurf (2,807,886 records) dominating. Rare attacks like Spy, Perl, and PHP are significantly underrepresented, which negatively impacts classifier performance due to class imbalance.
(b) *Corrected KDD:* To address redundancy and imbalance, the corrected version includes 311,029 records with better representation of all attack types. This helps mitigate bias and improves classification accuracy in evaluation studies.
(c) *10% KDD:* A smaller and widely used subset comprising 489,843 instances. It is derived from the full dataset and maintains representative distribution, making it suitable for training and testing IDS models while reducing computational load. The different variants of the KDDCUP99 dataset are summarized in Table 5).

3. *PU-IDS:* Panjab university-intrusion dataset (PU-IDS) [17] is subset of NSL-KDD dataset, which is widely used but is very old and has single level of labelling, so a new dataset known as PU-IDS with two levels of labelling is generated by taking the basic characteristics of NSLKDD dataset. Singh et al's work can thus be viewed as a traffic generator for generating PU-IDS. Approximately 200,000 data points are in PU-IDS and that have similar format and attributes as NSL-KDD. In PU-IDS, a total of 198,533 number of instances are present, out of which 92,727 are normal and 105,806 are anomalies. There are 30 Normal and 68 Anomaly protocol-service-profiles, and there are 72 Normal and 201 Anomaly network traffic activity profiles.
4. *MAWI:* Measurement and Analysis on WIDE Internet (MAWI) dataset [18] contains real traffic data between japan and the USA and is freely available to the public. MAWI traffic is used to extract the information of flows like IP length, frame length, IP version, flags, TTL window size, destination IP, source IP etc. It contains both attack and normal flows. In MAWI dataset, every trace includes the traffic captured for 15 min in a particular day. A project done on MAWI archive is MAWILAB, which includes labels of data. Four classifiers do the labelling of data, i.e., PCA (principal component analysis), KL (Kullback–Leibler), Hough transform, and Gamma distribution. In MAWILAB, metadata anomalies are reported in CSV format. Table 6 shows the work who used MAWI dataset.
5. *LBNL:* Lawrence Berkeley National Laboratory and ICSI - 2004/2005 (LBNL) dataset [19] was created in 2004 and has full header network traffic, which was recorded at a medium-sized site and is without payload. The main purpose of generating the LBNL dataset was to analyse the characteristics of the network traffic within organizational networks instead of publishing the data on intrusion

Table 5 KDDCUP99

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[144]	2025	Random forest	N/A	97.99	N/A	N/A	N/A
[145]	2025	Ensemble	N/A	99.88	99.9	99.85	99.87
[146]	2024	XGBoost	N/A	99.98	N/A	N/A	N/A
[147]	2023	Fuzzy optimized V-detector	N/A	99.85	98.71	94.20	96.40
[148]	2018	Stacked NDAES	NDAES	97.85	99.00	97.00	98.00
[149]	2018	LSSVM	MVMIFS	94.31	N/A	93.78	N/A

Table 6 MAWI

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[150]	2019	RNN	N/A	99.76	100	100	100
[151]	2017	GS-ANN and GSPSO-ANN	N/A	73.28	N/A	73.28	N/A

Table 7 LBNL

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[152]	2020	AMSCA with WEDMS	N/A	N/A	N/A	N/A	N/A
[153]	2018	Two step cluster analysis	N/A	N/A	N/A	N/A	N/A

Table 8 Kyoto

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[154]	2018	MPNN and SMO	N/A	95.73	N/A	96.92	N/A
[155]	2017	LSSV IDS	N/A	99.91	N/A	98.76	N/A
[156]	2015	DT (J48)	N/A	97.23	96.50	97.20	96.30

detection. As per the authors, the security researchers can still use this anonymised data set for background traffic as it has almost normal user behaviour, and contains a network traffic of more than 100 h, and is available for download at its website. Table 7 shows the work that used the LBNL dataset.

6. *Kyoto 2006:* The Kyoto 2006+ [20] dataset was given by the Kyoto University, which is available to the public, and is generated using "4" tools like darknet sensors, honeypots, a web crawler, and e-mail server. In the Kyoto 2006+ dataset, real network traffic and a limited range of realistic user behaviour can be observed. It took 3 years to capture the data. Since the dataset was recorded for such a long period, there are about 93 million sessions. As the IDS Bro converted traffic based on packets into a "session" which is a new format, so Kyoto 2006+ was categorized as other than packet and flow based. There are 24 features in each session, out of which 14 are conventional features derived from KDD CUP 99 data set and gives statistical information, and 10 remaining features are based on flow, like as anonymized IP addresses, durations, and ports. The presence of attacks is indicated by label. Table 8 shows the work who used Kyoto 2006+ dataset.
7. *CAIDA:* The CAIDA dataset [21], developed by the Center for Applied Internet Data Analysis, includes several subsets such as CAIDA DDoS (2007), Internet Traces (2016), and RSDoS Attack Metadata. The CAIDA DDoS subset provides 5-min pcap files and 1 h of DDoS traffic captured passively at the Equinix-Chicago exchange point, showcasing flood-based attacks that disrupt legitimate traffic. The RSDoS metadata, collected via the UCSD Network Telescope, includes randomly spoofed backscatter packets from denial-of-service attacks. A key limitation of CAIDA is its lack of attack diversity and full-network feature representation, making it difficult to clearly differentiate between normal and abnormal traffic flows. Table 9 shows the work who used CAIDA dataset.
8. *Twente:* In 2008, one of the first intrusion detection dataset based on flow was released by Sperotto et al in Twente University and is available to the public. The Twente dataset [22] have traffic from a honeypot server which provides FTP, web,

Table 9 CAIDA

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[157]	2022	Random forest	N/A	95.60	N/A	N/A	N/A
[158]	2015	Ensemble	N/A	99.89	100	99.00	N/A
[159]	2014	Fuzzy QL	N/A	88.77	93.10	N/A	N/A

Table 10 NSLKDD

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[160]	2025	Graph Attention Network	N/A	98.00	98.00	98.00	98.00
[161]	2025	SOM-XGBoost	N/A	87.24	N/A	61.96	60.63
[162]	2024	RF	N/A	99.50	99.00	99.00	99.00
[163]	2024	ResNet18	N/A	98.91	100	100	100
[164]	2024	Novel Fast hyper DLM	PCA and SVD	99	100	100	100
[165]	2024	KNN	RF	98.24	97.99	97.91	98.00
[166]	2021	AE	N/A	90.61	86.83	98.43	92.26
[167]	2020	IG, GR, CA	RNN	99.60	99.00	N/A	N/A

and SSH services for "6" days. The dataset only contains honeypot network traffic, and nearly all flows are malicious, with no normal user behaviour. For labelling flows in this dataset, the authors examined packet-based traffic and log files. Because of privacy concerns, IP addresses were taken out from the dataset. The dataset suffers from attack diversity and a lack of volume.

9. *NSL-KDD:* This widely used dataset was collected in 2009. Tavalle et al performed the analysis on KDDCUP99 and pointed out some problems like redundancies, so an improved version of the KDDCUP99 dataset was created, known as NSLKDD [23]. The creators of NSL-KDD eliminated duplicates from KDDCUP99, and more sophisticated subsets were created. Now the new data set has approximately 150,000 data points and is split into testing and training sets for intrusion detection (ID) methods. The NSL-KDD belongs to a category other than packet and flow, and shares similar attributes to KDD-CUP-99. The dataset is open to the public. There are approximately 4,900,000 single connection vectors in the KDD training set, each having "41" features and labelled as attack or normal, with only "1" particular attack type. The simulated attacks belong to one of 4 types: Probing attack, (U2R) User to Root, (DoS) Denial of Service, and (R2L) Remote to Local. Types of attacks in the train set are 24, and types of attacks in the test set are 14. Table 10 shows the work that used the NSL-KDD dataset.
10. *CDX:* Cyber Defence Exercise (CDX) dataset [24] was collected in 2009. Sangster et al put-forward an idea for generating network-based datasets from network warfare competitions and thoroughly explain the benefits and downside of such a methodology. In 2009, a network warfare competition was held for 4 days, and CDX dataset contains traffic from that event. The traffic is set-down in packet-based format and is open to the public. CDX contains both normal user behaviour and various kinds of attacks. Additionally, there is a plan, that describes metadata about the network structure as well as IP addresses, but no individual packet is labelled. In addition, there is an availability of log files based on host and warnings of IDS. This dataset has many shortcomings like a lack of volume and diversity of

Table 11 CDX

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[168]	2013	Naive Bayes	N/A	99.90	N/A	N/A	N/A

Table 12 DARPA2009

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[169]	2018	DFF	N/A	99.63	99.80	99.40	99.60
[170]	2015	Decision tree	N/A	99.90	N/A	N/A	N/A

Table 13 UNIBS

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[171]	2021	ABDNNA	STEF	97.10	76.50	N/A	68.90
[172]	2020	GMMS	SFS	98.03	N/A	N/A	68.70
[173]	2015	GMM	SFS	98.70	N/A	N/A	N/A

enterprise networks. Also, it does not have real attack traffic. Table 11 shows the work who used CDX dataset.

11. *DARPA 2009:* The DARPA 2009 dataset [25], developed at MIT Lincoln Lab, simulates traffic between a /16 subnet (172.28.0.0/16) and the Internet over a 10-day period (Nov 3-12, 2009). It includes background SMTP, DNS, and HTTP traffic, along with various attack scenarios such as DoS, worms with varied propagation, probing, user-to-root (U2R), and remote-to-local (R2L) attacks. The dataset comprises 7000 pcap files (totaling 6.5TB), each under 1000MB, covering 1–2 min of traffic. Analysis was conducted using tools like tcpdump and Argus, which aggregated traffic into daily flows and generated statistics. The dataset contains both labelled and unlabelled records for intrusion detection evaluation. Table 12 shows the work who used DARPA2009 dataset.
12. *UNIBS 2009:* The UNIBS 2009 dataset [26], like LBNL, was not designed for the detection of intrusion. It is still added to the list because it is mentioned in other works. To identify applications network traffic based on their flow (like Skype, mail clients, or web browsers), Gringoli et al. used the dataset. UNIBS 2009 includes approximately 79,000 flows that do not exhibit malicious behaviour. Network traffic is not classified as attack or normal because the labels only define the flow's application protocols. As a result, the property label is set to NO in the categorization scheme. The dataset is freely accessible to the public. Table 13 shows the work that used the UNIBS dataset.
13. *Botnet:* [29] The ISOT botnet dataset combines several publicly available malicious and non-malicious datasets (ISOT, ISCX 2012, CTU-13). Information regarding the behavioural pattern of the user as well as botnets are components of the resulting dataset. The Botnet data set is divided into "8.5" GB test data having 16 types of botnets and "5.3" GB training data having 7 types of botnets, both in packet-based format. In the training set, the different types of botnet include Virut, Zeus, Zeus control (C and C), SMTP Spam, NSIS, Rbot, and Neris. In the testing set the different types of botnet include Sogou, Neris, Menti, Rbot, etc. In terms of botnet

Table 14 BOTNET

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[174]	2020	Decision tree	RFE	99.99	99.00	99.00	99.00

Table 15 ISOT-CID

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[175]	2022	MP	Riemann chunking scheme	95.28	N/A	N/A	95.20
[176]	2021	DT	Riemann rolling	100	100	100	100
[177]	2020	Double DQ-N/W	N/A	96.87	NA	NA	NA

Table 16 CSIC-HTTP

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[178]	2020	DT J48	Fine tuned	N/A	94.00	94.00	93.90
[179]	2018	DBN	N/A	96.57	96.40	93.20	94.00
[180]	2018	Isolation forest	Stacked AE	88.32	80.00	88.30	84.10

topologies, they can be randomized, centralized, or distributed (e.g., P2P). Table 14 shows the work who used Botnet dataset.

14. *ISOT-CID 2010:* The ISOT Cloud Intrusion Dataset [27] is the first publicly available cloud-specific dataset, created by combining malicious traffic from the French Honeynet Project with normal traffic from Ericsson Research (Hungary) and LBNL. Data was collected for several days (1–2 h per day) across different OpenStack layers-network, guest hosts, and hypervisor-and included system logs, network traffic, memory dumps, and CPU metrics. The dataset totals 8 TB, with 55.2 GB of network traffic comprising 22,372,418 packets (15,649 malicious in Phase 1) and 11,509,254 packets (2,006,382 malicious in Phase 2). It contains both inside and outside attacks and is stored in packet format for detailed analysis. Table 15 shows the work who used ISOT-CID dataset.
15. *IEEE 300-bus power test system:* IEEE Test Systems Task Force created IEEE 300-bus test case [28] in 1993, led by Mike Adibi." There are 60 LTCs, 69 generators, 195 loads, and 304 transmission lines in this IEEE 300-bus system. This dataset contains an electrical and a topological structure of the power grid, which is particularly useful in the smart grid for detecting false data injection attacks. The system has a degree $(k, >)$ of 2.74 and 411 branches on average.
16. *CSIC HTTP 2010:* The CSIC HTTP 2010 dataset [30], developed by the Information Security Institute of CSIC (Spain), contains HTTP traffic generated from an e-commerce web application and is publicly available. It includes 36,000 normal and 25,000 anomalous HTTP requests, all labelled. Attacks in the dataset include SQL injection, buffer overflow, XSS, file disclosure, CRLF injection, parameter tampering, and more. Anomalous requests are categorized as static (accessing hidden resources), dynamic (modifying valid parameters), and unintentionally illegal (non-malicious but abnormal). Attack traffic was created using Paros and W3AF tools. The dataset is structured into three sets: a training set (normal traffic) and two test sets (one with normal and one with malicious traffic). Table 16 shows the work who used CSIC HTTP 2010 dataset.

Table 17 LWSNDR

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[181]	2019	HLMCC	HAP	94.78	93.50	95.50	94.50
[182]	2016	Boosting (DT)	N/A	99.00	99.00	99.00	99.00

Table 18 SSENET-2011

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[183]	2016	OCC	DPC, LFI	97.00	98.00	97.38	96.00
[184]	2015	PCC, NB	ICC	99.90	99.98	99.90	99.00

17. *LWSNDR 2010:* Sutaharan et al. [31] introduced the LWSNDR dataset to address the lack of properly labeled synthetic datasets in Wireless Sensor Networks (WSNs). It was created using a real sensor network with both single-hop and multi-hop configurations, employing Crossbow TelosB motes to collect temperature-humidity readings every five seconds over 6 h. Controlled anomalies were introduced to simulate irregularities at various levels, including individual sensor readings, network traffic features, and patterns across neighboring or grouped nodes. The dataset is fully labeled and publicly available for anomaly detection research in WSNs. Table 17 shows the work who used LWSNDR dataset.
18. *SSENet-2011:* Using a controlled environment, dataset SSENet-2011 [32] was generated for 4 h in 8 different geographically separated areas in Tamil Nadu, India, using the Tstat tool in 2011 and is available publically. The traffic was captured in pcap (packet-based) format and then converted to data points using the Tstat tool. Every data point has 23 attributes that are represented as 4 connection-based and 19 network attributes. Participants browsing activities generate normal user behaviour. There are 3 types of attacks in it and they are flooding, probing and privilege escalation attacks. Table 18 shows the work that used the SSENet-2011 dataset.
19. *TUIDS:* The dataset TUIDS [33] was generated at Tezpur University campus in 2011 and contains 3 parts TUIDS Intrusion data set, TUIDS coordinated scan data set, and TUIDS DDoS data set. Data were produced using the virtual environment that includes about 250 clients. Information about Traffic was collected in a format that is bidirectional flow-based and packet-based. Every subset covers 7 day period, and all 3 subsets are comprised of 250,000 flows. The features are of 24 types and are classified in 3 groups: basic, connection based and time window-based features. This dataset has normal user behaviour and attacks such as Denial of Service Using Targa, Probing Using nmap, Coordinated Scan Using rnmap, User to Root Using Brute Force SSH, Distributed Denial of Service Using Agent-handler Network, and Distributed Denial of Service Using IRC Botnet. Table 19 shows the work who used TUIDS dataset.
20. *ISCX-IDS 2012:* Shiravi et al. [34] developed the ISCX dataset, capturing seven days of labeled network activity (both normal and malicious), including Brute Force SSH, DDoS, HTTP DoS, and internal network infiltration attacks. A dynamic methodology was adopted using beta profiles (normal user behaviors like web browsing and emailing) and alpha profiles (attack scenarios). The dataset is available

Table 19 TUIDS

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[185]	2014	MLH-IDS	MI	98.26	99.60	99.70	98.80
[33]	2012	NADO	IPFix	99.29	98.00	98.00	98.00
[186]	2012	Tree CLUS and Clus lab	MMIFS	98.26	98.00	98.00	98.00

Table 20 ISCX-IDS-2012

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[187]	2021	Fine tree	N/A	100	99.94	99.94	99.94
[188]	2020	EFS and GOA	EFS and GOA	99.13	99.10	99.20	99.20
[189]	2019	IG-PCA	IGE	99.01	99.10	99.10	99.20
[190]	2018	BO	BO	99.92	99.00	99.00	99.00
[191]	2013	KML+NBC	N/A	99.00	98.00	98.00	98.00

Table 21 Malicia

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[192]	2021	Densenet	Densenet	89.48	89.30	89.20	89.20
[193]	2013	Firma	Firma	98.80	100	97.70	98.80

in both packet-based and bidirectional flow-based formats. Protocol behavior profiles were generated from real traces of IMAP, HTTP, SMTP, SSH, FTP, and POP3. However, the dataset lacks HTTPS traffic-despite HTTPS comprising 70% of modern network traffic-and its attack distribution does not reflect real-world statistical frequencies. Table 20 shows the work who used ISCX-IDS 2012 dataset.

21. *Malicia 2013:* The Malicia dataset [35], publicly available and covering March 7, 2012 to March 25, 2013, consists of five components, with a MySQL database as the core. It includes eight tables-most notably, the MILK table (malware events: time, landing URL) and the FILES table (malware binaries by SHA1, classification, traffic, icons, screenshots, and family labels). The LANDING IP table stores exploited server details: IPs, exploit kits, AS numbers, and country codes. Additional files include a schema diagram, 11,363 malware binaries (.exe/.dll), 5777 extracted icons, and a Snort IDS signature file generated via FIRMAtool. Table 21 shows work on the Malicia dataset as follows:
22. *Booters 2013:* The Booters dataset [36], published by Santanna et al., captures network traces of 9 distinct booter (DDoS-for-hire) attacks executed against a null-routed IP address, totaling 250 GB of unlabelled packet-based traffic. Each attack trace is stored in a separate file. Candidate Booter URLs were identified using a weekly crawler and Google Custom Search with keywords like "Booter," "Stresser," and "DDoS-as-a-Service." Each URL was manually verified to confirm its relevance. Data collection and URL investigation began in July 2013. The dataset is available upon request.
23. *CTU-13:* The CTU-13 dataset [37] was captured in 2013 in a network of CTU University at the Czech Republic and contains normal, background, and botnet traffic. Traffic was labelled in 3 stages. With all the traffic to and from infected hosts labelled as botnet, normal traffic matches particular filters, and consequently, the

Table 22 CTU-2013

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[194]	2020	CNN-LSTM	N/A	99.00	99.40	98.50	99.00
[195]	2019	LSTM	N/A	96.20	95.60	94.60	95.60
[196]	2017	NN	AGGR time window	89.38	92.50	85.70	88.00

Table 23 ADFA

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[197]	2017	SC2.2	Markov Chain + Bayesian Infer + Adaboost	65.20	N/A	71.40	74.60

Table 24 SSH cure

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[39]	2014	HMM + SSHCure	IPFIX	99.70	N/A	N/A	N/A

background traffic can be labelled as malicious or normal, and the dataset is divided into test and training subsets. Using 13 botnet scenarios, malicious traffic was generated. The dataset is based on "3" different formats, including unidirectional flow, bidirectional flow, and packet. Table 22 shows the work who used the CTU-13 dataset.

24. *ADFA2013:* Creech and Hu proposed the ADFA2013 dataset [38] at University of New South Wales in 2013 that employs payloads and vectors to attack the Ubuntu operating system. The payloads include "add new superuser", "C100 Webshell", " Java-based Meterpreter", "Linux Meterpreter Payload", and "password brute-force". This dataset includes 3 data types: attack data, training data, and validation data. 4373 traces are in the training dataset. There are 833 traces in the normal validation data. Each vector in the attack data has ten attacks. Disadvantage of the ADFA is a lack of attack diversity and behaviour of attacks are not well separated from normal behaviour. However, ADFA2013 is publicly available dataset. Table 23 shows the work on the ADFA dataset.
25. *SSHCure:* For detecting SSH attacks, Hofstede et al propose the SSHCure tool. Two datasets [39] (each for one month) were collected from the University of Twente's campus network to assess their work in November and December 2013 and January and February 2014, respectively. The network has a routable /16 IPV4 address block from which 25k addresses are actively used. The resulting data sets are accessible to the public and contain only SSH traffic. The authors gave additional log files based on the host that could be used to determine whether or not SSH login attempts were successful, instead of labelling flow-based network traffic directly. The number of honeypots, servers, workstations, and attacks in dataset1 are 13, 0, 0, 632, and in dataset2 are 0, 76, 4, 10716. Table 24 shows work on the SSHCure dataset.
26. *Gas pipeline:* Collected at Mississippi State University [40], this dataset includes labeled RTU telemetry for normal, command injection, and data injection attacks in a gas pipeline system. Attacks include Negative Values, Burst Values, and Illegal Setpoints. It captures PLC-related parameters and system states and includes timestamps, CRCs, and class imbalance, requiring careful metric evaluation.

27. *SANTA:* The SANTA (Scenarios for Analysis, Network Traffic, and Attacks) dataset [41], created in 2014 by the University of California, Santa Barbara, provides real-world ISP traffic labeled manually in session format, similar to NetFlow but enhanced with additional packet-based features. It supports external attack detection and is publicly available.
28. *SSENet-2014:* Derived from SSENet-2011, SSENet-2014 [42] includes over 1.2 million packets collected over 10 h in a simulated environment with 10 virtual machines. The traffic includes benign and attack data (e.g., port scans, DoS, buffer overflow). Each packet has 45 features; 28 are derived for analysis, aligned with KDD CUP 99. The dataset contains 200,000 labeled samples, split into training and testing sets, and is balanced.
29. *Android validation:* Proposed by Gonzalez et al. [43], this dataset includes 72 original Android apps and 792 transformed variants. Features include N-grams from the.dex files and APK metadata. It introduces relationships like Stepsiblings, Twins, and False Siblings to analyze structural app similarities.
30. *Power system:* Developed at Oak Ridge National Laboratory [44], this dataset contains 37 scenarios covering normal, disturbance, and cyberattack conditions in an electric transmission system. It includes 116 PMU-based features from four units and logs from Snort, control panels, and relays. Attacks include relay trip injection, relay disabling, and SLG fault replay. Data supports binary, 3-class, and multi-class classification.
31. *Gas pipeline and water tank:* Created by Morris and Gao [45], this dataset features attacks on two ICS testbeds: gas pipeline and water storage tank. Data was collected over Modbus (serial and TCP), covering 35 attack types and 214,580 Modbus packets, of which 60,048 were malicious. Attack types include command/response injection, reconnaissance, and DoS. Data is provided in ASCII and ARFF formats.
32. *AWID2:* The Aegean WiFi Intrusion Dataset [46], developed in 2015, focuses on wireless IDS. Captured from an 802.11 network with 11 users, it includes 37 million packets and 156 features per packet. It contains both normal and malicious traffic, split into training and test sets, and documents 16 attack scenarios, as summarized in Table 25.
33. *IRSC (Indian River State College):* The IRSC [47] dataset was collected at Indian River State College, Florida, USA in 2015 and reflects the network traffic from real real-world environment. Two types of datasets are created by collecting (FPC) full packet capture (which has all data from the network) and network flows (which have an aggregated unidirectional summary of network traffic between two networked devices). The attack traffic includes uncontrolled attacks (real attacks not created by the team) and controlled attacks (intentional attacks created by the team). Attacks from the internet and normal user behaviour were captured using real network

Table 25 AWID2

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[198]	2022	CNN	N/A	99.71	N/A	N/A	99.73
[199]	2022	Shallow and DL	N/A	99.48	98.94	95.87	97.35
[200]	2022	FFN	N/A	99.28	99.24	99.28	99.25

Table 26 UNSW-NB15

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[201]	2025	FedLLMGuard	N/A	96.47	96.27	96.61	96.43
[202]	2020	Integrated rule based model	IG	84.83	N/A	90.30	68.13
[203]	2020	Decision tree	XGBoost	90.85	83.30	98.38	88.45
[204]	2019	Bagging	Univariate FSM	99.47	99.00	99.00	99.00
[205]	2016	Decision tree	Gain ratio	85.56	N/A	N/A	N/A

Table 27 KENT

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[206]	2020	LSTM	Dram Algo + Log based FE	N/A	94.70	94.00	94.00

traffic. In addition, Manual attacks were carried out. Labelling was done using manual inspection and IDS SNORT. Data is being captured on a 24-h basis.

34. *UNSW-NB15 2015:* The UNSW-NB15 [48] dataset was generated in 2015 with the help of a tool known as IXIA Perfect Storm in a low-range virtual environment in the Cyber range lab of the ACCS and contains 31 h of malicious and normal network traffic based on packet format. It includes "9" distinct types of attacks, such as exploits, backdoors, fuzzers, DoS, and worms. There is also a format based on flow with extra attributes available for the dataset. The UNSW-NB15 is divided into two sections: training and testing splits. There are 45 distinct IP addresses in the dataset, which is freely accessible to the public. Table 26 shows the work that used UNSW-NB15 dataset.
35. *Kent 2016:* KENT [49] dataset was compiled over the course of "fifty-eight" days on the Los Alamos National Laboratory network in 2016. This includes approximately 130 million network traffic based on unidirectional flows, and several log files based on the host. For privacy reasons, heavily anonymized network traffic is used. This dataset is unlabelled and is available for download from the website. Authentication events are gathered from computers as well as Active Directory domain controller servers. From individual Windows computers the start and stop events of processes are collected, DNS lookups are gathered from internal DNS servers, key routers are used to gather network flow data and a series of well-defined red teaming events were conducted within the past 58 days that displayed bad behaviour. Using the "5" data elements, the dataset measures 12GB in size and has 1,648,275,307 events associated with 62,974 processes, 17,684 computers, and 12,425 users. Table 27 shows the work that used the Kent 2016 dataset.
36. *NDSec-1 2016:* This dataset is noteworthy due to its creation as a network security attack composition. Based on these results, the authors claim that existing network traffic can be salted using overlay methodologies such as NDSec-1 [50], which were collected in 2016 in a format based on packets. This data set is freely available to the public. It includes more syslog and Windows event log data. NDSec-1 attacks include botnets, (DoS) denial of service (SYN flooding, HTTP flooding, UDP flooding), brute force attacks (against SSH, HTTP, and FTP), exploits, spoofing, XSS/SQL injection, and port scans. Table 28 shows the work that used the NDSec-1 dataset.

Table 28 NDSec-1

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[207]	2022	CNN-4	N/A	95.50	95.50	94.50	95.40

Table 29 NG-IDS

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[208]	2021	ELM	MSAGOA	90.23	78.80	89.50	77.70
[209]	2020	GMM-Out	FRAR	95.48	N/A	99.55	N/A
[210]	2020	ERT	WEM	100	100	100	100

Table 30 UGR-16

Ref.	Year	Classification technique	Feature selection	Performance Metrics (%)			
				Acc	Prec	Rec	F1
[211]	2020	GAN	N/A	99.97	99.97	99.91	99.94
[212]	2020	DT	N/A	99.97	99.00	98.00	99.00
[213]	2019	GAN	N/A	99.97	99.97	99.91	99.94

37. *NG-IDS 2016:* NG-IDS [51] dataset includes host-based log files as well as packet-based network traffic. A virtual environment was created by generating normal behaviour of the user and attacks using the IXIA Perfect Storm tool (such as DoS attacks or worms) from "7" different attack families. As a result, IXIA Perfect Storm hardware primarily determines the quality of the created data. The labelled dataset includes about 1 million packets and is freely available to the public. The NGIDS-DS is made up of 5 different kinds of files, all of which could be downloaded from the NGIDS-DS download, 2016: (i) NGIDS.pcap network packets; (ii) 99 csv host log files; (iii) ground-truth.csv; (iv) readme.txt; and (v). feature-descr.csv. Table 29 shows the work that used the NG-IDS dataset.
38. *UGR-16 2016:* UGR16 [52] is a dataset with flows in only one direction. The dataset's primary goal is to capture timely effects in an ISP environment. This includes "16,900" million single directional flows and it lasts for "4" months. Anonymized IP addresses are used, and flows are classified as background, attack, or normal. Within that data set, the creators explicitly carried out many attacks (DoS, port scans, and botnets). The corresponding flows have been labelled as attacks, and some other attacks have been found and labelled as attacks manually. Traffic that matches a fixed pattern and normal user behavior is injected and labeled as normal. Yet, the majority of traffic is classified as background, which can be an attack or normal. The dataset is freely accessible to the public. Table 30 shows the work that used the UGR-16 dataset.
39. *Tor-nonTor 2016:* The Tor-nonTor [53] dataset was created by Arash Habibi Lashkari et al and is split into "2" categories: binary Tor-nonTor classification and multi-class Tor traffic classification. Data samples of Tor and non-Tor traffic are given for the binary classification, while Tor traffic from "8" different apps (chat, audio streaming, browsing mail, file transfer, video streaming, VOIP, and peer-to-peer) for the multi-class classification were collected, making use of Wireshark and Tcpdump. To extract all important information from the traffic data ISCXFlowMeter was used. Table 31 shows the work that used the Tor-nonTor dataset.

Table 31 TOR-NONTOR

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[214]	2023	AN-DCN	N/A	88.00	N/A	N/A	N/A
[215]	2020	ACOPE	N/A	81.12	73.50	61.08	66.70
[216]	2019	DSCA	CFC	86.92	80.00	73.00	76.30
[217]	2019	SVM	L1 Norm	96.00	N/A	N/A	N/A

Table 32 URL-2016

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[218]	2023	Light GBM	Denoising AE	97.76	97.70	97.70	97.70
[219]	2021	CADM	LASSO	99.85	100	100	100
[220]	2017	RF	N/A	86.90	N/A	80.50	N/A

Table 33 VPN-NonVPN

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[221]	2022	NIN	DL	97.90	98.30	97.80	97.90
[222]	2021	NB+DRL+Fuzzy K Means	N/A	97.50	98.80	96.00	97.30
[223]	2020	Deep packet	N/A	98.00	98.00	98.00	98.00
[224]	2019	DCGAF	N/A	78.00	N/A	N/A	N/A
[225]	2018	LSTM	NTSA	98.00	98.00	98.00	98.00

40. *ISCX-URL 2016:* Mamun et al. proposed the URL [54] dataset, which includes "5" different types of URLs: I. phishing URLs, II. Spam URLs, III. Benign URLs, IV. Defacement URLs and V. malware URLs. From Alexa's top website, 35,300 URL's which are benign, are collected in the Benign URL. From the WEBSPAM-UK2007 dataset, 12,000 URLs that are spam are gathered in Spam URL. From the OpenPhish repository of active phishing sites, 10,000 URLs that are phishing are gathered in the phishing URL. From DNS_BH, which is a maintained list of malware sites, 11,500 URLs are gathered in the malware URL. From Alexa, ranked trusted websites that host hidden or fraudulent links, 45,450 URLs are gathered in the Defacement URL. Table 32 shows the work that used the URL-2016 dataset.
41. *VPN-nonVPN 2016:* Draper-Gil proposed the VPN-nonVPN [55] dataset, which collected a session over a Virtual Private Network (VPN) and a regular session. At the Canadian Institute of Cyber Security, this dataset is publicly available. This dataset contains 15" commonly used applications, such as YouTube, Facebook, Netflix, and Vimeo, that have been encrypted using various encryption protocols. The VPN-nonVPN dataset is composed of labelled network traffic such as email (SMTPS), web browsing (Firefox), chat (Skype), file transfer (SFTP), streaming (e.g., YouTube), peer-to-peer (P2P) (uTorrent), and VoIP (Hangouts voice calls). Table 33 shows the work that used the VPN-nonVPN dataset.
42. *DDOS 2016:* DDoS 2016 [56] dataset was collected as there was no existing dataset that has modern DDoS attacks like SIDDOS, HTTP flood, etc. It was published by Alkasassbeh et al in 2016 using the network simulator NS2 and is packet-based. The dataset comprises of four types of DDoS attacks: smurf, UDP flood, SIDDOS,

Table 34 DDOS

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[226]	2023	Fuzzy inference	IGFMS	96.00	99.40	91.70	95.40
[227]	2021	Fuzzy logic	IGFMS	91.10%	99.34	91.10	95.05
[228]	2018	HMM	HMM	97.41	90.38	84.13	87.14

Table 35 WSN-DS

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[229]	2023	DT	Gini FSM	99.50	99.50	99.50	99.50
[230]	2019	GXGBoost	Genetic	99.70	99.00	99.00	99.00

Table 36 Urban IOT DDOS

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[231]	2023	DNN	N/A	88.60	69.40	82.40	75.30
[232]	2022	LSTM	Cauchy distribution	89.00	89.00	100	94.20

and HTTP flood. The dataset is labelled, and there are 2.1 million packets and 27 features. Table 34 shows the work that used the DDOS-2016 dataset.

43. *WSN-DS 2016:* Almomani and colleagues [57] developed the WSN-DS dataset to evaluate the performance of Intrusion Detection Systems (IDS) in Wireless Sensor Networks (WSNs). Using the NS-2 network simulator, they modeled a WSN composed of numerous sensor nodes operating under the LEACH (Low Energy Aware Cluster Hierarchy) protocol, a hierarchical MAC protocol tailored for WSNs. During the data collection phase, each node monitored communications from five neighboring nodes and reported observations to a central sink. To replicate malicious behavior, four types of Distributed Denial of Service (DDoS) attacks were simulated: Grey-hole (selective traffic dropping by a malicious node), Black-hole (complete traffic dropping), Scheduling attack (forcing simultaneous transmissions from all nodes), and Flooding attack (overloading the network with excessive routing messages). The dataset comprises 374,661 labeled records, with 23 extracted features including sent/received packet counts, node ID, and WSN topology-related information. Table 35 shows the work that used the WSN-DS dataset.
44. *Urban-IoT-DDoS 2016:* Hekmati et al. [58] introduced the Urban-IoT-DDoS dataset to address the lack of IoT-specific DDoS data. Collected from a real-world urban IoT deployment, it includes one month of data from 4060 spatially distributed event-triggered sensors, recording binary activity every 30 s. Each entry contains a timestamp, geolocation (latitude, longitude), and node ID. A provided script simulates DDoS attacks by altering activity status (normal: "0", attack: "1"). The dataset and script are publicly available in CSV format. Table 36 shows the work that used the Urban-IOT-DDOS dataset.
45. *CIC-IDS 2017:* This CIC-IDS 2017 [59] dataset was proposed by Sharafaldin et al and includes the network traffic based on packet and flow generated over a period of 5 days from 2017-july-3-friday to 2017-july-7-Monday. It is publicly available. There are 80 features of network flow that are generated from network traffic

Table 37 CIC-IDS-2017

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[235]	2026	XGBOOST	N/A	99.89	99.89	99.89	99.89
[236]	2025	DNN	ANOVA	97.62	65.29	88.58	67.16
[237]	2024	CNN	RF	N/A	98.8	99.8	98.7
[238]	2023	CNN	N/A	98.61	N/A	N/A	N/A
[239]	2021	CNN	CNN	99.00	98.00	99.00	99.00
[240]	2020	Instance based learning	CFS	99.82	99.80	99.80	99.80
[241]	2019	Adaboost	PCA and EFS	81.83	81.80	100	90.01

Table 38 CIDDS-001

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[242]	2025	Hawks GBM system	PCA	99.1	99.6	99.8	99.3
[243]	2023	CNN	N/A	99.13	N/A	N/A	N/A
[244]	2020	KNN	Hybrid FSM	99.40	97.30	99.70	98.50
[245]	2018	Random forest	SMRT-VAE	99.99	100	100	100
[246]	2018	LSTM	N/A	84.83	85.14	88.30	86.70

Table 39 CIDDS-002

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[247]	2021	Bagging	N/A	99.71	100	99.00	100
[248]	2021	Segmented FL	N/A	98.42	97.00	88.00	92.00
[249]	2020	CART	N/A	99.64	100	100	100

extracted using the CICFlowMeter tool. The attacks present are SSH, Heartbleed, DOS, Brute-force, web attack, botnet, and DDOS, Brute force FTP, and infiltration [233, 234]. Furthermore, based on protocols like FTP and HTTPS dataset takes out the abstract behaviour of "25" users. Table 37 shows the work that used the CIC-IDS-2017 dataset.

46. *CIDDS-001. 2017:* The CIDDS-001 dataset [60] was captured in a simulated small business network over four weeks of unidirectional NetFlow traffic. It uniquely includes an external server, regularly accessed by clients, which was exposed to real internet-based attacks. Malicious behaviors such as port scans, brute force, and DoS were simulated. The dataset is publicly available, and labeling was straightforward due to known attack origins, targets, and timestamps. The external server offered HTTP and Seafile services and was intentionally made vulnerable by assigning it a public IP address outside the OpenStack environment. Table 38 shows the work that used the CIDDS-001 dataset.
47. *CIDDS-002, 2017:* Based on scripts of CIDDS-001, CIDDS-002 [61] is a port scan dataset. This dataset includes unidirectional network traffic based on the flow of "2" weeks from a simulated small business environment. CIDDS-002 includes both a variety of port scan attacks and normal user behaviour. The dataset's additional meta-information, including anonymized external IP addresses, is included in the technical report. The dataset is freely accessible to the public. Table 39 shows the work that used the CIDDS-002 dataset.

48. *TRAbID 2017:* In 2017 TRAbID [62] dataset was proposed by Viegas et al. For evaluating intrusion detection systems TRAbID database include "16" types of scenarios. Every scenario was recorded in a virtual environment (100 clients and 1 honeypot server). The traffic was captured for 30 min and few attacks were carried out in each scenario. The authors labelled the network traffic using the clients' IP addresses. Every client uses Linux system. The majority of clients only performed user requests, which were normal to the honeypot server while some clients only perform attacks. HTTP, SSH, SMTP, and SNMP traffic are examples of normal user behaviour, whereas malicious network traffic includes DoS attacks and port scans. TRAbID is readily available. Table 40 shows the work who used TRAbID2017 dataset.
49. *Unified Host and Network DataSet 2017:* The Unified Host and Network [63] Dataset has data based on host and network collected in a real-world setting, the Los Alamos National Laboratory (LANL) enterprise network. In bidirectional network traffic based on flow files, Timestamps and IP addresses were anonymized for privacy reasons. For 90 days, the network traffic was collected with no labels. This dataset is freely available to the public.
50. *Android Adware 2017:* This Android Adware [64] dataset comprises "1900" apps of three types: Adware (250 applications), Benign (1500 applications), and General Malware (150 applications). Lashkari et al. go over the specifics of the Android adware dataset. The adware category includes the following well-known families: Shuanet, Airpush, Mobidash, Kemoge, and Dowgin. To verify the relationships between categories of each app (general adware, benign, and malware), Droidkin is used, which is a lightweight detector of Android app. TO run apps, the Android smartphones (NEXUS 5) are used, and a gateway is used to capture generated traffic, which is labelled into three types (adware, general malware, and benign). This dataset is publicly available.
51. *Android malware dataset 2017:* CICAndMal2017 [253] is the name of this dataset, and was introduced by Shiravi et al., which contains both benign and malware applications. It includes applications from the Google Play Market published in 2015, 2016, and 2017. In the CICAndMal2017 dataset, the malware is divided into "4" types: (1) SMS Malware (e.g., Zsone, Mazarbot, FakeInst, BeanBot, etc), (2) Scareware (e.g, AndroidSpy, Penetho, VirusShield, AndroidDefender, etc), (3) Ransomware (e.g, LockerPin, Jisut, WannaLocker, Pletor, Charge, etc), and (4) Adware (e.g, Shuanet, Ewind, Dowgin, Selfmite..). Furthermore, the dataset CICAndMal2017 includes more than 80 features of the traffic network in packet format (.pcap files). Table 41 shows the work that used the Android malware dataset.
52. *CAN-OTIDS 2017:* Lee et al. [66] developed the CAN-OTIDS dataset in 2018 for in-vehicle network security research, focusing on CAN bus nodes. Attack messages were injected via the OBD-II port, and CAN traffic was recorded periodically.

Table 40 TRABID

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[250]	2024	Gaussian NB	Correlation	66.4	38.3	71.1	49.8
[251]	2020	MLP	N/A	99.80	99.78	99.73	99.75
[252]	2020	Modest Adaboost	N/A	98.50	98.50	99.75	99.12

Table 41 Android malware

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[253]	2018	DT	CFS, Info gain	N/A	85.00	88.00	86.50
[254]	2018	Rotation Forest	N/A	88.26	88.16	88.00	88.40

Table 42 CAN-OTIDS

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[255]	2024	Stacking ensemble	N/A	99.9	99.9	99.9	99.9

Table 43 CIC-IDS-2018

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[256]	2026	Random forest	TreeSHAP	85.38	N/A	N/A	79.30
[257]	2025	J48	GWO	99.0	99.8	99.0	99.4
[237]	2024	CNN	N/A	N/A	99.1	95.3	97.2
[258]	2023	XGBOOST	PCA	99.76	92.07	98.87	94.93
[259]	2020	DNN	PCA	76.47	50.00	36.00	42.00
[260]	2020	Random forest	PCA	99.00	99.00	99.00	99.00

The dataset includes 2,369,868 normal messages and attack instances: 656,579 DoS, 356,271 spoofing, 591,990 fuzzy, and 995,472 impersonation attacks. A novel spoofing type was introduced by analyzing CAN logs. OTIDS simulates realistic conditions by transmitting remote frames and receiving messages. Each record includes timestamp, CAN ID (hex), DLC (0–8), and DATA [0–7] bytes. This dataset is vital for enhancing CAN bus intrusion detection. Table 42 shows the work that used the CAN-OTIDS dataset.

53. *PUF 2018:* Using only DNS connections, Sharma et al. collected flow-based PUF [67] data on a campus network over three days. Out of 298,463 are single direction flows, 38120 are malicious, whereas the rest are normal user activity. An intrusion prevention system log is used to label all flows. IP addresses have been eliminated from the dataset for privacy reasons. The creators made PUF available to the public.
54. *CIC-IDS 2018:* The Communications Security Establishment (CSE) and the Canadian Institute for Cybersecurity (CIC) proposed the dataset CSE-CIC-IDS 2018 [68]. This dataset consists of 7 types of attack: Heartbleed, DDoS, brute-force, DoS, botnet, web attacks, and network infiltration from inside. "50" machines are included in the attacking framework, while the victim organisation contains "5" departments and "420" machines, and "30" servers. "80" features of network flow, which are generated from network traffic, were extracted using the CICFlowMeter tool. Table 43 shows the work that used the CIC-IDS-2018 dataset.
55. *DS2OS 2018:* Pahl et al gave the dataset DS2OS [69] in 2018, which is an IIOT dataset. DS2OS dataset have attacks on applications and sensors. It has details of several anomalies and attacks in IIOT applications like smart buildings, homes, and factories, etc. The dataset DS2OS includes "13" features and "357,952" samples and has "10,017" anomalous data values and "347,935" normal data values with eight classes. The attacks present are data type probing, DOS, malicious control, wrong

Table 44 DS2OS

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[261]	2025	Ensemble	N/A	N/A	98.26	98 98	98
[262]	2024	Random forest	information gain	99.42	99	99	99
[263]	2023	Voting classifier	N/A	99	99.1	99	N/A
[264]	2021	Random forest	Correlation coefficient	99.99	99.90	99.80	99.90
[265]	2021	XGBoost	NHFRA	99.43	99.80	95.60	97.30

Table 45 N-BIOT

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[267]	2025	Autoencoder	Fisher's score	99.96	100	99.93	99.96
[268]	2025	CNN-LSTM	Fisher's score	99.99	99.98	99.96	99.97
[269]	2024	Random forest	Fisher's score	99	90	79	79
[270]	2021	RF-Smote	N/A	99.98	100	99.97	99.98
[271]	2020	WOA-V-ET	WOA	99.98	99.90	99.80	99.90
[272]	2020	DPE-TIDD	DPE	100	99.90	100	99.90

setup, malicious operation, scan, spying. Table 44 shows the work who used DS2OS dataset.

56. *N-BaIOT 2018:* The traffic for N-BaIOT [70] dataset was collected from IOT devices in which access points were connected via WI-FI that are then connected to the router through a wired central switch. Port mirroring on the switch is performed for sniffing the network traffic, and data was recorded using Wireshark in pcap format. Twenty-three features were extracted from five-time windows of the recent 100 ms, 500 ms, 1.5 s, 10 s, and 1 min. There are 2 types of attacks in this dataset, like BASHLITE and Mirai. In BASHLITE, there are attacks like scan, junk, UDP, TCP, and COMBO. In Mirai, the attacks present are scan, Ack, SYN, UDP, and UDPplain. Recently, Rezaei et al. [266] evaluated a federated RNN-based intrusion detection framework using the N-BaIoT dataset under adversarial poisoning attacks. The study introduced the MPLFA attack and the SNCOC defense mechanism, demonstrating the usefulness of N-BaIoT for assessing the robustness of federated intrusion detection systems in IoT environments. Table 45 shows the work that used the N-BaIOT dataset.
57. *Survival Analysis 2018:* Han et al. [71] proposed a dataset covering three in-vehicle attacks-Flooding, Fuzzy, and Malfunction-captured during 5-second attack injections every 20 s. Flooding uses CAN ID 0x000 to overwhelm the bus, while Fuzzy injects random CAN messages at 0.0003-second intervals with IDs from 0x000 to 0x7FF. Malfunction attacks target specific IDs-0x316, 0x153, 0x18E-linked to HYUNDAI YF Sonata, KIA Soul, and CHEVROLET Spark. An Attack-Free State represents normal driving. Each record contains Timestamp, CAN ID (HEX), DLC (0–8), DATA [0–7] bytes, and a Flag indicating injected (T) or normal (R) messages.
58. *BLEBeacon 2018:* Sikeridis et al. [72] introduced the BLEBeacon dataset to support research on Bluetooth Low Energy (BLE) beacons-compact, low-power devices that periodically broadcast information. In their study, 46 participants carried Gimbal Series Ten Beacons emitting BLE signals every second. These signals were captured by 32 Raspberry Pi-based gateways deployed across a 3-story campus

Table 46 Kitsune

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[273]	2024	LSTM	N/A	93.00	93.83	93.53	93.58
[274]	2021	Ensemble boosted tree	N/A	99.80	99.70	98.10	99.20

Table 47 OPCUA

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[275]	2025	Fuzzy K Means	Autoencoder	100	100	100	100
[276]	2024	CNN	N/A	99.3	N/A	N/A	N/A

building. Gateways forwarded the received messages to a central server. The dataset, collected from September 15 to October 17, 2016, includes timestamps, Received Signal Strength Indication (RSSI), and records of beacon entry and exit times from each gateway's coverage zone.

59. *Kitsune network (Mirsky et al. 2018):* Mirsky et al. collected dataset kitsune [73]. The dataset presents real attacks detected in an IP camera surveillance network with two deployments, each housing four HD cameras powered by PoE. Attacks target video uplink availability and integrity, including man-in-the-middle attacks like video injection, and SYN floods. Various attack types such as Recon, DoS, and Botnet Malware (Mirai) are included, captured via packet capture technology. Additionally, an IoT-populated Wifi network with Mirai-infected security cameras is used to assess Kitsune's performance in a noisy environment. Table 46 shows the work who used the dataset.
60. *OPCUA 2018:* Pinto introduced the OPCUA dataset [74], capturing OPC UA protocol traffic within a CPPS testbed. The setup includes seven Raspberry Pi nodes using Python FreeOpcUa-two production units (three devices each) and one MES (Manufacturing Execution System) node. Devices act as both OPC UA servers and clients, while the MES functions solely as a client subscribing to all device variables. A malicious node simulates attacks such as Denial of Service (DoS), Man-in-the-Middle (Eavesdropping), and Spoofing (Impersonation). Tshark records the OPC UA traffic, exporting it as a CSV dataset that includes both normal and anomalous behavior. Table 47 shows the work that used the dataset.
61. *TCP FIN FLOOD 2019:* Stiawan et al. [75] generated this dataset from a testbed with sensor nodes (MQ2, DHT22, etc.) and a WeMos D1 controller. The testbed included Apache server, MySQL, and Snort IDS. Using Hping3, TCP FIN flood attacks were simulated. Data from normal and attack scenarios were captured in pcap format over 5-min intervals per run.
62. *DIOT 2019:* Nguyen et al. [76] collected communication data from 33 IoT devices (e.g., smart plugs, cameras, sensors), grouped into 23 types. A lab network with Kali Linux was used as a gateway and access point. Device traffic was captured using tcpdump, reflecting both lab and real-world usage.
63. *Bot-IOT 2019:* A new dataset, Bot-IoT [77] was proposed by Koroniotis et al, which differs from older datasets in the IOT environment. Bot-IOT dataset contains records over "72,000,000", including DoS, DDoS, Keylogging, OS and Service Scan, and Data Extraction attacks. To reproduce the network behaviour

Table 48 BOT-IOT

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[277]	2026	MC-AIDA	Genetic algo	98.5	N/A	98.3	98.4
[278]	2025	SVM	Random forest	99.60	N/A	N/A	N/A
[279]	2024	SVM	Chi-squared + RF	99.99	99.99	99.99	99.99
[280]	2021	LSTM-RNN	N/A	99.80	99.00	99.00	99.00
[281]	2020	Naive Bayes	Bijective soft set	99.79	99.00	98.00	99.00

Table 49 CUPID

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[282]	2025	ETC	Extra tree classifier	97.21	97.54	97.03	96.38
[283]	2022	RF	Anova, Chi square, MI	99.90	99.70	99.60	99.70

of IoT devices Node-red tool were used. To connect machine-to-machine (M2M) communication, the dataset used the MQTT protocol, which is a light-weight communication protocol. The testbed employs "5" IoT scenarios: weather station, remotely activated garage door, motion-activated lights, smart fridge, and smart thermostat. Table 48 shows the work that used the BOT-IOT dataset.

64. *CUPID (Colorado University Penetrating Intrusion Dataset) 2019:* CUPID [78] is a publicly available dataset featuring human-guided traffic generated by 10 ethical penetration testers. Each tester produced benign traffic for 1 h, followed by malicious activity. Traffic was labeled based on the Kali instance's IP-'0' for benign and '1' for malicious. A baseline sample collected on April 22, 2019, contains 4,346,077 packets and is 3.3 GB in size; the complete dataset is around 50 GB. The sample includes traffic from 179 external hosts and primarily comprises TCP (75%), alongside DCE/RPC, ICMP, HTTP, Kerberos, SMTP, NetBIOS/SMB, and LDAP protocols. UDP accounts for 7.5%, including DNS and NTP, with additional ARP and 802.1Q traffic. SSL/TLS packets form 0.6% (25,341 packets). CUPID reflects real enterprise traffic with diverse protocol usage. Table 49 shows the work that used the CUPID dataset.
65. *Car hacking dataset 2020:* Car-hacking datasets [79] encompass a variety of intrusions, including fuzzy attacks, Denial-of-Service (DoS) attacks, drive gear mimicry, and RPM gauge emulation. These datasets were created by recording Controller Area Network (CAN) traffic from a real vehicle's OBD-II port during message injection attacks. Each dataset includes 300 intrusion instances, each lasting between 3 to 5 s, resulting in approximately 30 to 40 min of CAN traffic. The DoS attack involves injecting '0000' CAN ID messages every 0.3 ms, resulting in 3,665,771 messages, with 3,078,250 normal and 587,521 injected messages. In the fuzzy attack, random CAN ID and DATA values are injected every 0.5 ms, yielding 3,838,860 messages, with 3,347,013 normal and 491,847 injected messages. Spoofing attacks on drive gear and RPM information inject one message per millisecond, resulting in 4,443,142 and 4,621,702 messages, respectively, with normal message counts of 3,845,890 and 3,966,805, and injected message counts of 597,252 and 654,897. Lastly, in the GIDS Attack-free dataset, there are 988,987 messages, 988,872 of which are normal, and none are injected. Each dataset includes attributes like timestamp, CAN ID, DLC, DATA, and Flag to distinguish between

Table 50 CAR HACKING

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[284]	2026	RF	N/A	99.99	99.99	99.99	99.99
[285]	2025	Deep Q-Network (DQN)	N/A	91.00	88.00	95.00	91.00
[286]	2025	LETNN	N/A	99.98	99.97	99.98	99.98
[287]	2024	IVN-ViT	PSO	99.84	99.91	99.52	99.71
[288]	2023	Autoencoder	N/A	99.98	99.96	99.85	99.96
[289]	2021	Reduced inception-Resnet	N/A	N/A	100	99.89	99.95

Table 51 CIRA-CIC-DOBHRW

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[290]	2025	AutoRoC-DBSCAN	N/A	N/A	N/A	100	N/A
[291]	2025	MTL-DOHTA	N/A	N/A	N/A	N/A	98.37
[292]	2025	XGBOOST	ACO	99.99	100	100	100
[293]	2022	Stacked RF	N/A	99.99	99.91	99.92	99.91
[294]	2020	RF and GB	N/A	99.99	100	100	100

normal ('R') and injected ('T') messages. Table 50 shows the work who used CAR HACKING dataset.

66. *CIRA-CIC-DoHBrw-2020:* The CIRA-CIC-DoHBrw-2020 dataset [80], publicly available, was introduced by Montazeri Shatoori et al. in their study on detecting DNS over HTTPS (DoH) tunnels using time-series classification. DoH, a protocol by IETF, encrypts DNS queries to enhance privacy and defend against man-in-the-middle attacks. The dataset captures Malicious-DoH, Benign-DoH, and non-DoH traffic using four DoH servers (Cloudflare, AdGuard, Quad9, Google DNS) and five browsers/tools (Mozilla Firefox, Google Chrome, DNS2TCP, Iodine, DNSCat2). A two-layer detection model is proposed: the first layer uses statistical features to classify DoH vs. non-DoH traffic, while the second uses time-series classification to differentiate malicious from benign DoH traffic. Table 51 shows the work who used CIRA-CIC-DOHBrw dataset.
67. *INSDN 2020:* The InSDN dataset [81], publicly available for research, focuses on attacks specific to Software-Defined Networks (SDNs). It is categorized into three groups: normal traffic, attack traffic targeting a Metasploitable-2 server, and attack traffic on the OVS (Open vSwitch) machine. Traffic traces were captured using TCPdump at both the SDN controller and target machines, and CICFlowMeter was used to extract flow features. The dataset includes 343,939 instances-68,424 normal and 275,515 attack samples. Over 80 features are organized into 8 groups: Packet-based, Network identifiers, Bytes-based, Interarrival time, Flow timers, Flag, Flow descriptors, and Subflow descriptors. Source and destination IPs were used for labeling. Table 52 shows the work that used the INSDN dataset.
68. *IOT-23 2020:* In January 2020, Stratosphere Laboratory in Czechia made IOT-23 [82] dataset available initially and is captured in a real environment. It contains network traffic that has 20 malware and 3 benign sets. Number of flows in benign is 30,858,735. Number of flows in malicious captures are: Part-Of-A-Horizontal-Port Scan (213,852,924 flows), Okiru (47,381,241 flows), Okiru-Attack (13,609,479 flows), DDoS (19,538,713 flows), C and C-HeartBeat (33,673 flows), C and C (21,995

Table 52 INSDN

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[295]	2025	Ensemble	Hybrid	99.9	N/A	N/A	N/A
[296]	2025	CNN	N/A	97.14	N/A	N/A	N/A
[297]	2024	CenterNet ResNet152V2	N/A	99.65	99.56	99.81	99.68
[298]	2022	DAE	CNN	99.70	99.80	99.80	99.80
[299]	2021	Hybrid CNN-LSTM	N/A	96.32	97.60	97.24	97.42
[300]	2022	LSTM-AE	RF, IG	99.50	99.30	99.60	99.45
[81]	2020	RF	N/A	N/A	99.42	99.69	99.55

Table 53 IOT-23

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[301]	2025	Hybrid tree	N/A	99.99	99.99	99.99	99.99
[302]	2024	GRU + LSTM + MLPNN	PCA	99.3	99.1	98.7	98.9
[303]	2024	Random forest	N/A	96.81	N/A	N/A	N/A
[304]	2021	Random forest	PCA	99.00	95.00	94.00	92.00
[305]	2021	Random forest	N/A	92.96	95.80	94.80	94.80
[306]	2020	Stacking EL	DSAE	99.70	100	95.00	98.00

Table 54 LITNET

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[307]	2024	EPSO DT	N/A	99.86	99.15	98.37	98.72
[308]	2023	ILSTM	N/A	93.97	88.61	92.55	90.61
[306]	2020	DNN + LSTM	DSAE	100	100	100	100

flows), Attack (9398 flows), C and C (888 flows), C and C-Heart Beat Attack (883 flows), C and C-File download (53 flows), C and C-Tori (30 flows), File download (18 flows), C and C-HeartBeat File Download (11 flows), Part-Of-A-Horizontal-PortScan Attack (5 flows), C and C-Mirai (2 flows). There are 21 features present in IOT-23 dataset. The attributes are mixed, some are nominal, some are numeric, and some take time stamp values. Table 53 shows the work that used the IOT-23 dataset.

69. *LITNET-2020:* The LITNET-2020 [83] was collected in real network for 10 months. The traffic is captured in nfcad binary format files, a single nfcapd file is collected per week for 2 capture periods. The size of the files is 1.3 GB. There are 19 attacks types. Networks IP addresses are anonymized. There are total 85 attributes, 49 attributes specific to the NetFlow v9 protocol, the extended dataset has 15 attributes, and the generator has 19 attributes for the recognition of attack type. Table 54 shows the work that used the LITNET dataset.
70. *NetML-2020:* NetML [84] dataset was released in "open challenge network traffic analytics using machine learning workshop sponsored by Intel Corporation" in 2020. This dataset was created by obtaining 30 traffic data from Stratosphere IPS. In JSON format, flow features are extracted and are listed in the output file line by line. For each flow, metadata features are extracted, and if the flow sample has packets for the protocol, then DNS, HTTP, and TLS features are extracted. To obtain the NetML dataset, a unique number is assigned to identify every flow and

Table 55 NETML

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[306]	2020	DNN + LSTM + Meta classifier	DSAE	100	100	100	100

Table 56 DAD

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[309]	2021	AdaBoost	N/A	99.98	99.54	99.35	99.17

label information from the raw traffic packet capture file and is appended to the output JSON file. There are 484,056 flows and 48 feature attributes. IP addresses of the source and destination are replaced by an IP masked string to mask the IP. The dataset is divided into training, test-std, and test-challenge sets. Table 55 shows the work that used the NETML dataset.

71. *LoED: LoRaWAN Dataset 2020:* The dataset LoED (LoRaWAN at the Edge), developed by Bhatia and colleagues [85], is a publicly available dataset containing network data sourced from a LoRaWAN sensor network located in the urban area of London, U.K. The authors collected LoRaWAN packets over a period of four months in 2019 and 2020. This involved capturing data from nine strategically positioned gateways in different urban environments, including 5 outdoor and 4 indoor locations. Each packet recorded in the dataset contained diverse information, including signal modulation characteristics, sender's address, and raw payload. The researchers also performed an initial analysis to glean insights from the recorded traffic. This analysis involved examining metrics such as daily or per-node packet count and the distribution of specific traffic attributes like frequency, packet type, or LoRa spreading factor.
72. *GHOST-IoT 2020:* Developed within the GHOST EU project [86], the GHOST-IoT dataset captures network traffic from a real Smart Home setup with nine devices using IP, Z-Wave, ZigBee, RF869, and Bluetooth protocols. The dataset, recorded over 10 days, includes 3,811,419 raw packets. Normal behavior was recorded for nine days, followed by seven attack scenarios on the tenth day. These include device tampering, unknown device injection, and network flooding. Although attacks are not explicitly labeled, supplementary files provide flow-level and statistical features.
73. *DAD 2020:* Vigoya et al. [87] created the DAD dataset to support anomaly detection in IoT networks. Sixteen virtual IoT temperature sensors sent MQTT messages every 5 min over seven days. Attack scenarios were introduced by Interception, Modification, and Duplication of data on five days. The dataset comprises 101,583 packets, with 63.3% being MQTT and 16% labeled as malicious. Table 56 shows the studies utilizing the DAD dataset.
74. *MedBIoT 2020:* Guerra-Manzanares et al. [88] addressed the scarcity of large, diverse IDS datasets for ML-based IoT security by introducing MedBIoT. This annotated dataset includes 83 IoT devices-3 physical (1 smart bulb, 2 smart switches) and 80 emulated via Docker. Attack traffic was generated using BASHLITE, Mirai, and Torii malware, focusing on early botnet stages such as propagation and Command and Control (C&C) communication. The network was divided into three segments: a monitoring network, the Internet, and an IoT LAN for controlled malware spread.

Table 57 MEDBIOT

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[310]	2024	CNN	N/A	N/A	98	97	97
[311]	2023	Random forest	N/A	98.44	93.62	85.35	89.08
[312]	2022	Decision tree	SBS	99.00	99.00	99.00	99.00
[313]	2020	Fast gates RNN	N/A	99.99	99.99	99.99	99.99

Table 58 MQTTSet

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[314]	2025	CNN-LSTM	N/A	83	87	55	61
[315]	2024	Decision tree	N/A	92.5	N/A	N/A	92.38
[316]	2023	MAOA-AL-DBN	LSTM	99.96	5.48	100	1.096
[317]	2023	Decision tree	RFE	99.32	99.93	99.30	99.29

Data was captured at all endpoints and servers, and ML features were extracted using the Kitsune autoencoder tool, producing 100 statistical features from raw PCAPs across various time windows. Table 57 summarizes the studies utilizing the MedBIoT dataset

75. *MQTTset 2020:* Vaccari and colleagues [89] created the MQTTset dataset, a publicly available dataset specifically designed for the MQTT protocol. The researchers employed a network comprising eight Smart Home sensors that measured variables such as humidity, temperature, and smoke. These sensors were emulated through the use of the IoT-Flock, IoT traffic generator, and were connected to an MQTT broker using Eclipse Mosquitto. Both legitimate activities and attacks targeting the MQTT protocol are included in the complete traffic. The traffic was captured at the broker over the course of a week. The dataset contains 34 features and is available in either CSV Format or raw packet capture. These features include a label that indicates whether the traffic is classified as malicious or legitimate. Table 58 shows the work who used the MQTTset dataset.
76. *IOTID20:* Ullah and Mahmoud [90] introduced the IOTID20 dataset, developed in a smart home testbed using Wi-Fi cameras (EZVIZ, SKT NGU) as vulnerable IoT devices connected to a Wi-Fi router, along with tablets, smartphones, and laptops used to simulate attacks. Attack types include Mirai (HTTP flooding, brute force, UDP flooding), DoS (SYN flooding), scanning (OS, host port), and MITM (ARP spoofing). Network traffic was captured via pcap files and processed using CICFlowMeter, yielding a CSV file with 80 features and 3 labels (binary, category, sub-category). This dataset reflects real IoT network behaviors and is a rare publicly available resource for IoT intrusion detection. Table 59 shows the work that used the dataset.
77. *MQTT-IOT-IDS 2020:* This dataset [91, 320] pioneers the simulation of an MQTT-based network, offering a comprehensive exploration of network behaviors and associated attacks. Generated through a simulated MQTT network architecture, the dataset comprises an attacker, a broker, twelve sensors, and a simulated camera. It records five scenarios: aggressive scan, normal operation, UDP scan, MQTT brute-force attack, and Sparta SSH brute-force. Three levels of abstracted features are extracted from the raw pcap files: packet characteristics, one-way

Table 59 IOTID20

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[318]	2025	HIDS-IoMT	PSo	99.92	99.91	99.99	99.95
[302]	2024	LASTM + GRU + MLPNN	PCA	98.2	97.9	97.1	97.5
[319]	2023	Random forest	N/A	99.09	98.40	98.00	97.80
[90]	2020	Decision tree	N/A	88.00	88.00	88.00	88.00

Table 60 MQTT-IOTIDS2020

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[321]	2024	Naive Bayes	N/A	N/A	99	97	98
[322]	2023	Random forest	N/A	99.98	99.98	99.98	99.98
[323]	2023	TNN-IDS	N/A	99.99	99.99	99.99	99.99

Table 61 NF-UQ-NIDS

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[324]	2025	CNN-BiLSTM	N/A	95.56	96.39	87.44	91.7
[325]	2024	DNN	N/A	98.32	97.99	94.73	96.33
[92]	2021	ETC	N/A	97.25	N/A	95.60	94.00

flow attributes, and two-way flow attributes. The dataset includes csv feature files tailored for Machine Learning (ML) applications, facilitating predictive analysis. Table 60 shows the work who used the dataset.

78. *NF-UQ-NIDS 2020:* Sarhan et al. [92] introduced the NF-UQ-NIDS dataset, a unified network intrusion detection dataset combining BoT-IoT, UNSW-NB15, CSE-CIC-IDS2018, and ToN-IoT within the University of Queensland's NF-collection. It standardizes network-security data for enhanced interoperability and large-scale analysis. An additional label denotes the source dataset for each NetFlow record. Attack types are grouped into broader categories like DoS, DDoS, brute-force, and injection attacks. The dataset includes a wide variety of attacks such as MITM, ransomware, XSS, worms, theft, password, scanning, and more. Table 61 shows the work who used the dataset.
79. *IOTKeeper 2020:* Hafeez Ibbad et al. presented dataset IOTKeeper [93]. The test environment utilizes IOT-KEEPER as its network gateway, employing tcpdump to record all incoming and outgoing traffic through both wired and wireless interfaces on the gateway. When an IoT device communicates with the Internet through an IoT hub using protocols such as Weave or ZigBee, its device-to-internet (D2I) communications are observed by capturing network traffic at the IoT hub. The dataset includes traffic data from both harmless and harmful network actions; harmful activities are mimicked using Raspberry Pis and IoT devices infected with Mirai. To minimize potential hazards, the gateway is set up to block all unfiltered outgoing traffic, thus averting the propagation of malicious behavior onto the public Internet. Table 62 shows the work that used the dataset.
80. *IoT-DeNAT 2020:* Meidan et al. [94] recorded traffic from IoT and non-IoT devices behind NAT. A Cisco switch separated home and telco networks using VLANs, with NetFlow data collected by nProbe and mirrored traffic captured via Wireshark.

Table 62 IOTKeeper

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[93]	2020	IOT Keeper	CFS	99.00	N/A	N/A	N/A

Table 63 IOT-DENAT

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[326]	2024	LGBM	N/A	98.00	95.00	96.00	95.00

Table 64 ROAD

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[327]	2022	IDS-IVN	N/A	99.27	89.36	98.41	93.67

Devices from seven vendors (e.g., Amazon, Samsung) were used to study NAT-based behavior and deNAT analysis. Table 63 summarizes related work.

81. *IoT-Finder 2020:* Perdisci et al. [95] introduced a dataset of DNS traffic from 53 IoT devices (e.g., speakers, cameras, thermostats, consoles) over 1.5 months. Devices operated in a lab setting simulating real-life interaction, with DNS data logged for behavior analysis.
82. *ROAD Dataset 2020:* Verma et al. [96] developed the ROAD dataset, hosted on Zenodo, featuring 30 min of CAN-bus attack data (33 unique attacks) and 3 h of ambient data across 12 captures. Data was collected via SocketCAN and Kvaser Leaf Light V2 on a Linux system through a vehicle's OBD-II port, with anonymization applied. The dataset includes real-road and dynamometer scenarios with unusual behaviors (e.g., driving with doors open) to support IDS development. It provides signal-translated data using the CAN-D algorithm for 17 attack and all ambient captures. Attack types include Fuzzing (random ID injection), Fabrication, Masquerade (targeting signals like speed, temperature, lights), and Accelerator Attacks (locking vehicle speed). Stealthy attacks use the flam technique, embedding spoofed messages within normal frames. Obfuscated yet realistic, the dataset includes metadata on attack impact and driving context, addressing key gaps in CAN-based IDS research. Table 64 shows the work who used the ROAD dataset.
83. *X-IIoTID 2021:* Al-Hawawreh et al. [97] introduced X-IIoTID, a comprehensive intrusion detection dataset tailored for heterogeneous Industrial IoT (IIoT) environments. It captures diverse attack types, protocols, and multiview features, supporting end-to-end traffic collection from physical field devices to edge gateways, and further to cloud and enterprise systems. Network traffic was recorded for up to 2 h per session using the dumpcap tool on a Raspberry Pi B+ edge gateway, while SAR tools monitored system resources and OSSEC logs captured security alerts. Spanning from December 5, 2019, to March 23, 2020, the dataset includes both benign and attack traffic. Attack scenarios-conducted between January 7 and March 27, 2020-cover vectors such as weaponization, reconnaissance, exploitation, command and control, lateral movement, tampering, crypto ransomware, RDoS, brute force, and exfiltration, making it suitable for advanced IIoT security research. Table 65 shows the work who used the dataset.

Table 65 X-IIOTID

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[328]	2025	MSDL	N/A	N/A	99.72	99.84	99.78
[329]	2025	hybrid CNN-Bi-LSTM	N/A	98.1	97.5	98.3	97.9
[330]	2025	Novel SConv1D-Attention	bPSO	99.86	N/A	99.74	99.64
[331]	2024	SABiLSTM	N/A	96.54	90.94	83.79	88.90
[97]	2021	Decision tree	N/A	99.54	99.50	99.50	99.50

Table 66 CAN signal extraction and translation

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[332]	2020	ML	N/A	N/A	86.6	22.50	N/A

Table 67 CAN-FD

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[333]	2025	GRU NADAM	N/A	98	98	98	98
[334]	2024	LSTM	N/A	99.64	99.58	99.97	99.78
[99]	2024	ACDM	N/A	99.99	99.96	99.97	99.97

84. *CAN signal extraction and translation 2021:* Song et al. [98] developed this dataset to support research in CAN signal extraction and translation. It contains 40 CAN traffic logs collected via OBD queries during controlled driving, with each file named after the queried PID. Key powertrain parameters, like vehicle speed, show valid values only when the vehicle is in motion. CAN traffic was recorded using a Kvaser CAN interface, Raspberry Pi, or Arduino through the unauthenticated OBD-II port. Both diagnostic responses and normal CAN messages were captured concurrently at 200 ms intervals to avoid disrupting normal traffic. Each log includes approximately 127–128k CAN messages, with 300 diagnostic responses, and attributes such as Timestamp (seconds), CAN ID (hex), DLC (0–8), and DATA [0–7] (byte values). Table 66 shows the work who used CAN SIGNAL EXTRACTION AND TRANSLATION dataset.
85. *CAN-FD:* Gao et al. [99] introduced the CAN-FD dataset to advance automotive security by capturing real-world CAN-FD traffic from 2021 vehicles, including both normal and injected attack messages. The dataset contains 1 h of normal driving data and 500 s of injected attacks, with each record comprising Timestamp, Arbitration ID (hex), DLC (0-64 bytes), Data[0-64], and a Label (R for normal, T for attack). Three attacks are included: Flooding (AID 0x000 every 0.1 ms), Fuzzing (random messages every 0.2 ms), and Malfunction (targeting ID 0x125 every 1 ms). Data was collected on a city route near Korea University using physical CAN-FD access. The dataset is essential for IDS development in CAN-FD networks. Table 67 shows the work who used CAN-FD dataset.
86. *M-CAN intrusion dataset 2021:* Guan et al. [100] The M-CAN Intrusion dataset consists of normal and abnormal CAN traffic data, designed for studying attacks on in-vehicle communication networks. M-CAN, a bus-type topology for vehicle multimedia and navigation systems, was used to collect normal data from a Genesis G80 during a 36-min around Korea University. The abnormal dataset includes DoS

Table 68 M-CAN

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[100]	2024	PTIDS	N/A	90.14	N/A	N/A	N/A

Table 69 B-CAN

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[100]	2024	PTIDS	N/A	49.81	N/A	N/A	N/A

and fuzzing attacks, which were simulated on a local computer by injecting attack packets into the previously collected normal data. The dataset attributes include a timestamp, CAN ID, DLC (data length code), payload, and labels indicating normal or injected messages. The DoS attack involves injecting fixed CAN ID messages ('0x00000000') at 0.25-millisecond intervals, while the fuzzing attack involves random CAN ID, DLC, and payload values injected every 0.1 ms. Table 68 shows the work that used the M-CAN dataset.

87. *B-CAN intrusion dataset 2021:* The B-CAN Intrusion Detection dataset [100] contains normal and attack traffic data from the Genesis G80, focusing on low-speed communication systems such as BCM lights, power windows, and smart key modules. The normal dataset was recorded over 24 min during a city drive around Korea University, while the attack dataset was generated by injecting DoS and fuzzing attacks into the normal data via a local computer. The dataset attributes include a timestamp, CAN ID (in HEX), DLC (data length code), payload (byte values), and a label indicating whether a message is normal (0.0) or injected (1.0). The DoS attack consists of injecting messages with CAN ID '0x00000000' and Payload '00 00 00 00 00 00 00 00' every 0.25 ms for 4 s, with 25 attacks in total. The fuzzing attack injects random CAN ID, DLC, and payload values (ranging from '00 00 00 00 00 00 00 00' to 'FF FF FF FF FF FF FF FF') every 0.1 ms for 1 s, with 10 attacks in total. The normal drive route was a 1-h round trip through city roads, used to collect both B-CAN and M-CAN datasets. Table 69 shows the work that used the B-CAN dataset.
88. *HIKARI-2021:* The HIKARI-2021 dataset [101], captured between March 28 and May 4, 2021, is a recent IDS dataset reflecting evolving network traffic trends. Each capture session lasted 3–5 h and included both encrypted traces and ground truth data. It comprises 555,278 entries with synthetic benign and malicious traffic. Based on CICIDS-2017, it incorporates 80 original features plus 4 additional ones: source IP, destination IP, source port, and destination port. Packet traces were anonymized (e.g., payload, IPs), and features were extracted. The dataset includes packet traces, documentation, and processed feature sets for IDS research. Table 70 shows the work who used the HIKARI dataset.
89. *TON-IoT 2021:* The TON-IoT dataset [102] is a comprehensive, next-generation dataset featuring heterogeneous telemetry data from network traffic, OS logs (Windows 10/7, Ubuntu 14.04/18.04), and IoT/IIoT sensors. Developed at UNSW Canberra's IoT Lab, it was captured in a realistic testbed simulating NVF, SDN, and SO communication across fog, edge, and cloud layers. It includes over 22 million records across 23 CSV files (extracted with Zeek), with 43 features and 2 class labels

Table 70 HIKARI

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[335]	2025	Random forest	MF+RFE	100	100	100	100
[336]	2025	GPC-KOS-DFS	VE-DQN-MAFS	95.86	N/A	99.30	96.06
[337]	2024	Neural network	N/A	94	81	99	89
[338]	2022	KNN	N/A	98.00	98.00	98.00	98.00
[101]	2021	Random forest	N/A	99.00	99.00	99.00	99.00

Table 71 TON-IOT

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[339]	2026	Temporal-spatial attention fusion (TSAF)	N/A	99.0	99.1	98.2	98.3
[340]	2025	HYBRID ACBSLTM	Meta heuristic	99.8	99.8	99	99.4
[341]	2024	XGBoost	Chisquare	99.84	99.88	N/A	99.90
[342]	2023	Stacking Ensemble	Spearman rank, Chisquare	99.87	99.80	99.80	99.80
[343]	2022	Densenet multicategory	N/A	100	100	100	100
[344]	2022	XGBoost	Chisquare	98.20	95.40	99.60	97.50
[345]	2021	XGBoost	Chisquare	98.20	95.90	98.90	97.40

Table 72 USB-IDS-1

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[346]	2024	Decision tree	Genetic Algo	58	60	61	59
[347]	2023	Autoencoder	XAI-SHAP	98	N/A	98.00	100
[348]	2022	Deep AE	XAI-SHAP	100	N/A	N/A	N/A

each, split into training and testing sets. The dataset contains labeled normal and malicious events across 9 attack categories: DoS, DDoS, scanning, ransomware, backdoor, XSS, injection, MITM, and password cracking. Table 71 shows the work who used the TON-IOT dataset.

90. *USB-IDS-1 2021:* USB-IDS [103] dataset was developed at the University of Sannio at Benevento (USB), Italy, in 2021. It is a multilayer dataset in which both network traffic and application-level facets like server-side performance, configuration, and defences are included. This dataset has attacks based on the DoS protocol. Data is in the form of comma separated values (CSV) files. There is a labelled bi-directional network flow, 84 values per record, including the label are present. Flows are obtained using CICFlowMeter. In USB-IDS-1, Only attacks are present; it does not record benign network traffic profiles over a large network. The attacks present in this dataset are Hulk, TCPFlood, Slowloris, and Slowhttptest. Table 72 shows the work who used the USB-IDS-1 dataset.
91. *AWID3 2021:* AWID3 [104] adds to the applicability of the widely known AWID2 by capturing traces of attacks received at IEEE 802.1X Extensible Authentication Protocol (EAP) systems. A significant contribution to intrusion detection is expected from AWID3. Raw cleartext pcap files for AWID3 are publicly available. “254” features, i.e., “253” generic, plus “1” more for labeling, in CSV format, were manually taken out. This "AWID3-CSV" dataset adds to the original one given in pcap format. The taken-out features spread across both the MAC and application

Table 73 AWID3

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[349]	2025	Decision tree	RFE	99.82	99.82	99.82	99.82
[350]	2025	CNN	RF	97.28	97.71	99.37	98.53
[351]	2024	Boosted tree	GR and IG	99.10	99	99	N/A
[352]	2023	KNN	Pearson correlation	95.70	N/A	N/A	N/A
[353]	2022	Bagging	N/A	98.73	91.08	85.85	88.07

Table 74 Automotive ethernet network

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[355]	2022	Convolutional AE	Convolutional layer	94	95	95	95
[105]	2021	2DCNN	CNN	99.95	99.13	99.40	99.27

Table 75 WUSTL-IIOT

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[356]	2025	MLP	N/A	99.99	99.99	99.80	99.99
[357]	2024	SVM	N/A	99.94	99.94	99.94	99.94
[358]	2024	CNN	N/A	99.90	99.90	99.90	99.90
[359]	2023	Isolation forest	PCC	99.28	91.09	99.80	95.20
[360]	2022	DeepIIOT	N/A	99.99	99.99	99.99	99.99

layers of the recorded pcap files, and are separated based on the corresponding layer. Table 73 shows the work that used the AWID3 dataset.

92. *Automotive ethernet intrusion dataset:* Jeong et al. [105] introduced this dataset capturing benign AVTP packets from a physical automotive Ethernet testbed and replay attack scenarios. The attack involves injecting 36 AVTPDUs to simulate a single video frame on an AVB listener, demonstrating how attackers can replicate frames using pre-captured data. Four benign PCAP files are also included: *driving_01* (10 min), *driving_02* (16 min), *indoors_01* (24 min), and *indoors_02* (21 min). Replay attack outcomes are stored in *_injected.pcap files [354]. Table 74 shows the work that used the Automotive Ethernet Intrusion dataset.
93. *WUSTL-IIOT-2021:* Zolanvari et al. [106] introduced the WUSTL-IIOT-2021 dataset to address the lack of IIoT-specific security datasets. They developed a testbed simulating a typical Industrial Control System (ICS) focused on water level monitoring, featuring a PLC, three sensors, four actuators, and four multifunctional devices-one acting as an attacker. Communication used the Modbus protocol, and network activity was recorded for 53 h under both normal and attack conditions. Four attack types were simulated: reconnaissance, command injection, DoS, and backdoor installation. The publicly available dataset, in CSV format, includes 41 features plus one attack label, providing a comprehensive view of IIoT network traffic. It is used in research involving Explainable AI for IIoT security enhancement. Table 75 shows the work who used WUSTL-IIOT dataset.
94. *CCD-INID-V1 2021:* Liu et al. [107] introduced the CCD-INID-V1 dataset to address limitations of intrusion detection in dynamic, resource-constrained IoT environments. Captured from real network traffic in smart lab and smart home

Table 76 CCD-INID-V1

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[361]	2025	E-ResGraphSAGE	N/A	N/A	N/A	N/A	96.77
[362]	2023	Decision tree	N/A	N/A	N/A	N/A	100
[107]	2021	RCNN	RF	95.15	95.00	96.00	96.00

Table 77 NSS-Mirai

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[110]	2021	Random forest	N/A	99.00	N/A	N/A	99.00

settings, the dataset features data collected via HTTPS from four Raspberry Pi devices equipped with sensors-recording temperature in the smart home and both temperature and pressure in the smart lab. Five simulated attacks were executed: UDP Flood, ARP Poisoning, Hydra Bruteforce, ARP DoS (via Asterisk), and SlowLoris. Entries are labeled as normal or malicious. NFStream was used to extract 83 features, including timestamps, packet/byte counts, MAC addresses, and IP information. Although comprehensive, the dataset is not publicly available and must be requested directly from the authors. Table 76 shows the work who used CCD-INID-V1 dataset.

95. *IoT-environment 2021:* YOO et al. [108] presented traffic data from devices like EZVIZ, NUGU, Hue, TP-Link, and Google Home Mini. Attacks include port scanning, HTTP flooding, and OS/service detection. 125,182 packets were collected over 34 min using Wireshark.
96. *ELEGANT (DOS and MITM) 2021:* Sousa et al. [109] presented a dataset covering DoS and MiTM attacks, collected from Fed4Fire+ testbeds (Virtual Wall2 and Grid5000). Using OpenPLC v3, Modbus TCP packets were captured via dumpcap at the bridge node during PLC communications. File sizes range from 10 to 20 MB and are publicly available.
97. *NSS MIRAI 2021:* Kalupahana Liyagage et al. introduced the NSS MIRAI dataset [110], simulating a Mirai-style IoT botnet attack in an OpenStack-managed network. The setup included 65 IoT devices, seven gateways, a security manager, a victim, two external bots, a C&C server, a loader, and a port scan generator, all operating on Ubuntu 18.04 or custom Linux images. Over a 60-min experiment, it recorded activities like network/port scanning, brute-force logins, malware loading, C&C communication (successful/failed), scan-out behavior, and DDoS attacks (volumetric and reflective), along with false alerts. Publicly available, the NSS dataset supports in-depth analysis of IoT botnet behaviors in controlled settings. Table 77 shows the work who used the dataset.
98. *IoT-BDA botnet analysis 2021:* Trajanovski et al. [111] introduced IoT-BDA, based on 4077 IoT botnet samples from honeypots. Using sandboxing, static, behavioral, and network features were extracted. Virustotal and AVClass identified malware families. The dataset includes ELF binaries, system call logs, and PCAPs.
99. *IoT-CSIDDS 2021:* IoT-CSIDDS (IoT-Cold Storage Intrusion Detection Dataset) [112] is an intrusion detection dataset generated from a real IoT cold-storage monitoring testbed. It contains labeled records collected from IoT sensor

Table 78 VHS

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[114]	2022	Random forest	N/A	97.00	97.00	93.00	95.00

Table 79 ECU-IOFT

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[363]	2025	Random forest	N/A	92	98	88	93
[364]	2024	RFC	N/A	99.70	99.70	99.60	99.70

nodes under normal and attack scenarios, with 20 network and sensor-related features describing device and communication behavior.

100. *Malicious traffic PCAPs 2021:* Rosa et al. [113] released PCAPs from real malware attacks in a smart home testbed (Cyber-Trust). Attacks include Mirai, Zeus, Java-RMI backdoors, Tomcat exploits, brute-force SSH/FTP, and more. Normal traffic (e.g., file transfers, streaming) was also captured using tcpdump.
101. *VHS-22:* The VHS-22 [114] data set contains over 27 million flows, approximately out of which twenty million are regular traffic and seven million are network attacks. Flows with only one packet were identified in both of these groups and were referred to as zero-duration flows. They are responsible for a total traffic of 45% and total attacks of 83% (mainly DoS-related). It was discovered that the UDP protocol was used in 62% of flows, whereas others use TCP. In Zero-duration flows, there are 66% of UDP protocol flows and 38% of TCP protocol flows. Flows longer than 160 s are also included in the dataset. The length of majority of flows is less than 200k packets, but the length of the longest flow is nearly 11.7M packets. According to the distribution of attacks, 5.8% Botnet and 93.9% DoS related attacks cause the greatest number of network flows labelled as attacks. Whereas remaining attack related traffic are 0.15% Malware, 0.05% web attacks, and 0.8% brute force attacks. Table 78 shows the work that used the VHS dataset.
102. *ECU-IoFT 2022:* Ahmed and team [115] developed the dataset ECU-IoFT, which captures the wireless Drones network traffic, specifically the Tello drone. The drone was linked to access point of WI-FI and user controls it by corresponding smartphone application to compile this dataset. At the same time, a perpetrator initiated assaults on the drone, executing three distinct types of attacks: a WPA2-PSK Wi-Fi cracking, Tello API exploit, and Wi-Fi deauthentication. The wireless traffic produced during these actions was captured and stored on the attacker's device, resulting in a dataset spanning approximately 3 min and comprising 54,492 packets. The dataset is organized in a CSV file which has 10 features of traffic recorded. whether the traffic is normal or linked to an attack is indicated by Label. Table 79 shows the work who used ECU-IOFT dataset.
103. *CoAP-DoS 2022:* The latest dataset under consideration is the CoAP-DoS by Mathews et al [116] which is publicly available. Specifically tailored for IDS based on Machine Learning, this dataset stands out as the sole one identified that addresses the CoAP protocol, focusing on DoS attacks against this protocol. The authors installed measuring instruments on four devices, comprising a server, which is a Raspberry Pi, and three Computers, among which two functioned as

Table 80 COAP-DOS

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[365]	2024	CNN+LSTM	N/A	99.93	99.94	99.93	99.93
[366]	2023	Random forest	N/A	99.99	99.53	100	99.76
[116]	2022	SVM	N/A	99.88	N/A	N/A	N/A

Table 81 EDGE-IIOT

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[367]	2026	LEDEC	Fox optimizer (FO-FS)	97.6	96.8	97.2	97.0
[368]	2024	CNN-GRU + XGBOOST	N/A	N/A	99.5	99.7	99.6
[369]	2024	LSTM	N/A	99.85	91.8	99.1	97.8
[343]	2022	Inception time	N/A	99.51	99.50	N/A	99.50

Table 82 KNX

Ref	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[370]	2024	Random forest	N/A	99	N/A	N/A	N/A
[118]	2023	Neural network	N/A	100	100	100	100

malicious entities. These devices generated both normal and Distributed Denial of Service (DDoS) traffic, which was meticulously recorded over a span of 16 h. The dataset can be obtained in two formats: total of 661,304 extensive captures of packets, CSV file containing 17 extracted traffic features. JSON files contains traffic which is malicious. Table 80 shows the work who used CoAP-DoS dataset.

104. *Edge-IIoT 2022:* Ferrag et al. [117] developed the Edge-IIoT dataset using a hybrid testbed covering all IoT/IIoT layers-from orchestration services to perception-layer sensors and actuators (e.g., sound, temperature, water level, DC motors, servos). Designed to support Federated Learning, it contains benign and malicious traffic across various IP-based and application-layer protocols (DNS, MQTT, Modbus, HTTP). Over a 50-day period, 14 attack types were simulated. The dataset includes full packet captures (PCAP format) and 61 extracted features with labeled attack data. Table 81 shows the work who used Edge-IIOT dataset.
105. *KNX 2022:* Graveto et al. [118] provided PCAP traffic from a KNX-based smart home (lighting, heating, alarms) over 5 weeks. CSV metadata from ETS project file enriches 379,875 valid KNX packets. Table 82 lists studies using this dataset.
106. *IDS-MANET 2022:* Prasad et al. [119] generated a custom MANET intrusion detection dataset using the NS-3.25 network simulator with the AODV routing protocol under different network configurations containing 20–50 mobile nodes moving according to the Random Waypoint mobility model. The dataset captures traffic generated during simulated blackhole and wormhole attacks and consists of eight sub-datasets (four for each attack type) with varying numbers of normal and malicious nodes. A total of 20 features (12 basic and 8 derived) were extracted from packet captures, including duration, protocol, packet size, hop count, sequence numbers, message type, neighbor count, average flow, failed connection rate, and other routing-related characteristics. Each packet was labeled as normal or attack,

Table 83 IDSAI

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[120]	2023	XGBoost	N/A	94.80	94.97	94.80	94.83

Table 84 STPID

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[121]	2023	ST-DBSCAN	ST-DBSCAN	94.98	N/A	N/A	N/A

Table 85 TII-SSRC-23

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[122]	2023	XGBoost	FIA	99.99	99.98	99.54	99.76

producing lightweight datasets ranging from 876 to 118,074 samples (75.1 KB to 9.7 MB), specifically designed for supervised intrusion detection in MANETs.

107. *IDSAI 2023:* Fernando et al. [120] introduced the IDSAI dataset to evaluate supervised machine learning methods for intrusion detection in IoT network traffic. Created in a real-world attack environment, it contains 1,000,000 samples-500,000 intrusion and 500,000 non-intrusion-forming a balanced dataset. After preprocessing, 24 original attributes were reduced to 19 variables and 2 labels (intrusion type and binary class), yielding a 1,000,000 $\times$ 21 matrix. The intrusion class includes 10 attack types, each with 50,000 samples: SYN/ACK and RST Flooding, ICMP Echo Request/Ping Flood, ARP Spoofing, SYN Flooding (fast), DDoS MAC Flood, IP Fragmentation, Brute Force SSH, TCP Null, and UDP Port Scan. The dataset is publicly available for research use. Table 83 shows the work who used IDSAI dataset.
108. *STPID 2023:* Pitafi et al. [121] curated the STPID dataset to assess machine learning techniques for intrusion detection in perimeter intrusion detection systems (PIDS) using real network traffic from security cameras. The dataset captures authentic intrusions under varied conditions-day, night, and rain-over a 15-day period, with variations in zoom, roll, and yaw and cameras set to autofocus. From 10 days of footage, 30 h containing intrusion events were selected, and 17,000 frames were manually extracted for analysis. The dataset is publicly available to support further research in enhancing PIDS effectiveness. Table 84 shows the work who used STPID dataset.
109. *TII-SSRC-23 2023:* Herzalla et al. [122] introduced the TII-SSRC-23 dataset to support IDS research, available on Kaggle. It consists of 27.5 GB of raw traffic in PCAP format and a corresponding CSV file with extracted features. The dataset includes 32 traffic subtypes (26 malicious, 6 benign) across 8 categories: background, audio, text, brute-force, video, DoS, Mirai botnet, and information gathering. Each subtype incorporates variations in traffic parameters. Despite a class imbalance favoring malicious samples, the dataset is a valuable resource for ML-based intrusion detection. Table 85 shows the work that used the TII-SSRC-23 dataset.

Table 86 ROSIDS23

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[371]	2024	CMHA-SConv	N/A	97.07	96.79	95.73	96.75
[372]	2024	Robot-NIDS	N/A	98.2	98.6	98.2	98.4

Table 87 CICIoV2024

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[266]	2025	FL-RNN + SNCOC	N/A	93.26	72.71	66.81	69.63
[373]	2025	Adapt-LFA-CNN	N/A	N/A	57.28	62.26	61.47

110. *ROSIDS23 2023:* Deugirmenci et al. [123] introduced the ROSIDS23 dataset, derived from pcap-format network traffic collected from an autonomous robotic system. Using CICFlowMeter, 83 traffic features were extracted per entry, along with a timestamp and a label indicating one of five classes: benign, DoS, unauthorized publish, subscriber flood, and unauthorized subscribe. The dataset includes 62,511 benign, 31,000 DoS, 30,064 subscriber flood, 7817 unauthorized publish, and 5289 unauthorized subscribe entries. While DoS is a general attack, the other three are specific to the Robot Operating System (ROS). Both raw and processed files are provided, and attack start times are documented in a *_attacktimes.txt file. A limitation of the dataset is its focus on only four attack types, primarily within the UDP/TCP and HTTP protocols of ROS, which may limit applicability to systems using other or custom protocols. Table 86 shows the work who used ROSIDS23 dataset.
111. *CICIoV2024:,neto2024ciciov2024* The CICIoV2024 dataset was introduced to support intrusion detection research in Internet of Vehicles (IoV) environments. The dataset contains both benign and malicious vehicular network traffic collected from connected vehicle communication scenarios. It includes multiple attack categories relevant to intelligent transportation systems and provides labelled data suitable for machine learning and deep learning-based IDS evaluation. The dataset was designed to address the growing need for realistic vehicular cybersecurity benchmarks and has recently been used in studies involving federated learning, adversarial attacks, and distributed intrusion detection systems. Table 87 shows recent work that used the CICIoV2024 dataset.
112. *X-CANIDS dataset 2024:* The X-CANIDS dataset [125], introduced in IEEE Transactions on Vehicular Technology (2024), offers raw CAN messages and 688 deserialized signals collected from a 2017 Hyundai LF Sonata e-VGT using a Kvaser Memorator in real-world conditions. The benign dataset includes six driving and one idling scenario with 62–64 arbitration IDs, while the intrusion dataset contains fuzzing, fabrication, masquerade, suspension, and replay attacks targeting ECUs. Attacks involve techniques like random payload injection and replay, with balanced labeling and attack parameters embedded in filenames. Both raw and signal-level data formats are available for evaluating message- and signal-based IDS models. Table 88 shows the work that used the X-CANIDS dataset.
113. *Linux-APT 2024:* Karim et al. [126] introduced the Linux-APT Dataset 2024 to support research on detecting Advanced Persistent Threats (APTs) and sophisticated

Table 88 X-CANIDS

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[287]	2025	IVN-ViT	Particle swarm optimization	99.03	98.89	98.77	98.94
[374]	2025	KG-ID	N/A	N/A	94.22	100	99.45
[375]	2024	CANival	N/A	N/A	N/A	N/A	86.6

Table 89 LINUX-APT

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[376]	2025	GNN	N/A	99.86	99.77	99.52	99.65

Table 90 CAN-MIRGU

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[377]	2025	DTREE	N/A	99.60	99.60	99.60	99.60

Linux-based attacks. Spanning October 1, 2023, to January 7, 2024, the dataset is divided into 17 files (10,000 records max per file) and also provided in combined and processed forms. It includes 125,898 records featuring both benign and malicious traffic, covering simulated APTs (e.g., APT29, APT41, APT28, Turla), malware, keyloggers, Apache Struts vulnerabilities, and Linux privilege escalation payloads. Data is available in raw, XML, and CSV formats, capturing both qualitative (APT/malware selection) and quantitative (logs, traffic) information. Publicly hosted on Zenodo and Mendeley, it enables evaluation of complex, time-intensive attack behaviors in realistic Linux-based environments. Table 89 shows the work who used LINUX-APT dataset.

114. *CAN-MIRGU 2024:* Rajapaksha et al. [127] developed the CAN-MIRGU dataset to support IDS research for autonomous electric vehicles by capturing both benign and malicious CAN traffic from a 2016 EV with autonomous features. Benign data was collected over six weeks on UK roads, while attacks were conducted under controlled conditions via a high-speed 500 Kbps CAN interface directly connected to the CAN gateway. Simulated attack types include DoS (high-priority ID 0x000 at 0.001s), fuzzing (random/valid IDs with full payloads), replay (out-of-context signal injection), spoofing (targeting brake, steering, FCA warnings using flam/time-based methods), suspension (dropping legitimate frames), and masquerade (removing frames before spoofed ones). Attacks targeted various ID frequencies and included timing analyses to show impacts on frame intervals. These insights, including visualizations, reveal the complexity of detecting advanced attacks and support the development of time- and frequency-aware IDSs. Table 90 shows the work who used CAN-MIRGU dataset.
115. *LSPR23 2024:* [128] The LSPR23 dataset is a high-quality intrusion detection dataset derived from the Locked Shields cyber defense exercise. It consists of approximately 16 million network flows, including about 1.6 million malicious flows, and captures realistic cyberattack and defense activities. Its accurate labeling methodology and diverse attack scenarios make it a reliable benchmark for evaluating intrusion detection systems. Table 91 shows the work who used LSPR23 dataset.

Table 91 LSPR23

Ref	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[128]	2024	RF	N/A	N/A	99.90	99.40	99.70

Table 92 CICIoMT2024

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[129]	2024	RF	N/A	99.60	97.10	95.10	96.10

Table 93 Farm-flow

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[131]	2025	DNN	N/A	92.67	93.55	92.67	92.63

116. *CICIoMT2024:* [129] The CICIoMT2024 dataset is a comprehensive network traffic dataset developed to support cybersecurity research in the Internet of Medical Things (IoMT) environment. It captures realistic network communication generated by interconnected medical devices operating in a smart healthcare environment under both normal and attack conditions. The dataset contains labeled network traffic collected from IoMT devices communicating through common healthcare protocols such as Wi-Fi, MQTT, and Bluetooth. It includes both benign traffic and multiple categories of cyberattacks, including DoS, spoofing, reconnaissance, and data manipulation attacks, which were conducted in a controlled testbed to simulate real-world threats against resource-constrained medical devices. The captured network traffic is provided in PCAP format and processed into CSV files containing flow-based statistical features. Table 92 shows the work who used CICIoMT2024 dataset.
117. *Windows-APT 2025:* Windows-APT 2025 [130] is a publicly available host-based intrusion detection dataset developed to support research on Advanced Persistent Threat (APT) detection in Windows environments. The dataset contains telemetry collected from 36 APT-inspired attack scenarios executed on Windows 10 virtual machines using the MITRE Caldera adversary emulation framework, with monitoring performed through Sysmon and Wazuh. It includes approximately 102,013 log records (63,620 malicious and 38,393 benign) across 17 log files, covering 45 MITRE ATTACK techniques and 9 enterprise tactics.
118. *Farm-Flow 2025:* The Farm-Flow dataset [131] is a publicly available benchmark for intrusion detection in smart agriculture (AG-IoT) networks. It simulates a realistic AG-IoT environment with various IoT devices and protocols, including 8 attack types (e.g., ARP Spoofing, DDoS variants, MQTT Flood, Port Scanning) and normal traffic. Traffic was captured using Zeek and Flowmeter, converted to structured flow data in CSV format. The dataset contains 1,309,887 instances (1,282,429 malicious and 27,458 normal), supports both binary and multiclass classification, and is 532 MB in size with 101 features. Key features include `missed_bytes`, `fwd_pkts_per_sec`, and `flow_pkts_per_sec`. Data was annotated, cleaned, and engineered for ML-based intrusion detection and is hosted on Zenodo. Table 93 shows recent work that used the Farm-Flow dataset.

Table 94 Gotham

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[378]	2026	RF	N/A	100	98.00	99.00	99.00
[132]	2025	Deep neural network	N/A	89.00	90.00	89.00	87.00

Table 95 UAV-NIDD

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[133]	2025	FNN	ANOVA	97.18	97.76	97.18	95.20

119. *Gotham 2025:* The Gotham dataset [132] is a large-scale IoT dataset for smart city cybersecurity research. It includes traffic from diverse IoT devices (e.g., cameras, meters, vehicles), annotated for normal and anomalous activities. Attacks include DoS, Remote Code Execution, Telnet Brute Force, CoAP Amplification, and others. It supports multiple protocols (MQTT, CoAP, HTTP), and includes PCAPs, CSV/JSON logs, device metadata, anomaly labels, and environmental context. Structured for reproducibility, it aids research in intrusion detection, anomaly detection, and smart city IoT security. Table 94 shows recent work that used the Gotham dataset.
120. *UAV-NIDD 2025:*Hassan Jalil Hadi et al. developed UAV-NIDD (Unmanned Aerial Vehicles-Network Intrusion Detection Dataset) [133], a dynamic cybersecurity dataset designed specifically for intrusion detection in UAV networks. UAV-NIDD was created using a real-time UAV testbed consisting of physical UAVs, ground control stations (GCSs), access points, and monitoring devices. The dataset captures multi-layer interactions between UAVs, GCSs, and access points, thereby providing a comprehensive representation of UAV network behavior under both benign and attack conditions. It includes traffic generated through UAV-specific communication protocols such as MAVLink and DJI SDK, making it suitable for developing intrusion detection systems tailored to UAV networks. The dataset covers ten cyberattack categories, including Scanning, Reconnaissance, DoS, DDoS, GPS Jamming and Spoofing, Man-in-the-Middle (MITM), Replay, Evil Twin, Brute-Force, and Fake Landing Packet attacks. Data were collected in two phases: normal flight operations and identical flight operations under cyberattack scenarios, ensuring the availability of both benign and malicious traffic for machine learning and security analysis. Table 95 shows recent work that used the UAV-NIDD dataset.
121. *UAVIDS-2025:* Qingli Zeng et al. introduced UAVIDS-2025 [134], a benchmark dataset specifically designed for intrusion detection in UAV networks. The dataset was generated using the NS–3.24 network simulator on an Ubuntu 20.04 platform to emulate realistic UAV ad hoc network conditions, including dynamic topology changes, wireless communication constraints, and UAV-specific routing behavior. The simulation employed the IEEE 802.11ac wireless standard, the AODV routing protocol, and the BOID mobility model to capture the cooperative movement of multiple UAVs. UAVIDS-2025 contains 122,171 flow-level records representing both benign and malicious traffic. Four network-layer attacks were implemented: Black Hole, Flooding, Sybil, and Wormhole attacks. The dataset includes 22 extracted flow-level features covering connection, traffic volume, and performance

Table 96 UAVIDS-2025

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[134]	2025	XGB	N/A	96	96	96	96

Table 97 5GCID

Ref.	Year	Classification technique	Feature selection	Performance metrics (%)			
				Acc	Prec	Rec	F1
[135]	2026	DR-mLSTM	Distinguishing and redundancy (DR)	98.50	N/A	N/A	98.50

characteristics, and all samples are fully labeled to support both binary and multiclass intrusion detection tasks. Table 96 shows recent work that used the UAVIDS-2025 dataset.

122. *BigFlow-NIDS 2026:* The BigFlow-NIDS dataset [379] is designed for intrusion detection in large-scale network environments and is based on NetFlow data with around 66.9 million flows and 55 features, available in CSV and Parquet formats for big data processing. It includes raw and processed data with labeled benign and attack traffic, supporting binary and multi-class classification with 32 attack types, and is built by combining datasets like NF-UNSW-NB15, NF-ToN-IoT, NF-BoT-IoT, and NF-CSE-CIC-IDS2018. The dataset reflects real-world class imbalance (more benign than attack traffic), is publicly available for research, and is suitable for scalable intrusion detection using frameworks like Apache Spark.
123. *5GCID 2026:* The 5GCID dataset [135] is designed for intrusion detection in 5 G Core Network and includes both control-plane and user-plane traffic from a 3GPP-based testbed. It provides structured data (flow/packet level) with features like IP details, protocol information, and labeled normal vs attack traffic for machine learning tasks. The dataset mainly contains DoS attacks targeting AMF, SMF, and UPF using protocols like SCTP, GTP-U, UDP, and HTTP, along with normal Ng interface traffic from a simulated environment. Table 97 shows recent work that used the 5GCID dataset.
124. *Cubesat cybersecurity 2026:* The CuCD-ID dataset [136] is designed for intrusion detection in CubeSat cyber-physical systems and includes telemetry and command-level data from a high-fidelity simulated environment. It is provided in structured tabular format with raw and augmented data, containing multiple features, engineered attributes, and labeled attack categories suitable for machine learning. The dataset covers attacks like command flooding, false data injection, storage exhaustion, and defense impairment, along with normal operations, and is publicly available for research.

5.3 Dataset readiness for modern AI-driven IDS

Recent advances in artificial intelligence have expanded intrusion detection research beyond conventional supervised machine learning and deep learning models. Emerging approaches based on foundation models, Graph Neural Networks (GNNs), Federated Learning (FL), Continual Learning (CL), Large Language Models (LLMs), and Adversarial Machine Learning (AML) impose additional requirements on IDS datasets. These requirements include large-scale and heterogeneous data, relational information, natural data partitions, chronological traffic streams, semantically meaningful security

information, and support for realistic adversarial manipulation. Consequently, datasets that are suitable for conventional classification tasks may not necessarily provide the characteristics required for evaluating modern AI-driven IDS architectures. Table 98 summarizes the major dataset requirements, recent examples of surveyed datasets used with these approaches, and the corresponding readiness limitations.

The analysis indicates that the use of a dataset in a modern AI-driven IDS study does not necessarily imply that the dataset was originally designed for that learning paradigm.

Table 98 Dataset requirements and readiness considerations for modern AI-driven IDS architectures

AI paradigm	Dataset requirements	Recent studies and datasets	Current readiness limitations
Foundation models	Large-scale and heterogeneous pretraining corpora, multiple traffic environments, diverse attack scenarios, multimodal security data, and reusable representations supporting downstream security tasks	Existing network, IoT, IIoT, vehicular, host-based, and multi-source datasets provide potential sources for cross-domain security pretraining	Most IDS datasets are independently collected, domain-specific, differently formatted, and relatively limited in scale. Differences in feature spaces, labeling schemes, traffic representations, and collection environments restrict direct integration into standardized pretraining corpora
Graph neural networks	Relational information among hosts, devices, flows, CAN messages, or communication events; network topology; entity identifiers; and temporal interactions that permit meaningful graph construction	Recent GNN-based IDS studies have used Car-Hacking, CICIDS2017, and the HCRL CAN Intrusion Dataset [380, 381]	Most conventional IDS datasets represent traffic as independent tabular records and do not provide explicit graph structures. Researchers must construct graphs using communication relationships, temporal proximity, or other similarity measures, and the resulting graph topology may substantially influence IDS performance
Federated learning	Natural partitions representing devices, vehicles, users, organizations, or network sites; heterogeneous local traffic distributions; sufficient per-client samples; and realistic non-IID data characteristics	UNSW-NB15, Car-Hacking, CICIDS2017, and KDDCUP99 have been used in recent federated IDS studies [382, 383]	Most benchmark datasets are centrally collected and do not provide natural federated-client partitions. Existing studies therefore commonly partition centralized datasets artificially among clients, which may not adequately represent realistic client heterogeneity, non-IID traffic distributions, or communication constraints
Continual learning	Chronologically ordered traffic streams, timestamps, evolving normal behavior, emerging attack classes, distribution shifts, and sequential tasks suitable for evaluating adaptation and catastrophic forgetting	Recent continual-learning IDS studies have used UNSW-NB15, CICIDS2017, ToN-IoT, and NSL-KDD [384, 385]	Many IDS datasets are static benchmarks with predefined or randomly generated training and testing partitions. Although such datasets can be reorganized into sequential learning tasks, they were generally not designed to capture long-term concept drift, attack evolution, or naturally occurring task boundaries
LLM-assisted IDS	Semantically meaningful logs, alerts, event descriptions, security reports, threat intelligence, contextual metadata, and multimodal information that can support language-based reasoning and explanation	Recent LLM-assisted IDS studies have used CICIDS2018, LID-DS2019, RT-IoT2022, and synthetic security-event datasets [386, 387]	Most traditional IDS datasets primarily contain numerical packet- or flow-level features and provide limited semantic context for language-based reasoning. Consequently, LLM-assisted IDS frameworks often require external knowledge sources, textual logs, retrieval mechanisms, or additional explanation modules
Adversarial machine learning	Modifiable traffic representations, realistic perturbation constraints, repeatable attack-generation procedures, support for poisoning and evasion scenarios, and preservation of valid network semantics after adversarial manipulation	Recent adversarial IDS studies have used CIC-IoT2023 and NF-ToN-IoT-v2 to evaluate adversarial attacks and defense mechanisms [388, 389]	Many benchmark datasets provide static tabular features rather than raw traffic-generation environments. Although adversarial samples can be generated mathematically, feature perturbations may violate protocol constraints or produce unrealistic network traffic, limiting the validity of adversarial robustness evaluations

For example, conventional benchmark datasets can be transformed into graph structures, artificially distributed among federated clients, reorganized into sequential tasks, or perturbed to generate adversarial samples. However, such transformations introduce experimental assumptions that may influence the validity and generalizability of the resulting evaluations.

GNN-based intrusion detection depends on meaningful relationships among network entities. Existing flow-based and CAN datasets can support graph construction when source-destination interactions, message sequences, or temporal relationships are preserved. However, the absence of explicit topology information requires researchers to define nodes, edges, and neighborhood relationships during preprocessing. Consequently, different graph-construction strategies may produce different structural representations, affecting the reproducibility and comparability of GNN-based IDS evaluations.

Federated IDS research requires datasets representing naturally distributed security environments. Although existing benchmarks have been used to evaluate federated learning architectures, centralized datasets are commonly divided among simulated clients. Such artificial partitioning may not capture realistic non-IID distributions, device behavior, client availability, network conditions, or privacy constraints. Future datasets should therefore preserve device-, user-, organization-, or site-level partitions and document the statistical heterogeneity among participating entities.

Continual-learning IDS methods require temporally ordered observations that allow models to adapt to evolving network conditions while retaining previously acquired knowledge. Existing datasets can support such experiments when chronological traffic or timestamps are available, but many were designed for static classification. Future datasets should include longer observation periods, explicit concept drift, emerging attacks, evolving normal behavior, and documented sequential evaluation protocols to enable realistic assessment of adaptation and catastrophic forgetting.

Foundation models and LLM-assisted intrusion detection introduce additional requirements for dataset scale, heterogeneity, and semantic richness. Foundation models require large and diverse pretraining corpora supporting reusable representations across security tasks, while LLM-assisted IDS architectures benefit from contextual information such as system logs, alerts, attack descriptions, incident reports, and threat intelligence. Most existing IDS datasets remain isolated collections of numerical network features, limiting their suitability for large-scale pretraining and language-based security reasoning.

Adversarial machine-learning research requires datasets that support realistic modification of network observations while preserving protocol semantics and operational validity. Although adversarial examples are generated from existing IDS datasets, mathematically perturbed feature vectors may not correspond to feasible network traffic. Future benchmarks should therefore provide raw traffic, traffic-generation environments, domain constraints, and repeatable adversarial scenarios for realistic evaluation of poisoning, evasion, and other adversarial threats.

Overall, existing IDS datasets provide useful foundations for evaluating emerging AI-driven intrusion detection methods, but their readiness varies according to the requirements of the learning paradigm. Future dataset-development initiatives should move beyond static tabular benchmarks and provide richer relational structures, naturally

distributed data partitions, long-term temporal streams, multimodal security information, semantically meaningful context, and realistic adversarial experimentation environments. These characteristics would improve the reliability, reproducibility, and practical relevance of IDS evaluation under modern AI-driven security architectures.

5.4 Realism assessment of representative IDS datasets

Dataset realism is a multidimensional property and cannot be determined solely from whether a dataset is described as real, simulated, or emulated. In intrusion detection research, a realistic dataset should reflect the operational conditions under which an IDS is expected to function, including encrypted traffic, diverse benign activity, realistic user or device behaviour, temporal variation, evolving attack behaviour, and heterogeneous network environments. Therefore, this subsection assesses selected representative benchmark datasets using six realism dimensions: Encrypted Traffic Representation (ETR), Background Traffic Diversity (BTD), User or Behavioural Realism (UBR), Temporal Dynamics (TD), Attack Evolution Realism (AER), and Network Heterogeneity (NH).

The assessment uses four qualitative levels. *High* indicates clear evidence of strong representation of the corresponding realism dimension. *Moderate* indicates meaningful but partial, controlled, or limited representation. *Low* indicates narrow, synthetic, highly controlled, or substantially limited representation. *N/D* indicates that the available dataset documentation and supporting studies do not provide sufficient evidence to determine the corresponding dimension reliably. The assessment is based primarily on original dataset publications and is supplemented by recent application studies and survey papers where appropriate. The resulting comparative assessment is summarized in Table 99.

The comparative assessment reveals substantial differences in realism across benchmark datasets. Legacy datasets such as KDD Cup'99 and NSL-KDD provide limited realism because they inherit synthetic traffic-generation environments, restricted behavioural diversity, predefined attacks, and legacy network architectures. More recent datasets, including CICIDS2017, CSE-CIC-IDS2018, and ToN-IoT, provide richer background traffic and substantially greater network heterogeneity. Nevertheless, their traffic and user activities are still generated within controlled experimental environments, and the collection periods are generally insufficient to represent long-term temporal drift.

The analysis also shows that improvements in one realism dimension do not necessarily imply overall dataset realism. For example, N-BaIoT and Car-Hacking contain traffic collected from real IoT devices or vehicles, but their operational scope and environmental diversity remain limited. Similarly, Bot-IoT provides diverse attack scenarios and a heterogeneous testbed but relies substantially on generated normal traffic and simulated IoT activities. ISCX-IDS2012 provides comparatively rich background traffic and temporal characteristics, although its controlled behaviour profiles and limited collection duration restrict long-term realism.

A particularly important limitation is the insufficient characterization of encrypted traffic. For several widely used datasets, the available documentation does not provide sufficient evidence regarding the prevalence and diversity of encrypted traffic. Furthermore, none of the assessed datasets provides strong evidence of long-term concept drift or adaptive adversarial evolution. These findings indicate that future IDS datasets should move beyond short-term controlled experiments and incorporate representative

Table 99 Realism assessment of representative IDS benchmark datasets

Dataset	ETR	BTD	UBR	TD	AER	NH	Ref.
KDD Cup'99	*N/D:* encrypted traffic diversity and prevalence are not reported	*Low:* synthetic background traffic generated in a simulated LAN with limited resemblance to operational traffic	*Low:* controlled synthetic activity without natural user behaviour	*Low:* multi-week collection and temporal features exist, but realistic long-term evolution or drift is not demonstrated	*Low:* predefined simulated attacks without progression, adaptation, or evolving adversarial behaviour	*Low:* legacy simulated network architecture with limited modern operational heterogeneity	[16, 390]
NSL-KDD	*N/D:* inherited from KDD Cup'99; encrypted traffic diversity and prevalence are not reported	*Low:* KDD-derived synthetic traffic with no improvement in underlying traffic realism	*Low:* controlled synthetic benchmark traffic rather than natural user activity	*Low:* no demonstrated preservation of long-term temporal evolution or drift	*Low:* predefined simulated attacks without progression, adaptation, or evolution	*Low:* legacy simulated environment with limited operational network heterogeneity	[23, 390]
ISCX-IDS2012	*Low:* payloads are inspectable and substantial encrypted-traffic diversity or prevalence is not documented	*High:* rich normal traffic with bidirectional flows, protocols, services, payloads, and diverse flow characteristics	*Moderate:* realistic profile-driven activities, but no unconstrained users in a production environment	*Moderate:* timestamps and temporal features are available, but no long-term evolution or concept drift	*Moderate:* multiple executed attack scenarios, but attacks are predefined and non-adaptive	*Moderate:* multiple hosts, protocols, services, and communication patterns, but limited organizational heterogeneity	[34, 391, 392]

Table 99 (continued)

Dataset	ETR	BTD	UBR	TD	AER	NH	Ref.
UNSW-NB15	*N/D:* encrypted traffic diversity and prevalence are not clearly reported	*Moderate:* multiple services and protocols represented in a controlled cyber-range	*Low:* generated traffic with no natural, unconstrained user activity reported	*Moderate:* timestamps and two capture periods preserved, but no long-term drift	*Low:* diverse attack families, but no documented progression or evolution	*Moderate:* multiple network roles and components in a controlled cyber-range	[48, 393]
CICIDS2017	*N/D:* encrypted traffic prevalence is not systematically characterized	*High:* diverse benign activities, applications, services, and network interactions	*Moderate:* realistic but profile-based and scripted user activities	*Moderate:* five-day chronological collection, but no long-term drift	*Moderate:* broad executed attacks, but no adaptive attack evolution	*High:* heterogeneous operating systems, devices, infrastructure, protocols, and services	[59]
CSE-CIC-IDS2018	*N/D:* encrypted traffic diversity and prevalence are not systematically reported	*High:* large-scale enterprise-like traffic with diverse normal activities, services, and user profiles	*Moderate:* profile-based activities without natural, unconstrained operational user behaviour	*Moderate:* ten-day chronological collection, but no long-term evolution or concept drift	*Moderate:* broad contemporary attacks, but no documented adaptation or evolution	*High:* large enterprise-like network with numerous clients, servers, departments, attacker machines, and roles	[394, 395]
N-BaIoT	*N/D:* encrypted traffic diversity and prevalence are not reported	*Moderate:* genuine benign traffic from multiple IoT devices, but limited background diversity	*Moderate:* realistic device-specific behaviour, but limited unconstrained behavioural diversity	*Moderate:* time-dependent behaviour represented, but no long-term evolution or concept drift	*Low:* real Mirai and BASHLITE attacks, but no documented progression, adaptation, or evolution	*Moderate:* heterogeneous commercial IoT devices, but limited network-scale diversity	[70, 396]

Table 99 (continued)

Dataset	ETR	BTD	UBR	TD	AER	NH	Ref.
Bot-IoT	*N/D:* encrypted traffic diversity and prevalence are not reported	*Moderate:* multiple benign IoT services, but controlled traffic and limited benign activity	*Low:* generated normal traffic and simulated IoT activities without natural user behaviour	*Moderate:* timestamps and concurrent benign/attack traffic preserved, but no long-term drift	*Moderate:* diverse botnet scenarios, but no sustained adaptive attack evolution	*Moderate:* multiple systems, roles, services, and simulated IoT devices in a controlled testbed	[77]
ToN-IoT	*N/D:* encrypted traffic diversity and prevalence are not clearly reported	*High:* diverse legitimate activities across IoT/IIoT services, traffic, OS logs, and telemetry	*Moderate:* realistic device/service interactions, but limited natural human-user behaviour	*Moderate:* time-stamped events, but limited long-term evolution and concept drift	*Moderate:* diverse attack scenarios, but no adaptive adversarial evolution	*High:* heterogeneous physical/virtual systems, IoT/IIoT devices, services, and Edge–Fog–Cloud architecture	[397]
Car-Hacking	*Low:* unencrypted CAN traffic with no encrypted traffic diversity	*Low:* genuine vehicle traffic, but limited normal activity and operational diversity	*Moderate:* genuine vehicle/ECU behaviour, but limited driver and behavioural diversity	*Moderate:* time-stamped sequential traffic, but short collection periods and no long-term drift	*Low:* real injection attacks, but isolated scenarios without progression, adaptation, or evolution	*Low:* real CAN environment limited to a single vehicle and network architecture	[398, 399]

ETR encrypted traffic representation, *BTD* background traffic diversity, *UBR* user/behavioural realism, *TD* temporal dynamics, *AER* attack evolution realism, *NH* network heterogeneity, *N/D* not determined

encrypted traffic, naturally evolving user and device behaviour, long-duration traffic collection, adaptive and progressive attack campaigns, and heterogeneous operational environments.

5.5 Quality, bias, and evaluation risks of benchmark IDS datasets

The assessment is based on original dataset publications and subsequent studies that investigated dataset characteristics, labeling procedures, temporal properties, attack realism, duplicate records, feature redundancy, and potential sources of leakage, contamination, and evaluation bias. The analysis distinguishes documented dataset limitations from structural characteristics that may introduce benchmarking risks under inappropriate preprocessing or data-partitioning strategies. Accordingly, the reported

leakage and evaluation-bias risks do not necessarily indicate direct target leakage in the original datasets, but identify conditions that may lead to train–test contamination, shortcut learning, memorization, or optimistic performance estimates. The supporting references provided for each dataset collectively substantiate the corresponding quality assessment.

5.5.1 Dataset bias and benchmarking pitfalls

The assessment in Table 100 shows that benchmarking bias can arise from both intrinsic dataset characteristics and inappropriate evaluation protocols. Duplicate or near-duplicate records and highly correlated flows may appear across training and test partitions when record-level random splitting is used, allowing models to memorize dataset-specific patterns rather than learn generalizable attack behaviour. This concern is particularly important for datasets containing extensive duplication, repetitive attack flows, temporally correlated records, or traffic generated from a limited number of devices, sessions, or experimental scenarios. Although NSL-KDD reduces the extensive duplication present in KDDCUP99, it retains the synthetic traffic characteristics and legacy attack distributions inherited from the original benchmark.

Leakage and evaluation bias can also arise from dataset-specific identifiers, timestamps, addresses, ports, attack-schedule artifacts, and infrastructure-specific attributes that correlate strongly with class labels. Such attributes may allow learning algorithms to exploit shortcut patterns instead of the behavioural characteristics of attacks. The risk is particularly relevant when records originating from the same device, attack session, traffic-generation process, or temporal window are distributed across both training and test sets. Under these conditions, high detection performance may reflect recognition of dataset-specific patterns rather than generalizable intrusion-detection capability.

Preprocessing and experimental design constitute additional sources of evaluation bias. Performing normalization, feature selection, dimensionality reduction, missing-value imputation, or class-balancing procedures before train–test partitioning can transfer information from the evaluation data into the training process. Similarly, applying oversampling or synthetic sample generation before data splitting may introduce identical or highly related records into both partitions. Hyperparameter optimization or decision-threshold selection using the test set can further produce optimistic estimates of generalization performance.

The characteristics of individual benchmark datasets require corresponding evaluation strategies. Legacy datasets such as KDDCUP99 and NSL-KDD require careful consideration of duplication, synthetic artifacts, and outdated attack distributions. For ISCX-IDS2012, chronological or session-aware partitioning can reduce overlap among related traffic flows. UNSW-NB15 should be evaluated with careful control of duplicates, correlated flows, and preprocessing procedures. CICIDS2017 and CSE-CIC-IDS2018 require attention to metadata artifacts, correlated traffic, duplicate records, and temporal or attack-session overlap. Device-aware or time-aware splitting is particularly relevant to N-BaIoT, while chronological splitting can reduce contamination caused by periodic CAN patterns in the Car-Hacking dataset. Bot-IoT requires consideration of repetitive attack-session traffic, identifier features, and severe class imbalance, whereas ToN-IoT requires careful treatment of timestamps, identifiers, infrastructure-specific features, and heterogeneous data sources.

To reduce benchmarking bias, IDS evaluations should perform duplicate and near-duplicate analysis before data partitioning and fit all preprocessing, feature-selection, dimensionality-reduction, and resampling procedures exclusively on the training data. Researchers should remove or explicitly evaluate leakage-prone identifiers and metadata and prefer chronological, session-based, device-based, attack-scenario-based, or group-aware splitting when supported by the dataset structure. Model selection, hyperparameter optimization, and decision-threshold selection should be conducted using training and validation data without accessing the final test set. Performance should also be reported using class-wise precision, recall, F1-score, and appropriate imbalance-sensitive metrics rather than relying solely on aggregate accuracy.

Finally, cross-dataset validation and external evaluation should be used whenever feasible to determine whether improvements generalize beyond dataset-specific artifacts. Reporting duplicate-removal procedures, partitioning strategies, preprocessing order, random seeds, feature exclusions, and data-cleaning decisions is equally important for reproducibility. These practices reduce the likelihood that reported performance gains arise from train–test contamination, shortcut learning, memorization, or benchmark-specific biases rather than genuine improvements in intrusion-detection capability.

5.6 Dataset sustainability and reproducibility

The master comparison in Table 2 reports baseline reproducibility indicators for all surveyed datasets, including availability, metadata, labeling status, predefined splits, format, traffic type, feature count, duration, and attack categories. However, these fields do not fully capture dataset sustainability. A dataset may be marked as available at the time of survey construction, but its download link may later become inactive, access may require a private request, documentation may be incomplete, or the feature-generation process may be difficult to reproduce. Therefore, this subsection complements Table 2 by examining sustainability and reproducibility from a practical research-use perspective.

Five dimensions are considered: current accessibility, documentation quality, labeling transparency, feature-pipeline reproducibility, and long-term hosting stability. Current accessibility refers to whether the dataset can be accessed through a working public source, registration process, or private request. Documentation quality evaluates whether the dataset paper or official repository explains the collection environment, traffic generation, file formats, features, and attack scenarios. Labeling transparency indicates whether the procedure used to assign benign and attack labels is clearly described. Feature-pipeline reproducibility assesses whether raw traces, feature-extraction tools, scripts, or processing steps are available or sufficiently documented. Long-term hosting stability considers whether the dataset is maintained by a stable institutional repository, public dataset portal, or a less reliable temporary/personal source.

The assessment uses four qualitative levels: *High*, *Moderate*, *Low*, and *N/D*. *High* indicates strong evidence of sustainability or reproducibility in the corresponding dimension. *Moderate* indicates partial support, limited documentation, or some access constraints. *Low* indicates weak support, unstable access, limited documentation, or poor reproducibility. *N/D* indicates that sufficient evidence was not available. Table 101 summarizes the assessment for representative benchmark datasets.

The assessment shows that dataset reproducibility depends on more than basic availability. Datasets such as CICIDS2017, CSE-CIC-IDS2018, UNSW-NB15, Bot-IoT,

Table 100 Quality, bias, and evaluation-risk assessment of representative IDS benchmark datasets

Dataset	Label reliability	Temporal characteristics	Attack realism	Duplicate records	Leakage/contamination/evaluation bias risk	Feature redundancy	Supporting refs.
KDDCUP99	Predefined ground-truth labels	Limited temporal representativeness	Legacy attacks in simulated environment	Extensive duplication reported	Duplicate-induced train–test contamination, memorization, and optimistic evaluation risk under record-level random splitting	Redundant/irrelevant features reported	[23, 390, 393, 400]
NSL-KDD	Ground-truth labels retained from KDDCUP99	Limited temporal representativeness	Legacy attacks inherited from simulated environment	Duplicates removed during dataset construction	Duplicate-related bias is reduced, but inherited synthetic artifacts and benchmark-specific biases remain	Redundant/irrelevant features reported	[23, 390, 401, 402]
ISCXIDS2012	Controlled scenario-based ground truth	Seven-day chronological traffic collection	Profile-based realistic testbed traffic	Duplicate removal reported in later preprocessing studies	Similar or temporally correlated flows from the same traffic sessions may cause contamination and optimistic evaluation under record-level random splitting	Correlated features reported in later studies	[34, 391, 392, 403]
UNSW-NB15	Controlled scenario-based ground truth	Traffic collected during two capture periods	Contemporary attacks in controlled cyber-range	Training-set duplicates reported	Duplicate records, correlated flows, controlled traffic-generation artifacts, and inappropriate splitting or preprocessing may produce optimistic evaluation	Irrelevant/redundant features reported	[48, 203, 404, 405]
CICIDS2017	Controlled attack-schedule-based labeling	Five-day chronological traffic collection	Realistic user profiles and contemporary attacks	Duplicate records and extraction artifacts reported	Metadata shortcuts, temporally correlated flows, duplicates, and record-level random splitting may cause contamination and inflate reported performance	Constant/correlated features reported	[406–409]
CSE-CIC-IDS2018	Controlled attack-schedule-based labeling	Ten-day chronological traffic collection	Enterprise-scale testbed with contemporary attacks	Duplicate/header artifacts reported	Metadata shortcuts, duplicate or correlated flows, and random record-level splitting may cause train–test contamination and optimistic evaluation	Redundant/irrelevant features reported	[257, 394, 395, 408]
N-BaIoT	Controlled infection-based ground truth	Traffic collected before and after botnet infection	Real IoT devices subjected to Mirai and BASHLITE attacks	Duplicate feature vectors reported in later preprocessing studies	Device-specific patterns, overlapping statistical windows, temporally correlated records, and record-level random splitting may produce optimistic evaluation	Correlation/redundancy possible across multi-window statistical features	[396, 410, 411]
Car Hacking	Controlled attack-injection-based labeling	Sequential CAN traffic collection	Attacks executed on an actual vehicle CAN bus	Duplicate removal reported in later preprocessing studies	Periodic CAN patterns, temporally correlated messages, and random record-level splitting may place highly similar vehicle behaviour across training and test sets	Compact raw-message feature representation; limited evidence of redundancy	[398, 412]

Table 100 (continued)

Dataset	Label reliability	Temporal characteristics	Attack realism	Duplicate records	Leakage/contamination/evaluation bias risk	Feature redundancy	Supporting refs.
Bot-IoT	Controlled scenario-based ground truth	Chronologically captured scenario-based traffic	Contemporary attacks generated in controlled IoT testbed	Duplicate/repetitive records reported in later studies	Identifier features, repetitive attack-session flows, severe class skew, and random record-level splitting may produce dataset-specific shortcuts and optimistic evaluation	Invalid/redundant features reported	[77, 413–415]
ToN-IoT	Ground truth based on predefined attacks, timestamps, and attacker information	Timestamped heterogeneous data collection	IoT/IIoT cyber-range with modern attack scenarios	Duplicate removal reported in later preprocessing studies	Identifier and timestamp features, infrastructure-specific patterns, correlated records, and record-level random splitting may introduce shortcut learning and optimistic evaluation	Infrastructure-specific and redundant features reported	[397, 416, 417]

Leakage, contamination, and evaluation-bias risk refers to the possibility that duplicate or highly correlated records, temporally related flows, device- or session-specific patterns, metadata or identifier features, traffic-generation artifacts, and inappropriate preprocessing or data-partitioning strategies may allow models to exploit dataset-specific shortcuts or related information across training and test sets. A reported risk does not necessarily imply direct target leakage in the original dataset; rather, it indicates conditions that may produce optimistic performance estimates under inappropriate benchmarking protocols

Table 101 Sustainability and reproducibility assessment of representative IDS benchmark datasets

Dataset	Current accessibility	Documentation quality	Labeling transparency	Feature-pipeline reproducibility	Long-term hosting stability	Ref.
KDD Cup'99	*Moderate:* widely mirrored and historically accessible, but official access is legacy-dependent	*Moderate:* dataset structure and features are documented, but collection details remain limited by legacy design	*Moderate:* attack classes are labeled, but labeling procedure is not fully transparent by modern standards	*Low:* processed connection features are available, but full feature-generation pipeline is difficult to reproduce	*Moderate:* long-used public benchmark, but dependent on legacy hosting and mirrors	[16, 390]
NSL-KDD	*Low:* official availability has become problematic and users often rely on mirrors	*Moderate:* dataset motivation and redundancy removal are documented	*Moderate:* labels are inherited from KDD Cup'99 and retained in processed files	*Low:* sampling strategy is described, but full reconstruction from original traffic is not straightforward	*Low:* long-term access depends heavily on secondary sources	[23, 390]
ISCX-IDS2012	*Moderate:* publicly known benchmark with institutional origin, but access routes may vary	*High:* traffic profiles, attacks, flow data, and payload-based analysis are well described	*Moderate:* labels are available, but full labeling workflow is not always reproduced in later studies	*Moderate:* raw traffic and extracted features support reproducibility, but exact processing choices may vary	*Moderate:* institutional origin improves stability, though long-term access requires periodic verification	[34, 391, 392]
UNSW-NB15	*High:* dataset files are publicly distributed through an institutional source	*High:* cyber-range setup, traffic generation, feature groups, and attack categories are documented	*High:* normal and attack classes are clearly defined	*Moderate:* PCAPs and extracted features are available, but exact regeneration may require specific tools and settings	*High:* hosted through a university research source with strong long-term visibility	[48, 393]
CICIDS2017	*High:* official CIC access route is available	*High:* user profiles, attack scenarios, PCAPs, flows, and feature extraction are documented	*High:* benign and attack labels are clearly provided	*High:* PCAPs and CICFlowMeter-based feature extraction support reproducibility	*High:* maintained through an institutional CIC dataset portal	[59]
CSE-CIC-IDS2018	*High:* official access route and cloud-hosted files are available	*High:* large-scale network setup, attacks, features, and data files are documented	*High:* class labels are provided and attack categories are clearly defined	*High:* flow features and preprocessing are reproducible using documented CIC-style feature extraction	*High:* institutional hosting provides comparatively strong sustainability	[68, 394]

Table 101 (continued)

Dataset	Current accessibility	Documentation quality	Labeling transparency	Feature-pipeline reproducibility	Long-term hosting stability	Ref.
N-BaIoT	*High:* widely accessible public IoT benchmark	*Moderate:* device types and malware scenarios are documented, but operational details are limited	*High:* benign and botnet traffic labels are clearly provided	*Moderate:* extracted statistical features are available, but raw capture and full regeneration details are limited	*Moderate:* widely mirrored and used, but long-term stability depends on public hosting continuity	[70, 396]
Bot-IoT	*High:* publicly available through institutional dataset sources	*High:* testbed, simulated IoT services, attack scenarios, and feature sets are documented	*High:* attack and normal labels are provided	*Moderate:* processed features are available; complete regeneration requires matching testbed and tool configuration	*High:* institutional hosting and widespread adoption support long-term availability	[77]
ToN-IoT	*High:* publicly accessible through dataset repositories	*High:* IoT/IIoT testbed, telemetry, OS logs, network traffic, and attack scenarios are documented	*High:* labels and attack categories are clearly provided	*Moderate:* multiple data sources are available, but full reproduction requires complex testbed reconstruction	*High:* repository-based hosting and strong reuse support sustainability	[102]
Car-Hacking	*Moderate:* public dataset is available, but access depends on specific dataset portals	*Moderate:* attack files, CAN fields, and injection scenarios are described in later studies	*High:* injected and normal CAN messages are explicitly labeled	*Moderate:* raw CAN messages are available, but exact vehicle setup and injection process are not fully reproducible	*Moderate:* widely used vehicular benchmark, but hosting stability should be periodically checked	[79, 398]

Accessibility and hosting stability are time-dependent and should be periodically verified. This table complements the baseline availability, metadata, labeling, and split information already reported in Table 2

ToN-IoT, and Edge-IIoTset provide stronger reproducibility support because they combine public access, documentation, labels, and reusable feature representations. In contrast, legacy datasets such as KDD Cup'99 and NSL-KDD remain important for historical comparison but present weaker sustainability due to legacy hosting, limited feature-pipeline reproducibility, and dependence on mirrors or secondary sources.

The analysis also shows that raw data availability and feature reproducibility are different issues. A dataset may provide processed CSV files, but if the original packet captures, feature-extraction scripts, tool versions, or preprocessing steps are unavailable, exact reproduction of published features becomes difficult. This issue is particularly relevant for datasets that provide only processed flow records or compact feature tables.

Conversely, datasets that provide PCAP files, labels, feature descriptions, and extraction tools offer stronger support for reproducible IDS research.

These findings have direct implications for the SHIELD repository. Beyond listing dataset names and basic properties, SHIELD will be extended to maintain sustainability metadata, including official access links, last-verified access dates, documentation references, label-generation notes, feature-extraction tools, raw-data availability, license/access restrictions, and mirror status. Such information would help researchers identify not only which dataset is suitable for a specific IDS task, but also whether it remains accessible, well documented, and reproducible for future experimentation.

6 Classification techniques

In addition to understanding the properties of intrusion detection datasets, it is equally important to examine the types of classification algorithms commonly applied to them. The performance and suitability of a dataset are often closely linked to the learning approach it supports-whether supervised, unsupervised, or semi-supervised. These algorithms play a crucial role in evaluating the effectiveness of a dataset for intrusion detection tasks. Figure 2 provides an overview of various classification techniques typically employed in IDS research. In this section, we discuss these techniques and explore how different intrusion detection datasets align with each paradigm.

1. Supervised learning: It uses a Labelled training dataset to initially train the underlying algorithm, which is then applied to an Unlabelled test dataset for classifying the data into relevant groups. This method is well-suited for two types of problems: classification, where the output variable is discrete and can be grouped into different classes, and regression, where the output variable is a real value. In the subsequent sections, we will briefly outline the commonly used supervised machine learning algorithms for intrusion detection [418].

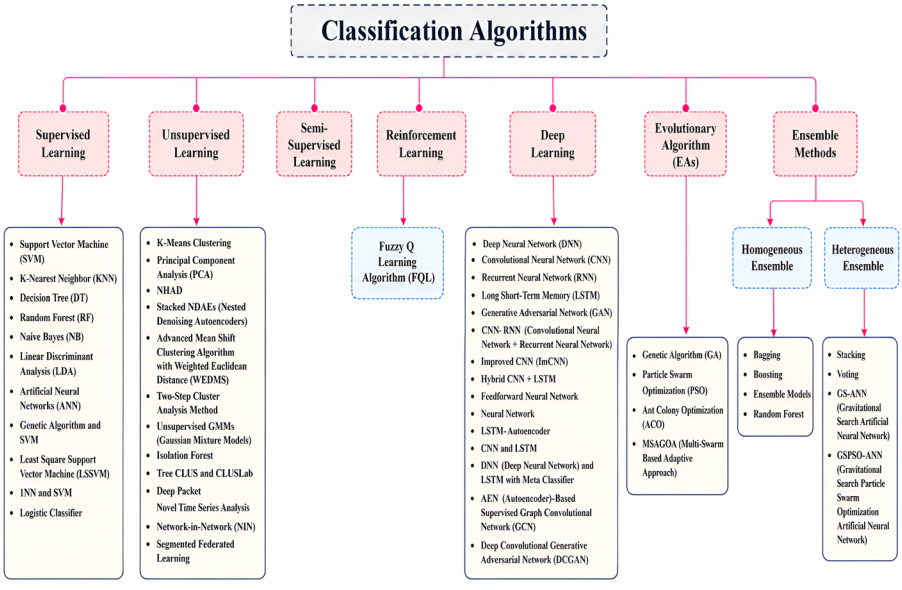


Fig. 2 Classification techniques

(a) Support vector machine (SVM): It is a supervised algorithm [419] used for the regression and classification of the data. Being a binary classification method, it can be employed for linear as well as non-linear data. Here, the input vector is initially mapped, and then the best-separating hyperplane is obtained in the higher-dimensional feature space. Instead of using the whole training sample, only the support vectors are used by SVM to determine the hyperplane, and thus, it is incredibly robust to the outliers. SVMs can be used effectively for detecting intrusion as they have good generalization ability and are capable of overcoming the curse of dimensionality [217, 278, 279, 420] have employed SVM for effective Intrusion Detection.

(b) K-Nearest neighbour (KNN): KNN, a supervised clustering-based algorithm [421], works on the principle that related data points lie in proximity in the vector space [422]. Here initially, the data is categorized into K number of clusters based on the distance between the different data points present in the dataset (calculated based on Euclidean distance) [423]. The closest points are grouped, and the centroid of the newly formed cluster is updated using the mean of their previous data points. The process is repeated until the centroid of the clusters does not change, i.e., all the data points have been assigned to their respective clusters. This algorithm is simple to implement, however, the main problem lies in determining the appropriate value of K [424]. It is one of the commonly used techniques for intrusion detection [425–427]. Using KNN, unlabelled data is separated into clusters of normal or attack classes. Since it uses the clustering-based approach, it can be used for the detection of unknown attacks. Moreover, it is scalable, simple to use, and effective for large datasets, which makes it the right choice for the detection of intrusion in the network.

(c) Decision Tree (DT): DT is an important classification method that uses the divide-and-conquer strategy [428]. A tree-like representation is used to solve a problem where internal nodes are used to represent the attributes present in the dataset, and the leaf nodes represent the class labels [429]. The process begins with the selection of the root node, and initially, the entire training set is considered as the root node. After the selection of the root node, the training set is split such that each subset contains the same value of the attribute which has been used for the split. This process is recursively repeated until leaf nodes are obtained, i.e., the attributes which belong to the same class are placed together [430], and the process stops. DT can be used for intrusion detection [431, 432], where a DT is constructed using the training data. Depending on the training, the model learns on its basis and classifies the future data as normal or attacked. The attack classes can further be classified into different categories based on the same approach. Moreover, decision trees are easy to implement as they reduce uncertainty and work well for large datasets.

(d) Random Forest (RF): RF is a combination of decision trees [433] where each tree produces an output, and the final output is obtained on the basis of majority voting. It has several applications as it is stable, has good generalization ability [434–436], and is capable of reducing the overfitting problem [286]. RF is used for feature selection, where decision trees rank the attributes based on their significance. According to this, the essential attributes in the intrusion detection system (IDS)

are taken into consideration while the irrelevant, unused attributes (out-of-bag samples) are left behind and removed as they do not contribute significantly to intrusion detection. Recent implementations of RF for intrusion detection have promising results [437, 438]. However, using RF, several noisy trees are produced [439], which affects the accuracy of the system.

(e) Naive Bayes (NB): NB, based on Bayesian learning, is used for learning tasks where instances are characterized by attribute conjunctions and the target function is chosen from a finite set. It controls the posterior probabilities of observing a class label from a group of normal and anomaly class labels. When applied to a test case, the predicted class is chosen as the one with the highest posterior probability [440]. NB is feasible to address the issues where the target function has a discrete output (yes/ no), attribute-value pairs represent instances, and the independent hypothesis of Naive Bayes is valid. There exist three NB algorithms: Gaussian NB [441], Bernoulli NB [442], and Multinomial NB [443]. Each of these NB algorithms works on a distinct probability calculation method. Furthermore, NB offers fast detection speed compared to other classifiers, as a result of which it enhances the performance of IDS [444, 445]. However, it relies on the assumption of feature independence, which might not be accurate for identifying various attacks.

(f) Linear Discriminant Analysis (LDA): It is a widely recognized statistical method employed for dimensional reduction in various machine learning and pattern recognition applications. It is commonly utilized during the pre-processing stage. LDA aims to project a dataset with n dimensions into a reduced k-dimensional dataset, where k is less than n. This reduction is achieved while preserving essential class-discriminating information. LDA possesses numerous advantages that render it suitable for developing an IDS model. Firstly, it is straightforward and can be easily implemented. Secondly, it exhibits efficiency and imposes minimal computational overhead. Lastly, its performance is comparable to many other well-established IDS models. It is a commonly used technique employed for feature selection case of IDS [446, 447].

(g) Artificial neural networks: ANN is a collection of nodes that are connected in different layers, namely input, intermediate hidden, and output layers, used for the exchange of information. These nodes are known as artificial neurons. Weights are associated with the connections, which represent the signal strength. Based on the input signal and the associated weights, an output is generated, based on which, the system learns [448]. During the learning phase, the actual output is compared with the desired output on the basis of which error in the hidden nodes is calculated, and then this error is propagated back to the system, and accordingly, the weights are updated [449]. This process is repeated until the error rate is reduced up to a certain threshold value, and accordingly, the entire process of learning is enhanced. ANN is capable of learning intricate non-linear relationships present in the model [450], due to which they can be effectively used for real-time intrusion detection [451, 452]. In the training phase, the model learns about the different attack classes from the input data. On the basis of this learning, the ANN classifies the unseen data as normal or intrusive during the testing phase. So, ANN has good generalization ability.

2. Unsupervised algorithms: It deals with unlabelled data, meaning that no output variable is present corresponding to a particular input value, allowing the algorithm to autonomously analyse and act upon the information. Here, the machine's objective is to group unorganized data based on similarities, patterns, and differences without any pre-existing knowledge or guidance from a teacher or labelled data. Consequently, the machine is solely responsible for discovering the underlying structure within the unlabelled data, relying on its capabilities and algorithms to uncover meaningful insights. So, it is opposed to that of the supervised learning algorithm [453].

   (a) K-Means clustering: K-means is a clustering-based algorithm that operates under the hypothesis that normal data is closer to their nearest cluster centroid, while anomalies are distant from their nearest cluster centroid. The algorithm begins by clustering the data into K clusters, with K data points initially assumed as the centroids of the diverse clusters. The remaining data is then assigned to clusters based on their proximity to the centroids. In the following step, the centroids are recalculated by averaging the data points within the cluster. This process is continued till a stopping criterion is met and there is no further change in the centroids. It has gained significant popularity among researchers working on IDS [454–456], who have integrated it with other classifiers.
   (b) Principal component analysis (PCA): PCA was introduced by Karl Pearson in 1901 [457]. It is a common technique used in machine learning for transforming the higher-dimensional feature space to a lower-dimensional space [458] to reduce the overfitting problem. Here, principal components are used to decrease the complexity and confusion of the model [459]. The number of principal components depends on the number of attributes; it can be less than or equal to the number of attributes present in the dataset. In the IDS, PCA can be used for feature extraction based on which the model is trained [459]. Due to the feature selection, the processing speed is enhanced, i.e., the time taken for training and testing of data is reduced [460] as there is a reduction in the dimensionality of the dataset. So, PCA is used in combination with other classifiers as it can be used to optimize the process of intrusion detection [461].

3. Semi-supervised learning: Supervised learning poses certain drawbacks, including the need for manual labelling by ML specialists or data scientists, which can be time-consuming and costly. Additionally, unsupervised learning has a limited number of applications. To address these issues, the concept of Semi-supervised learning has been introduced. This approach involves training the algorithm using a blend of both labelled/ unlabelled data. However, since labelled data availability is relatively small compared to unlabelled data, an unsupervised learning algorithm is initially used to cluster similar data, which aids in assigning labels to the unlabelled data. This approach acknowledges that acquiring labelled data is a more expensive process compared to acquiring unlabelled data. The objective of anomaly detection is to identify patterns or observations that deviate from the norm, appearing unusual or distinct.
4. Reinforcement learning: Reinforcement learning falls within the realm of machine learning and focuses on selecting actions that maximize rewards in specific situations. It is widely used by software and machines to determine the most favourable behaviour or path given a particular context. Unlike supervised learning, where the training data

contains the correct answers, reinforcement learning operates without explicit answers. Instead, the reinforcement agent learns how to accomplish a given task by making decisions and observing the outcomes, relying on its accumulated experiences rather than a predetermined training dataset. Reinforcement learning, often abbreviated as RL, revolves around the science of decision-making. Its objective is to acquire the optimal behaviour within an environment to attain the highest possible reward. Unlike supervised or unsupervised machine learning, RL systems gather data through a trial-and-error process, rather than relying on predefined datasets as input. It employs algorithms that leverage feedback from outcomes to make informed decisions about subsequent actions. Following every action, the algorithm gets feedback indicating if its choice was fit, neutral, or unfit. This technique proves valuable for automated systems that need to make numerous minor decisions independently, without human intervention. Owing to its ability to tackle issues associated with supervised and unsupervised learning, it can be used for intrusion detection [462–465]. By taking action against intrusion, its goal is to maximize rewards, effectively engaging in a process of learning by doing to attain optimal results.

5. Deep neural network: The DNN, which stands for Deep Neural Network, is an ANN that includes several layers positioned from input to output layers [466–468]. In a more specific manner, it can be described as a fully connected neural network structure resembling a multilayer perceptron (MLP). Within a connected DNN, the neurons in the lower layers establish connections with all the neurons in the upper layers. To accomplish supervised learning tasks with nonlinear functions, DNN utilizes backpropagation [466] to develop a distributed deep learning model for real-time processing and analyzing extensive datasets for an intrusion detection framework. [469] presented an IDS comprising a combination of a Recurrent Neural Network (RNN) with Gated Recurrent Units (GRUs), an MLP, and a softmax module. It is worth noting that DNNs can theoretically approximate any function [470]. Recently, other DNN-based IDSs were presented in [471].

   (a) Convolutional neural network (CNN): CNN is an ANN characterized by its shared-weight configuration that relies on convolutional kernels or filters. Taking inspiration from biological practices [472], CNN applies convolutional kernels across input attributes, enabling the extraction of translation-equivariant responses referred to as feature maps. CNNs and their associated architectures gained significant attention owing to their exceptional performance in computer vision tasks [473]. As a result, several CNN architectures such as LeNet-5 [474], AlexNet [475], and ResNet [476] were proposed. Apart from computer vision problems, CNN architectures have also shown promising results in IDS. [477–480] introduced an image transformation technique for the NSL-KDD dataset, where CNNs spontaneously learn graphic NSL-KDD transformations' features. In comparison to DNNs, CNNs excel in extracting local features, resulting in a reduced number of weights and computational complexity, thereby improving the prediction rate.

   (b) Recurrent neural network: RNN belongs to a category of artificial neural networks capable of exhibiting temporal memory behaviour. This dynamic behaviour is achieved through interconnections among nodes, forming a graph along a time sequence [481]. The core state of an RNN enables it to process an input series of

varying lengths. Based on the presence of loops within the constructed graph, it can be further categorized into finite or infinite impulse [482]. The former networks can be opened and switched with feedforward neural networks (FNNs) with strict causality, while it is not possible in the case of infinite-pulse recurrent networks. Additionally, RNN incorporates extra stored states, enabling the implementation of networks with interruptions or feedback loops (e.g., LSTM), further enhancing their capabilities. RNNs were introduced as a solution to address the challenge faced by DNNs in effectively capturing temporally changing data. Consequently, RNNs have played a crucial role in various domains, including natural language processing and action recognition [483]. In the context of IDS, which typically deals with continuous data streams, RNNs are increasingly being employed [484–487]. Conversely, one limitation of RNNs is their lack of specialized treatment for activation functions. This can lead to issues such as the vanishing gradient problem or even gradient explosion, particularly when the network has a large number of layers.

(c) Long short-term memory: LSTM addresses the challenge of gradient vanishing in traditional RNNs by incorporating supplementary storage states [488]. This approach efficiently mitigates the issue by utilizing a gate function as the activation function, enabling selective passage of information. Building upon the original LSTM design [488] integrates forgetting gates, enabling LSTM to reset its state and mimic the memory forgetting procedure. Recognized for its exceptional execution [489, 490], this LSTM architecture is widely regarded as the conventional approach. Taking inspiration from this [491] presented a gate recurrent unit (GRU) comprising reset and update gates, which preserves LSTM's performance while utilizing fewer parameters. Due to its ability to overcome the gradient vanishing issue in customary RNNs, LSTM has emerged as one of the most widely used variants of RNNs. Numerous IDS studies leverage LSTM networks [492, 493] due to their suitability for time-series data classification and prediction, with the forgetting procedure aligning well with data stream detection. As a result of its advantages, LSTM is employed for intrusion detection in recent research works. However, the classical LSTM architecture cannot inherently be trained in parallelly [494], leading to potential computational costs associated with running LSTM-based models.

(d) Generative adversarial network: GANs, a novel approach to semi-supervised and unsupervised learning, leverage the implicit modelling of high-dimensional data distributions. Introduced in 2014 [495], GANs involve competitively training a pair of networks. An analogy commonly used, especially for graphic data, is to consider the two networks as an art forger and an art expert, respectively. The generator (G), known as the forger, aims to produce realistic images as forgeries, while the discriminator (D), the expert, distinguishes between forgeries and authentic images. Both networks are trained concurrently and compete with one another. Importantly, the generator has no direct access to real images and relies solely on its interaction with the discriminator for learning. The discriminator, on the other hand, evaluates both synthetic and real images. Its error signal is based on determining whether an image is from an actual stack or a generator. This error signal can be utilized to train the generator, enabling it to generate higher-quality

forgeries. Typically, the generator and discriminator networks are implemented using multilayer architectures comprising convolutional or fully connected layers. These networks need to be differentiable, although direct invertibility is not a requirement [496, 497].

(e) Evolutionary algorithms (EAs): They are heuristic search methods that efficiently utilize Darwinian evolution principles, exhibiting robustness and flexibility in capturing global solutions for complex optimization problems [498]. By employing EAs, there is a high probability of discovering a near-optimal solution early in the optimization process. EAs do not rely on any fitness gradient information, making them easily parallelizable and capable of escaping local minima where deterministic optimization methods may fail or be inapplicable. The convergence of EAs remains unaffected by the continuity or differentiability of the target function, and they can not only serve as replacements for traditional optimization methods but also enhance them through hybrid approaches, resulting in improved overall performance. EAs excel in both single and multi-objective optimization, accommodating scenarios involving multiple disciplines. Particularly in multi-objective optimization problems, EAs offer significant advantages over traditional methods, as they can simultaneously handle integer, discontinuous, or discrete design variables, and they can adapt to different game strategies [499].

    (i) Genetic algorithm: GA belongs to the class of evolutionary algorithms [440]. It is an adaptive heuristic search algorithm, i.e., it adapts to the changes in the environment and uses the idea of natural selection and genetics [500] to find an optimal solution to a given problem. Initially, two individuals are chosen at random based on their fitness levels, as they can effectively adapt to the changes in the environment. After the selection of the individuals, a crossover between them is carried out to produce new individuals, i.e., offspring. The genes of the offspring are then modified, and this is done so that the diversity in the population is maintained, and the early convergence can be avoided. This process continues till convergence, i.e., when the termination condition is met, the process stops. On the basis of the survival of the fittest principle, GA selects an optimum solution from several possible solutions for the detection of intrusion. So, it is used in combination with other classification techniques to enhance the process of intrusion detection, thereby increasing the overall performance of the system and reducing the false alarm rate [501].

    (ii) Particle swarm optimization (PSO): It is an evolutionary, population-based, stochastic algorithm proposed by Kennedy and Eberhart [440] in 1995, which is used to find an optimal solution to a given problem. The algorithm is modeled after how birds in a flock shift their positions about one another to find food. Similarly, each particle in the system represents a possible solution to a given problem, and the information is exchanged among the particles based on which they adjust their position. Initially, all the particles are initialized with a random value, and based on the fitness function, their position is updated iteratively so that an optimal solution can be attained, and the resources are utilized efficiently [502]. PSO is used for feature selection in

intrusion detection, where the selected attributes are represented by 1, and the other attributes that are not selected are represented by 0. The selected attributes are then passed to the classifier, which calculates the fittest of the particles based on which the intrusion detection rate is obtained [503].

(iii) Ant colony optimization (ACO): It was proposed by Colorni, Dorigo, and Maniezzo [504] in 1992. The algorithm is inspired by observing the behaviour of ant colonies where the food is obtained by finding the shortest path using a chemical called pheromone, which is used for communication, i.e., the ants deposit this chemical on the way they take. Over time, pheromone evaporates, and the path having more amount of pheromones is the shortest one, which is ultimately taken by all the ants while other paths gradually diminish [505]. ACO is used for feature selection in IDS, where artificial ants represent the computational agents trying to obtain an optimal solution to the problem by cooperating through artificial pheromones and heuristic information [441]. Here, artificial ants are employed for constructing several paths using different feature subsets in the form of a graph, where each node represents features, and edges represent the next possible feature choice. Among these feature subsets, an optimal set of features is selected. If the optimal solution is not obtained, then the value of the pheromone is updated, and the process is repeated until convergence [442].

(f) Ensemble methods: To address the security issues, an ensemble-based technique is employed to significantly enhance the competence of an IDS, where the performance of the system can be improved using different classifiers. It combines several weak classifiers to attain a powerful classifier that can be employed for the classification of data. The classifier obtained is robust and has a better prediction rate and good generalization ability, i.e., it can be used effectively for the classification of unlabelled data. As it combines the merits of different algorithms, better results can be obtained using the ensemble approach, thereby reducing the error rate [443]. The ensemble is broadly categorized into homogeneous and heterogeneous [506]. In a homogeneous ensemble, the same classifiers are used to train and test the data, whereas, in a heterogeneous approach, diverse classifiers such as SVM, ANN, and RF are employed for analysing the system performance. Bagging [507, 508] and boosting [252, 509] are commonly used techniques for a homogenous ensemble while stacking [510, 511] and voting [437, 512, 513] are used for a heterogeneous ensemble.

7 Feature selection techniques

High-dimensional network traffic data can significantly impact the performance of intrusion detection systems (IDS). Not all features in a dataset are equally relevant-some may introduce noise or redundancy. Feature selection addresses this by identifying the most important attributes that contribute to accurate threat detection. It reduces computational cost, enhances detection accuracy, and improves overall model efficiency.

Over the years, numerous studies have explored the integration of feature selection techniques within IDS frameworks to enhance performance. Researchers have applied various approaches, such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Principal Component Analysis (PCA), and ReliefF, to identify the most influential

features across multiple intrusion detection datasets. These methods have consistently shown improvements in classification accuracy and significant reductions in training time. For instance, GA-based feature selection applied to datasets like NSL-KDD and CICIDS2017 has demonstrated improved detection rates and lower false positive rates when combined with classifiers such as SVM, Random Forest, and XGBoost.

In their study, Kaushik et al. [514] have improved the detection effectiveness of intrusion detection systems (IDS) by employing different feature selection methods, including chi-square and information gain. These techniques are utilized to identify the most relevant features for accurate detection. The researchers conducted their analysis using a large dataset of simulated intrusions in a military network domain. In their research, Amin and Kabir [515] introduced a new feature selection approach called the attribute ratio method. They investigated several popular threat classification methods, with a focus on six widely used machine learning (ML) techniques for intrusion detection: Random Forest, Naive Bayes, Artificial Neural Networks, K-nearest neighbour algorithm, Support Vector Machines, and bagging. The evaluation was conducted on the NSL-KDD dataset, and the results demonstrated an impressive accuracy of 99.86%. Krishnaven et al. [516] presented a methodology that combines feature selection techniques, including gain-ratio, chi-squared, and information gain, with ensemble classification methods such as SVM, LR, NB, and DT. The objective was to develop a robust intrusion detection system (IDS) using this integrated approach. The researchers evaluated the performance of their methodology on three datasets, namely Honeypots, Kyoto, and NSL-KDD. Yuyang Zhou et al. [517] introduced a novel framework for intrusion detection. They combined feature selection using the CFS-BA method with ensemble learning techniques, including C4.5, Random Forest, and Forest by Penalizing attribute. The framework aimed to enhance the effectiveness of feature selection by identifying a subset of features that are both significant and correlated, ultimately improving the accuracy of the classification process. Kasongo et al. [203] presented various feature selection methods in their study. These methods encompassed Xgboost and extra-tree classifier, which were applied to the UNSW-15 dataset with the implementation of multiple machine learning classifiers for performance evaluation. Ali et al. [518] utilized Correlation-based Feature Selection (CFS) along with the Naive Bayes (NB) classifier algorithms. They applied these techniques to diverse machine learning algorithms as a means of handling vast and diverse data, aiming to enhance performance. In their study, Iram et al. [519] employed a range of machine learning algorithms, including logistic regression, K-nearest neighbor, Support Vector Machine, Naive Bayes, Extra-tree classifier, Multi-layer Perceptron, random forest, and decision tree. The purpose was to classify data as either normal or malicious based on these algorithms. Anuradha et al. [520] presented a comprehensive examination of feature selection techniques, namely filter, wrapper, and embedded methods. These techniques were explored as a means of addressing the challenges posed by large-scale data, aiming to alleviate the curse of dimensionality and enhance classification performance. The study also investigated the utilization of various machine learning algorithms in this context. Pasupulety et al. [521] proposed a methodology for feature selection that involved employing an ensemble of multiple techniques. The primary objectives of this approach were to reduce training time and improve accuracy. Vinutha et al. [522] put forth a range of feature selection techniques, including Cfs, Chi-square, SU, Gain Ratio, Info Gain, and OneR. These techniques were employed to

identify relevant features. The selected features were then utilized in AdaBoost, Bagging, and Stacking algorithms. Chen et al. [523] introduced the concept of feature selection techniques and their practical implementation. The purpose was to mitigate the challenges posed by the curse of dimensionality and effectively handle class-related information within the dataset. Cia et al. [524] presented a range of feature selection methods with the aim of accelerating computation, improving learning accuracy, and enhancing the interpretability of the learning model for information extraction. Deng et al. [525] employed a combination of feature selection techniques, including wrapper, filter, embedded, and hybrid methods, in their classification approach. These techniques were utilized alongside multiple machine learning algorithms to enhance the classification performance. Kumar et al. [526] implemented three commonly used feature selection methods, namely Gain ratio feature evaluator, Information Gain feature evaluator (IGF), and Correlation-based Feature Selection (CFS). Subsequently, they evaluated the performance of different classifiers on the NSL-KDD dataset. Thakkar and Lohiya [527] highlighted the utilization of data availability for feature selection. The classification approach emphasizes the selection of features that facilitate sample classification, while the clustering approach focuses on selecting features that enhance the formation of clusters, without the need for labelled data or a specific learning algorithm. Maldonado et al. [528] presented various feature selection techniques that are based on supervised learning. These techniques include:

1. Filter-based feature selection approach.
2. Wrapper-based feature selection approach.
3. Interactive feature selection approach.
4. Embedded feature selection approach.

Ambusaidi et al. [529] introduced a filter-based feature selection technique specifically designed for constructing an intrusion detection system called LSSVM-IDS (Least Square Support Vector Machine-based IDS). Siddiqi and Pak [530] proposed a novel approach in filter-based feature selection by incorporating transformation techniques. Their method involved performing normalization and transformation as preprocessing steps prior to the classification process. Ullah and Mahmoud [531] introduced a filter-based feature selection method specifically designed for an anomaly-based intrusion detection system. In their approach, a set of features was estimated using criteria such as distance, consistency, information, and dependency. Kushwaha et al. [532] (IEEE Malaysia Section et al.) implemented an anomaly-based intrusion detection system using a filter-based feature selection approach. Their study specifically focused on applying this approach to the KDD-CUP 99 dataset. The goal was to identify and select the most relevant features for effective anomaly detection in the intrusion detection system. Almasoudy et al. [533] introduced a feature selection method for an intrusion detection system (IDS) based on the Differential Evolution technique. Their approach utilized a wrapper-based approach, where the feature selection process was embedded within the IDS itself. In their work, Zang et al. [534] presented an efficient feature selection method based on a wrapper approach. This method utilized the Bayesian Network classifier to evaluate the relevance of features. Li et al. [535] introduced a wrapper-based feature selection technique to develop a lightweight Intrusion Detection System (IDS). They utilized a modified version of random mutation hill climbing (RMHC) for the purpose of

searching and evaluating the feature space. Khammassi et al. [536] presented a wrapper-based feature selection approach that incorporated the searching strategy of a genetic algorithm along with logistic regression for learning. In order to enhance the performance of an intrusion detection system, Bonab et al. [537] proposed a wrapper-based feature selection approach that employed a hybrid technique. The approach involved selecting new features using a combination of the Fruit Fly Algorithm (FFA) and the Ant Lion Optimizer Algorithm (ALO). Maldonado et al. [538] proposed the interactive feature selection approach incorporates expert human intervention to enhance classification performance [538]. Liu et al. [107] introduced an embedded feature selection approach combined with convolutional neural networks (CNN) for classification on the New IoT dataset. The study presented two models: Random Forest with CNN (RFCNN) and extreme Gradient Boosting with CNN (XCNN). Hua [539] presented a highly effective approach for traffic classification, which incorporated an embedded feature selection method and LightGBM. Ayo et al. [540] introduced a rule-based hybrid feature selection method that leveraged a deep learning model for optimization. The proposed method consisted of three distinct steps: hybrid feature selection, rule evaluation, and rule detection. Singh et al. [541] introduced an Evolutionary Algorithm-based Feature Selection method for Intrusion Detection Systems (IDS). In their approach, a new hybrid technique combining Teaching Learning Based Optimization (TLBO) and Ensemble Feature Selection (EFS) was proposed. In the study conducted by Ranganathan Bala Saraswathi et al. [542], it was observed that the previously discussed methods for feature selection involved complex calculations, which resulted in inefficiency. As a result, a new category of algorithms, known as Bio-inspired Algorithms, was introduced. These algorithms draw inspiration from nature and are characterized by their simplicity and ease of implementation. Bio-inspired algorithms can be classified into three main groups:

1. Evolutionary algorithms
2. Swarm-based algorithms
3. Ecology-based algorithms

Almomani proposed [543] a wrapper-based method utilizing Particle Swarm Optimization (PSO), Grey Wolf Optimizer (GWO), Firefly Optimization (FFO), and Genetic Algorithm (GA) to improve the performance of the Network Intrusion Detection System (NIDS). The aim of this approach was to optimize the feature selection process by employing these metaheuristic algorithms. By leveraging PSO, GWO, FFO, and GA, the wrapper-based method aimed to enhance the effectiveness and efficiency of the NIDS. Alazaam et al. [544] presented a bio-inspired feature selection technique for Intrusion Detection Systems (IDS) based on the Pigeon Inspired Optimizer. This wrapper-based method utilizes the continuous Pigeon Inspired Optimizer and compares its performance to that of other Swarm Intelligent Algorithms. Yulianto et al. [241] introduced an improved Intrusion Detection System (IDS) based on the AdaBoost algorithm. The proposed approach incorporated three techniques: Synthetic Minority Oversampling Technique (SMOTE), Principal Component Analysis (PCA), and Ensemble Feature Selection (EFS). SMOTE was used to address class imbalance, PCA was employed for dimensionality reduction, and EFS facilitated the selection of relevant features. Chen et al. [545] presented a lightweight intrusion detection system based on Principal Component

Analysis (PCA). The proposed system utilized PCA for dimensionality reduction to reduce the computational complexity and memory usage. In addition to PCA, the C4.5 algorithm was employed for classification. In the study conducted by Hasan et al. [546], direct feature selection was performed using the Random Forest approach. This involved selecting attributes based on their importance. The Random Forest algorithm employs two important measures to determine feature importance: the Gini Importance Index (GI) and the Permutation Importance Index (PIM). Li et al. [547] presented an effective deep learning method called Auto Encoder IDS for constructing training and feature selection. This method utilizes the random forest feature selection technique to identify desirable features. Disha et al. [548] employed multiple machine learning models to introduce an efficient feature selection method known as Gini Impurity-based Weighted Random Forest (GIWRF). This method utilizes the Gini impurity criterion to weigh the importance of features within a random forest framework. The Institute of Electrical and Electronics Engineers (IEEE) and Hindustan Institute of Technology [549] highlighted the use of Random Forest (RF) in machine learning for classification problems. This algorithm operates in two steps:

- Creation of Forests: Random Forest constructs multiple decision trees using the given dataset.
- Classifier Prediction: Once the forests are created, the algorithm predicts the classification based on the collective decisions of the individual decision trees.

By utilizing this two-step process, Random Forest provides an effective approach for classification tasks in machine learning. Kasongo and Sun [203] described a feature selection process in which features are selected based on their measuring scores using a filter-based method with XGBOOST. This method involves evaluating the importance of each feature using XGBOOST and selecting the most significant ones. Devan et al. [471] proposed an efficient model for an intrusion detection system (IDS) that combines feature selection using XGBoost with classification using a deep neural network (DNN). In this approach, XGBoost is utilized to identify the most relevant features for intrusion detection. Sharma et al. [550] discussed the Extra Trees classifier as an updated version of Random Forest (RF) for attribute or feature selection. The Extra Trees classifier selects relevant attributes for each individual attack. Halim et al. [551] proposed a genetic algorithm-based feature selection approach for Intrusion Detection Systems (IDS). This method aims to identify a subset of features that not only optimizes the performance of the IDS but also preserves the distinct properties of the selected feature subset. Kamalov et al. [552] identified a gap in existing methods that did not differentiate between continuous and discrete features in feature selection. To address this issue, the authors proposed a new feature selection method based on correlation and mutual information. Specifically, they introduced the MICorr feature selection method, which combines Pearson correlation and mutual information techniques. Zhang et al. [553] put forth an Ensemble-based feature selection (EAFS) technique to enhance the efficiency and capabilities of Intrusion Detection Systems (IDS). In this technique, each feature is ranked individually using a base approach for feature selection. Kamalov and Thabtah [554] introduced a method for feature selection that combines and normalizes three prominent feature selection techniques: information gain, chi-square statistic, and inter-correlation.

Table 102 Traffic analysis tools

S. no.	Stage	Category name	Description
1	Data ingestion	Packet sniffers/synthesizers	Capture live packets or generate synthetic streams for testing
2	Flow assembly	Traffic aggregators	Merge individual packets into flow records (e.g., NetFlow/IPFIX), optionally computing basic statistics
3	Feature processing	Feature extractors	Enrich raw packets or flows with numerical attributes (e.g., byte counts, timing, header fields) suitable for ML
4	Central storage	Flow repositories	Collect, index, and store exported flow records in databases or files, often providing querying and aggregation
5	Deep inspection	Protocol analyzers	Drill down into packet and flow contents-parsing protocols, decoding payloads, and surfacing anomalies beyond simple statistics
6	Traffic labeling	Classification engines	Automatically tag traffic as benign or specific attack types based on signatures, heuristics, or ML models



Fig. 3 Traffic analysis tools

8 Traffic analysis tools

In building intrusion detection datasets, traffic can be acquired, transformed, examined, stored, and finally annotated, as summarized in Table 102 and illustrated in Figure 3.

We organise the tools by the part they play in that pipeline:

1. *Data ingestion: packet sniffers and synthesizers:* Tools in this group either sniff real network traffic off an interface or synthesize custom streams for controlled experiments. *Examples:* Wireshark/TShark, tcpdump (live packet capture) Ostinato, Scapy, D ITG (synthetic traffic generators)

   (a) *Wireshark:* Wireshark [555] is a free, open-source network protocol analyzer used to capture and inspect real-time data traffic across networks. It offers deep packet analysis, customizable filters, and visual tools like color coding to simplify diagnostics. Widely used by professionals, it's ideal for troubleshooting, security analysis, and learning network protocols.
   (b) *Tcpdump:* Tcpdump [556] is a lightweight, command-line packet analyzer used to capture and inspect real-time network traffic. It supports protocols like TCP, UDP, and ICMP, and uses BPF syntax for precise filtering. Though powerful, it lacks a graphical interface, making it better suited for advanced users comfortable with the terminal.
   (c) *Ostinato:* Ostinato [557] is a network traffic generator that allows users to create and send custom packets to test network performance and behavior. It supports various protocols and is ideal for simulating rare or specific traffic patterns. However, being stateless, it's less suited for tests requiring real user-like interactions or connection tracking.
   (d) *Scapy:* Scapy [558] is a powerful Python-based tool for crafting, sending, and analyzing network packets, widely used for testing, security research, and protocol development. It supports a wide range of protocols and allows custom packet creation with scripting flexibility. Ideal for advanced users, Scapy combines packet manipulation with network automation capabilities.
   (e) *D-ITG (Distributed Internet Traffic Generator):* D-ITG [559] is a tool designed to generate realistic network traffic for performance testing and benchmarking. It simulates various traffic types (e.g., VoIP, video, gaming) and measures metrics like delay, jitter, and packet loss. D-ITG is useful for evaluating network quality under different conditions.

2. *Flow Assembly: Traffic Aggregators* These utilities read streams of packets (from live capture or PCAP) and roll them up into bidirectional or unidirectional flows, reducing data volume and offering a flow centric view. *Examples:* nProbe, YAF, Softflowd, Argus

   (a) *nProbe:* nProbe [560] is a versatile tool that acts as both a flow exporter and collector, supporting protocols like NetFlow, IPFIX, and sFlow. It handles IPv4/IPv6 traffic, converts flow formats, and integrates with monitoring tools. The Pro version adds high-speed packet capture and application detection with nDPI.
   (b) *YAF:* The tool, YAF [561] captures traffic either live or from saved pcap files. It was originally developed as a test implementation for features being developed by the IETF IPFIX working group. YAF focuses on bidirectional flow representation, data archiving, and exporting structured data, including deep packet inspection (DPI) details (Fig. 3).
   (c) *Softflowd:* The tool, softflowd [562] is a network traffic analyzer that exports NetFlow data (versions 1, 5, or 9). It works by listening on a live network interface or reading from packet capture (pcap) files. It supports IPv6 and runs on Linux

and OpenBSD. It needs the libpcap library to build and run. Flows can either be exported to a standard NetFlow collector or summarized directly within softflowd.

(d) *Argus:* Argus [563] is a network monitoring tool that collects and analyzes flow data to provide insights into network traffic, performance, and security. It is useful for intrusion detection, traffic analysis, and forensic investigations. Argus offers scalability, real-time monitoring, and efficient resource usage, but lacks full packet capture.

3. *Feature processing: feature extractors* Once flows (or packets) are available, these tools compute or extract the numerical descriptors-timing, counts, flags, header fields-that feed machine learning pipelines. *Examples:* CICFlowMeter, ICSFlowGenerator, NFStream, LiteEx.

   (a) *CICFlowMeter:* CICFlowMeter [564] is an open-source network traffic flow generator and analyzer that extracts over 80 statistical features from bidirectional flows (Biflows) in PCAP files. It supports detailed analysis of both forward and backward traffic directions, making it valuable for intrusion detection and anomaly detection. With flexible feature selection and the ability to customize flow timeouts, CICFlowMeter is widely used in cybersecurity research and offers both command-line and GUI interfaces.

   (b) *ICSFlowGenerator:* ICSFlowGenerator [565] is a flexible tool for generating network flows from both offline PCAP files and real-time LAN sniffing. It supports flow annotation using true or predicted labels based on attack histories or pretrained IDS models. The tool offers customizable options for flow computation, port-based separation, and output, making it ideal for creating labeled datasets and performing anomaly detection.

   (c) *NFStream:* NFStream [566] is a high-performance Python framework designed for efficient analysis of both online and offline network traffic. It supports encrypted layer-7 visibility, statistical feature extraction, and kernel-level insights, while offering extensibility through NFPlugins and seamless machine learning integration. Its goal is to provide a unified platform for network analytics, ensuring reproducibility and facilitating real-time traffic analysis.

   (d) *LiteEx:* LiteEx [567] is a lightweight, graphical tool for extracting packet- and flow-level features from network traffic traces with an intuitive GUI. It processes pcap files, extracts features, and exports them to CSV for use in machine learning models. The tool includes packet- and flow-level modules with efficient feature extraction, multi-threaded flow reconstruction, and system resource management.

4. *Central storage: flow repositories:* In large environments with multiple exporters, a central collector ingests, deduplicates, and archives flow data into searchable stores or time series databases. *Examples:* nfdump/nfcapd (with NfSen frontend), SiLK, ManageEngine NetFlow Analyzer, Elastiflow

   (a) *nfdump:* nfdump [568] is an open-source command-line tool for processing and analyzing NetFlow data, enabling real-time network monitoring and performance analysis. It supports various NetFlow versions, provides detailed traffic reports,

and offers powerful filtering and export capabilities in formats like CSV, JSON, and XML. Commonly used for bandwidth monitoring, security threat detection, and network optimization, Nfdump is compatible with Linux/Unix systems and Windows through WSL.

(b) *nfcapd:* nfcapd [569] is the NetFlow capture daemon used with the nfdump toolset. It supports various versions of NetFlow (v1, v5, v7, and v9) as well as IPFIX (though IPFIX support is still experimental). nfcapd captures flow data from the network and saves it into binary files. The file format used for storage is independent of the specific version of NetFlow.

(c) *NfSen frontend:* NfSen [570] is a web-based graphical front end for the nfdump NetFlow tools, offering an intuitive interface to visualize and analyze network traffic data. It uses Round Robin Database (RRD) technology for efficient time-series data storage and supports alerts, custom plugins, and historical profiles for tracking network trends. NfSen enhances NetFlow data analysis, making it easier for administrators to monitor and gain insights into network performance.

(d) *SiLK:* SiLK [571] is a suite of tools designed for traffic analysis. Developed by CERT NetSA, SiLK supports the collection, storage, and analysis of network flow data. It also allows users to query large datasets of historical traffic data. SiLK can collect NetFlow v5, NetFlow v9, and IPFIX flow data and is compatible with Linux and other Unix-based systems.

(e) *ManageEngine NetFlow Analyzer:* NetFlow Analyzer [572] is a comprehensive network traffic analysis tool that supports various flow technologies, offering real-time insights into bandwidth usage and network behavior. It provides deep analysis of network performance, including jitter, latency, and QoS validation, and is compatible with devices from major vendors like Cisco, Juniper, and Huawei. With built-in database support and a user-friendly interface, it is ideal for both real-time monitoring and long-term network capacity planning

(f) *Elastiflow:* ElastiFlow [573] is a network observability solution that simplifies the collection and analysis of network traffic flows, providing deep visibility into network operations. It integrates with leading data platforms for performance monitoring, capacity planning, and security analysis, offering flexible deployment options. With intuitive dashboards and analytics, ElastiFlow enables effective network monitoring and optimization across various IT environments.

5. *Deep inspection: protocol analyzers:* Beyond counting bytes and packets, these frameworks parse application protocols, extract fine grained logs, and flag suspicious patterns at the packet or session level. *Examples:* Zeek (Bro), Suricata, Snort, Tranalyzer

(a) *Zeek (Bro):* Zeek Network Security Monitor [574] is an open-source framework for advanced network traffic analysis and security monitoring, offering deep insights into network behavior through protocol analyzers. Its flexible scripting language allows for customized monitoring policies, making it suitable for high-performance networks. Zeek is widely used in large-scale environments for real-time threat detection, traffic auditing, and retrospective analysis.

(b) *Suricata:* Suricata [575] is an open-source, multi-threaded network security engine that operates as an IDS, IPS, and NSM tool, offering deep packet

inspection, real-time threat detection, and broad protocol support for high-throughput environments.

(c) *Snort:* Snort [576] is an open-source intrusion prevention system (IPS) and packet logger that uses signature-based detection to identify a wide variety of network threats. It provides real-time threat detection and integrates with other tools to enhance security frameworks. Despite its strong community support, Snort may generate false positives and require fine-tuning during setup.

(d) *Tranalyzer:* Tranalyzer [577] is an open-source, lightweight flow generator and packet analyzer for live or offline pcap traffic, offering detailed insights from flows to individual packets. Its modular plugin-based architecture ensures high customizability for scalable network analysis.

6. *Traffic Labeling: Classification Engines:* Finally, automated classifiers assign ground truth labels-benign vs. malicious, or specific attack categories-using rule based or statistical methods. *Examples:* Zeek's notice framework, Snort, nDPI, FlowTransformer

(a) *Zeek's Notice Framework:* Zeek's Notice Framework [578] generates structured, context-rich alerts for suspicious network activity, with customizable severity and responses-from logging to automated mitigation-enabling real-time threat detection and seamless integration into security workflows.

(b) *nDPI:* nDPI [579] is an open-source deep packet inspection library that identifies network traffic by analyzing application-layer protocols, including encrypted or obfuscated flows, using packet sizes and flow patterns. Supporting over 200 protocols, it is extensible and integrates with network monitoring tools to enhance traffic visibility and threat detection.

(c) *FlowTransformer:* FlowTransformer [580] is a Python-based framework for developing transformer-based models for flow-based NIDS. Utilizing the sequential nature of network traffic, it enables detection of complex attacks through a modular pipeline for data ingestion, preprocessing, input encoding, and model customization. Built with TensorFlow and Keras, it supports reproducible experimentation with advanced ML techniques for network security.

9 Performance metrics

Performance metrics in the context of intrusion detection systems (IDS) serve the purpose of assessing how well these systems identify and react to potential security threats, measuring both their effectiveness and efficiency. These metrics enable organizations to gauge the precision and dependability of their IDS, empowering them to make informed choices regarding their cybersecurity tactics. The following are the principal performance metrics for intrusion detection systems:

1. Confusion matrix: The Confusion Matrix, an N X N matrix with N representing the number of predicted classes, serves as a valuable tool. This matrix comprises four distinct combinations of predicted and actual values and proves highly beneficial for assessing precision-recall, Specificity, Accuracy, and notably, AUC-ROC curves.

(a) True positive (TP): This term denotes the accurate identification by the IDS of a genuine intrusion or attack.

(b) False positive (FP): A false positive arises when the IDS erroneously categorizes normal or harmless network activity as malicious, potentially resulting in unnecessary alerts and resource wastage.
(c) True negative (TN): A true negative is registered when the IDS correctly discerns the absence of intrusion or attack, aligning with the expected outcome for legitimate traffic.
(d) False negative (FN): A false negative transpires when the IDS fails to detect an actual intrusion or attack, signifying a missed threat.

2. Detection rate (DR): Also known as True Positive Rate (TPR) or Sensitivity, this metric calculates the percentage of actual intrusions that the IDS successfully detects. It is calculated as

$$\mathrm{DR} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN}).$$

3. False positive rate (FPR): This metric measures the percentage of benign activities incorrectly flagged as intrusions by the IDS. It is calculated as

$$\mathrm{FPR} = \mathrm{FP}/(\mathrm{FP} + \mathrm{TN}).$$

4. Accuracy: Accuracy represents the overall correctness of the IDS. It is calculated as

$$\mathrm{Accuracy} = (\mathrm{TP} + \mathrm{TN})/(\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}).$$

5. Precision: Precision measures the proportion of detected intrusions that are genuine (true positives) among all detected intrusions. It is calculated as

$$\mathrm{Precision} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP}).$$

6. Specificity: Specificity gauges the ability of the IDS to correctly identify non-intrusive traffic (true negatives). It is calculated as

$$\mathrm{Specificity} = \mathrm{TN}/(\mathrm{TN} + \mathrm{FP}).$$

7. F1 score: The F1 Score is the harmonic mean of precision and recall (sensitivity). It balances the trade-off between precision and recall and is calculated as:

$$\mathrm{F1\text{-}Score} = 2 * (\mathrm{Precision} * \mathrm{Recall})/(\mathrm{Precision} + \mathrm{Recall}).$$

8. False alarm rate (FAR): FAR measures how often the IDS raises false alarms, and it is the complement of specificity. It is calculated as

$$\mathrm{FAR} = 1 - \mathrm{Specificity}.\ \mathrm{Or}\ \mathrm{FAR} = \mathrm{FP}/\mathrm{TN} + \mathrm{FP}.$$

9. Mean time to detect (MTTD): MTTD calculates the average time it takes for the IDS to detect an intrusion after it occurs.
10. Mean time to respond (MTTR): MTTR measures the average time it takes for an organization to respond to and mitigate an identified intrusion. False Discovery Rate

(FDR): FDR measures the proportion of alerts generated by the IDS that are false positives. It is calculated as

$$\mathrm{FDR} = \mathrm{FP}/(\mathrm{TP} + \mathrm{FP}).$$

11. PID coverage: It is defined as the number of PIDs analysed by an algorithm over the number of PIDs supported by a test vehicle.

$$\mathrm{PID\ Coverage} = \mathrm{Analysed\ P\ IDs/Supported\ PIDs}.$$

Assessing an IDS with these performance metrics empowers organizations to refine their security approaches, enhance IDS settings, and strengthen their defences against cybersecurity risks. The prioritization of specific metrics can vary among organizations based on their individual security needs and risk tolerance levels.

In analyzing performance metrics across various studies, accuracy was the most frequently used, with 166 papers relying on it to measure model performance. Precision and F1-score were also heavily utilized, appearing in 117 and 118 studies, respectively, to evaluate positive predictive value and balance between precision and recall. Recall was used in 106 studies, underscoring its importance in assessing the model's ability to capture all relevant instances. Detection rate appeared in 29 papers, highlighting the focus on the model's effectiveness in identifying true positives. The false alarm rate (FAR) was assessed in 13 studies, while false positive count (FP) and sensitivity were each used in 10 studies, emphasizing the importance of reducing erroneous classifications. False positive rate (FPR) was seen in 28 papers, and false negative rate (FNR) in 11, further reflecting the attention to specific error types. Other metrics, including true positives (TP, 4), true negatives (TN, 1), and specificity (8), were adopted less frequently, indicating more tailored analyses for individual study needs. More complex or specialized metrics such as the area under the ROC curve (AUC, 14), minimum mean squared error (MMSE, 3), and ROC (3) provided nuanced performance insights, while metrics like the geometric mean (GM, 4), MMC (7), and false alarms (4) demonstrated further specificity. Less common measures, including HTER (1), TTBM (1), negative predictive value (NPV, 2), and other unique metrics like AURPR (5), PID (1), and MCR (1), illustrate the diverse approach to performance evaluation. This distribution highlights that while accuracy remains the most popular metric, there is a broad and growing selection of metrics in use, showing the complexity and precision required for evaluating cybersecurity models in diverse contexts.

10 Challenges and future work

The comprehensive analysis of 124 intrusion detection datasets reveals several persistent challenges that impede the development and evaluation of robust IDS solutions. First, no dataset simultaneously achieves the ideal properties of being up-to-date, accurately labeled, publicly available, and representative of real-world traffic across a wide spectrum of attack scenarios. Researchers therefore often resort to combining multiple heterogeneous datasets, complicating experimental design, model evaluation, and comparison across studies.

Inconsistent formats, feature sets, and incomplete documentation further hinder reproducibility. Essential information concerning dataset provenance, collection

environments, traffic-generation procedures, labeling mechanisms, feature-extraction pipelines, evaluation protocols, and maintenance status is frequently missing or reported inconsistently. The absence of common documentation requirements also makes it difficult to assess dataset quality, reproduce preprocessing pipelines, and compare experimental results across IDS studies.

Realism and scalability pose another major challenge. Many datasets are collected in synthetic, emulated, or limited-scale testbeds that fail to adequately capture the heterogeneity, encrypted traffic prevalence, background traffic diversity, temporal dynamics, evolving attack behaviour, and operational conditions of production networks. These limitations are particularly evident in under-represented domains such as IoT and IIoT, where specialized protocols, device diversity, resource constraints, and rapidly changing deployment environments are not sufficiently represented.

Privacy and anonymization concerns further restrict the availability of high-fidelity network captures, forcing trade-offs between data utility and confidentiality. Labeling remains labor-intensive and error-prone and may rely on predefined attack schedules, automated security tools, or controlled experimental scenarios that can introduce uncertainty or bias.

Dataset bias and benchmarking practices present additional challenges. Duplicate or highly correlated records, temporally related flows, infrastructure-specific features, identifiers, traffic-generation artifacts, and inappropriate record-level random splitting may lead to train–test contamination, shortcut learning, memorization, and optimistic performance estimates. Similarly, preprocessing, feature selection, dimensionality reduction, or class-balancing procedures applied before data partitioning can introduce information leakage. These issues emphasize the need for transparent preprocessing pipelines, leakage-aware evaluation protocols, and chronological, session-based, device-based, or group-aware partitioning strategies where appropriate.

The rapid evolution of cyber threats also outpaces the release cycle of publicly available datasets. Emerging threats such as multi-stage intrusions, zero-day exploits, adaptive attacks, adversarial evasion techniques, and attacks against heterogeneous IoT, IIoT, vehicular, and cyber-physical environments are inadequately represented in many existing benchmarks.

Furthermore, modern IDS architectures based on graph neural networks, federated learning, continual learning, foundation models, large language models, and adversarial machine learning introduce additional dataset requirements, including graph structures, distributed partitions, temporal streams, semantic context, provenance information, and adversarial samples. Future dataset development must therefore consider both emerging threat landscapes and the requirements of modern AI-driven IDS architectures.

Although the present survey introduces comparative quality, realism, modern AI-readiness, bias, and sustainability assessments for representative benchmark datasets, developing a unified quantitative scoring framework across all surveyed datasets remains an important direction for future research. Such a framework requires explicit and consistently measurable criteria, defensible weighting strategies, and task-specific suitability definitions to avoid subjective or misleading dataset rankings. A single aggregate score may conceal important trade-offs among dataset properties; therefore, future scoring frameworks should combine dimension-level assessments with task-specific suitability profiles.

Table 103 Recommended minimum documentation standards for IDS datasets

Documentation category	Minimum required information	Purpose
Dataset provenance and scope	Creators, release date, version, intended application domain, collection objectives, citation information, and license or usage conditions	Establishes dataset identity, scope, ownership, and appropriate use
Collection environment	Real, simulated, or emulated setting; network topology; devices and systems; protocols and services; collection duration; traffic-generation tools; and experimental configuration	Supports assessment of dataset realism, representativeness, and reproducibility
Traffic and attack characteristics	Benign activity generation, attack categories, attack tools, attacker and victim roles, attack schedules, traffic distributions, and class distributions	Enables assessment of dataset coverage and experimental limitations
Labeling and ground truth	Label-generation procedure, label granularity, synchronization with attack events, validation method, and known labeling errors or uncertainties	Improves labeling transparency and reliability of supervised evaluation
Data and feature schema	Raw and processed data availability, file formats, feature definitions and units, feature-extraction tools, software versions, parameters, and preprocessing procedures	Supports feature interpretation and reproduction of processed dataset representations
Evaluation protocol	Recommended train, validation, and test partitions; chronological, device-, session-, or group-aware splitting strategies where appropriate; known duplicate and leakage risks; and recommended evaluation metrics	Reduces benchmarking bias and improves comparability across IDS studies
Accessibility and maintenance	Persistent official source, access restrictions, version identifiers, checksums where available, changelog, documentation source, maintainer contact, and last-update information	Supports long-term accessibility, integrity, version tracking, and dataset sustainability

To address these challenges, the following directions are identified for future research:

- *Continual dataset expansion:* Design extensible and versioned datasets that can be incrementally updated with new traffic captures, network environments, attack scenarios, and threat types. Dataset updates should preserve historical versions and document changes to enable longitudinal evaluation and maintain alignment with evolving threat landscapes.
- *Standardization of formats, metadata, and documentation:* Develop community-driven minimum documentation standards for IDS datasets to improve transparency, reproducibility, and fair comparison. Such standards should cover the complete dataset lifecycle, including dataset provenance, collection environments, traffic and attack characteristics, labeling and ground-truth generation, data and feature schemas, evaluation protocols, and accessibility and maintenance information. Table 103 summarizes the proposed minimum documentation requirements.
- *Centralized repositories and reproducible toolchains:* Develop and maintain shared platforms for organizing, discovering, comparing, and accessing IDS datasets, together with reproducible pipelines for data collection, transformation, feature extraction, labeling, anonymization, and evaluation. Repositories should maintain verified access information, dataset versions, documentation sources, processing tools, and maintenance status. The continued development of SHIELD represents a step toward such centralized dataset discovery and comparison.
- *Realistic and scalable data collection:* Expand dataset collection to large-scale and heterogeneous operational environments, including enterprise networks, IoT/IIoT infrastructures, vehicular networks, critical infrastructure, software-defined networks, and cloud–edge environments. Future datasets should capture

representative encrypted traffic, diverse background activity, realistic user and device behaviour, temporal drift, evolving attack campaigns, and multi-protocol and multi-layer communication.

- *Automated and semi-supervised labeling:* Investigate semi-supervised, unsupervised, active-learning, weak-supervision, and human-in-the-loop approaches to reduce manual labeling effort while improving label accuracy and scalability. Dataset publications should clearly document label-generation procedures, validation mechanisms, uncertainty, and known labeling errors.
- *Privacy-preserving dataset publication:* Advance anonymization, data transformation, federated data-sharing, and privacy-preserving publication mechanisms that balance confidentiality with research utility. Future work should investigate approaches that preserve important behavioural, temporal, and structural characteristics while protecting sensitive information.
- *Leakage-aware and bias-resistant benchmarking:* Develop evaluation protocols that explicitly address duplicate records, temporally correlated traffic, device- and session-specific patterns, identifier leakage, preprocessing leakage, and dataset-specific shortcuts. Deduplication should be performed before partitioning, all preprocessing and resampling procedures should be fitted exclusively on training data, and chronological, session-based, device-based, attack-scenario-based, or group-aware splitting should be preferred where appropriate.
- *Adversarial-resilient and evolving benchmarks:* Create datasets and benchmark suites that include adaptive adversaries, progressive attack campaigns, adversarial examples, evasion techniques, poisoning attacks, and changing attacker behaviour. Such datasets are required to evaluate IDS robustness under realistic and evolving threat conditions.
- *Dataset readiness for modern AI-driven IDS:* Develop datasets that support emerging IDS architectures, including graph neural networks, federated learning, continual learning, foundation models, large language models, and adversarial machine learning. Future datasets should provide structural relationships, distributed partitions, temporal streams, semantic information, provenance metadata, and adversarial scenarios where appropriate.
- *Unified evaluation protocols and metrics:* Define standardized benchmark scenarios, partitioning strategies, preprocessing procedures, and reporting requirements to enable consistent evaluation across IDS studies. Performance reporting should include class-wise precision, recall, F1-score, imbalance-sensitive metrics, computational requirements, uncertainty estimates where applicable, and external or cross-dataset validation.
- *Multi-dimensional dataset scoring and selection frameworks:* Develop transparent quantitative frameworks for assessing and ranking IDS datasets according to multiple dimensions, including attack diversity, labeling quality, temporal realism, traffic representativeness, feature richness, class balance, reproducibility, accessibility, and suitability for specific IDS tasks and application domains. Future research should investigate normalized scoring criteria, evidence-based weighting strategies, and sensitivity analysis to avoid arbitrary rankings and enable researchers to adjust dimension weights according to their experimental requirements. Rather than relying exclusively on a single aggregate score, such frameworks should provide dimension-

level scores and task-specific suitability profiles to expose trade-offs among dataset characteristics. Integration of scoring and recommendation mechanisms into repositories such as SHIELD could support more objective, transparent, and application-oriented dataset comparison and selection.

- *Long-term dataset sustainability:* Establish mechanisms for maintaining dataset accessibility, documentation, version histories, feature-generation pipelines, and persistent official sources. Dataset repositories should periodically verify access links, record last-verified dates, document access restrictions, maintain changelogs, and provide alternative sources where appropriate.

To provide concrete guidance for improving consistency in IDS dataset reporting, the following table proposes a minimum set of documentation requirements covering dataset provenance, collection environments, traffic and attack characteristics, labeling procedures, feature-generation information, evaluation protocols, accessibility, and maintenance.

The proposed documentation requirements provide a minimum baseline rather than a domain-specific specification. Specialized IDS environments may require additional information depending on their communication protocols, network architectures, devices, operational constraints, and threat models. Nevertheless, consistent reporting of these core elements would substantially improve dataset transparency, reproducibility, comparison, and long-term reuse.

These documentation requirements can also guide the continued development of SHIELD. As a continuously evolving repository, SHIELD can progressively incorporate information concerning documentation completeness, dataset versions, labeling procedures, feature-generation pipelines, recommended evaluation protocols, access restrictions, maintenance status, and last-verified dates. Future integration of multi-dimensional scoring and task-specific dataset suitability profiles could enable SHIELD to support more objective dataset selection while preserving transparency regarding the individual strengths and limitations of each benchmark. Aligning repository records with common documentation requirements and evidence-based assessment criteria would enable SHIELD not only to support dataset discovery and comparison, but also to encourage more consistent dataset reporting and improve the reproducibility, sustainability, and objective selection of datasets for IDS research.

11 Conclusion

This paper presents a comprehensive survey of intrusion detection datasets with a particular emphasis on their relevance to communication networks. By examining 124 publicly available datasets and characterizing them across nearly 20 critical properties, we provide a structured and unified resource for researchers and practitioners working in network security and intrusion detection. Our analysis explores how these datasets are applied in IDS research within communication environments, including detection methodologies, feature selection strategies, classification models, performance metrics, and the suite of tools used for traffic analysis and dataset construction.

We offer a taxonomy of traffic analysis tools and a comparative framework that supports informed dataset selection and experimental planning for intrusion detection in real-world networking scenarios. Our study also highlights pressing gaps-such as limited dataset realism, inconsistent documentation, and insufficient coverage of modern

threats-and puts forth actionable recommendations to improve dataset quality, standardization, sharing, and scalability.

By addressing these gaps and aligning future dataset development with the evolving needs of communication networks, the research community can accelerate progress toward more robust, reproducible, and scalable intrusion detection solutions. We hope this survey serves as both a reference and a catalyst for advancing IDS research in the context of modern and next-generation communication systems.

Author contributions
F.S.M. conceived the study and supervised the overall work. W.A. conducted the literature collection, dataset analysis, and initial drafting of the manuscript. F.S.M. and W.A. jointly refined the structure and technical content of the survey. All authors contributed to manuscript review, editing, and approved the final version.

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